

Online Appendix: Trade and Technology Compatibility in General Equilibrium

Weili Chen
Penn State

Jingting Fan
Penn State

Wenlan Luo
Tsinghua University

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Appendix A Theory

A.1 A Random-Sampling Microfoundation of the CES Sourcing Structure

This appendix presents the random supplier-sampling model of which the CES sourcing structure in Section 2 is the reduced form: with $\xi = 1 + \zeta$, the cost fixed point and the trade shares of the two formulations coincide, with the production cost $c_d^i(\theta)$ of the main text corresponding to the location parameter $C_d^i(\theta)$ below (up to a constant). The results stated here are proved in Appendix A.2.

Consider a firm v in country-sector (d, i) that has already chosen technology $\theta(v)$. This firm can access a random set of production *techniques*, denoted by $R(v)$. Each technique $r \in R(v)$ is characterized by (i) a production efficiency $A(v, r)$ and (ii) a set of potential suppliers drawn independently and uniformly from firms in country-sector pair (o, j) for each o and j . Denote the set of suppliers for technique r from (o, j) by $\Omega_o^j(v, r)$.

Under technique r , the firm's output is given by

$$y(v, r) = A(v, r)[l(v)]^{\gamma^{iL}} \prod_j [m^j(v, r)]^{\gamma^{ij}}, \text{ with } \gamma^{iL} + \sum_j \gamma^{ij} = 1,$$

where $l(v)$ is the labor hired by firm v , and $m^j(v, r)$ is the sector- j intermediate good. Since there is no love-of-variety in input sourcing, suppliers of input j from around the world engage in perfect competition and the cheapest one wins the order from firm v .

Denote the wage rate in d by w_d . The factory-gate price of firm v under technique r is

$$p(v, r) = \frac{1}{A(v, r)} \cdot [w_d]^{\gamma^{iL}} \cdot \prod_j [c^j(v, r)]^{\gamma^{ij}}, \quad (\text{A.1})$$

where $c^j(v, r)$ is the minimum cost for input j among the suppliers available for technique r :

$$c^j(v, r) = \min_o \min_{\omega \in \Omega_o^j(v, r)} \tilde{c}^j(v, \omega).$$

$\tilde{c}^j(v, \omega)$, the (compatibility-adjusted) cost of input j from supplier $\omega \in \Omega_o^j(v, r)$ is

$$\tilde{c}^j(v, \omega) = \tau_{do}^j \cdot p(\omega) \cdot \frac{1}{z(v, \omega)} \cdot t(\theta(v), \theta(\omega)), \quad (\text{A.2})$$

where τ_{do}^j is the iceberg trade cost, $p(\omega)$ is the factory-gate price of supplier ω , and $z(v, \omega)$ is an idiosyncratic match-specific efficiency draw. $t(\theta(v), \theta(\omega))$ is an increasing function of the distance between $\theta(v)$ and $\theta(\omega)$, capturing technology compatibility.

Since firms sell their products to downstream firms at their marginal production cost, their profits accrue solely from sales to final good producers. Firm v chooses among $R(v)$ the technique that maximizes these profits. Since firms charge a constant markup to final good producers under monopolistic competition, profit maximization is equivalent to minimizing the factory-gate price:

$$p(v) = \min_{r \in R(v)} p(v, r).$$

Because firms buy from and sell to each other simultaneously, the price distribution in any (d, i) depends endogenously on those in all other country-sector pairs. To characterize these distributions, we make the following assumption on the techniques available to firms.

Assumption 3. For any firm v in country-sector (d, i) :

1. For any $a_1 < a_2 \in (0, +\infty)$, the number of production techniques with $A(v, r) \in (a_1, a_2)$ follows an independent Poisson distribution with mean $(a_1 / A_d^i)^{-\lambda} - (a_2 / A_d^i)^{-\lambda}$, with $\lambda > 1$.
2. For any production technique r and each (o, j) , suppliers in $\Omega_o^j(v, r)$ are drawn independently and uniformly from intermediate-good firms in country o sector j .
The match-specific efficiency draws of these suppliers, $z(v, \omega)$, satisfy the following: for any $z_1 < z_2 \in (0, +\infty)$, the number of suppliers in $\Omega_o^j(v, r)$ with $z(v, \omega) \in (z_1, z_2]$ follows an independent Poisson with mean $z_1^{-\zeta} - z_2^{-\zeta}$, where for integrability $\zeta \in (\max\{\max_{ij} \gamma^{ij} \lambda, 1\}, \lambda)$.

Assumption 3 specifies both the productivity draws for each technique and the match quality draws for suppliers. The Poisson assumption, which follows [Boehm and Oberfield \(2020\)](#), can be viewed as an extension to the more familiar Pareto assumption ([Chaney, 2008](#); [Kortum, 1997](#)).¹ In Part 1, A_d^i determines the efficiency level of available techniques in (d, i) —higher A_d^i implying more high-efficiency techniques, which means country d has an advantage in i —while λ governs the heterogeneity among these techniques.² In Part 2, the uniform sampling of suppliers in $\Omega_o^j(v, r)$ implies that the technologies of potential suppliers follow those of all firms in (o, j) . Parameter ζ then governs the heterogeneity of match quality among these suppliers, similar to how λ does for available techniques.

We now characterize the distribution of factory-gate prices of firms in (d, i) conditional on the technology they choose:

Proposition A.1. Under Assumption 3, $p_d^i(\theta)$, the factory-gate price of a firm in (d, i) that has chosen technology θ , follows a Weibull (inverse Fréchet) distribution with c.d.f.

$$F_d^i(p; \theta) = 1 - \exp(-[p / C_d^i(\theta)]^\lambda). \quad (\text{A.3})$$

Its location parameter $C_d^i(\theta)$ is determined as the solution to the following fixed point problem:

$$C_d^i(\theta) = \frac{\Xi^i}{A_d^i} [w_d]^{\gamma^{iL}} \prod_j \left[\sum_o \int [\tau_{do}^j C_o^j(\tilde{\theta}) t(\theta, \tilde{\theta})]^{-\zeta} d\Theta_o^j(\tilde{\theta}) \right]^{-\frac{\gamma^{ij}}{\zeta}}, \quad (\text{A.4})$$

where Ξ^i is a sector-specific constant, and Θ_o^j is the probability measure that describes the distribution of chosen technologies across firms in (o, j) .³

Proof. See Appendix A.2. □

¹Draws satisfying Part 1 of Assumption 3 can be generated as follows: take any $x > 0$, draw the number of techniques with $A(v, r) > x$ from a Poisson distribution with mean $(x / A_d^i)^{-\lambda}$; for each technique, draw its efficiency independently from a Pareto distribution with position parameter A_d^i and tail parameter λ . Based on the thinning properties of Poisson, the limit of these process when $x \rightarrow 0$ satisfies Part 1.

²An intuitive way to see this is to consider the limit as $a_2 \rightarrow \infty$, which reveals that A_d^i shapes the number of available techniques exceeding any threshold a_1 and λ governs how quickly that number decays with a_1 .

³Specifically, $\Xi^i \equiv \left(\int_0^\infty \int_0^\infty \dots \int_0^\infty \mathbb{I}[\prod_j [m^j]^{\frac{\gamma^{ij}}{\zeta}} \leq \kappa] \prod_j [\Gamma(1 - \zeta / \lambda)]^{\frac{\gamma^{ij}}{\zeta}} \exp(-m^j) \lambda \kappa^{-\lambda-1} dm^1 \dots dm^S d\kappa \right)^{-1/\lambda}$, which is finite provided that $\max\{\max_{ij} \gamma^{ij} \lambda, 1\} < \zeta < \lambda$. As long as this inequality is satisfied, the exact value of λ does not affect our quantitative exercises.

To prove this result, we show that if supplier price distributions in each (o, j) are Weibull, then the induced distribution of factory-gate prices for outputs is also Weibull. Because technology compatibility affects input sourcing, firms with different θ have different production cost distributions. We denote by $C_d^i(\theta)$ the location parameter of the cost distribution for firms in (d, i) which have chosen technology θ . In equilibrium, each $C_d^i(\theta)$ depends on all $\{C_o^j(\tilde{\theta}) : \tilde{\theta} \in \mathbb{T}\}_{o=1, j=1}^{N, S}$ through the fixed-point relationship (A.4). The fixed point (A.4) has the same form as the CES cost fixed point (5) with $\zeta = \bar{\zeta} - 1$, and the sourcing decisions below coincide with the CES trade shares (6)–(8).

Corollary A.1. *For a firm in (d, i) with technology θ*

1. *The expenditure share of the firm's input j produced by firms in o with technology $\tilde{\theta}$ is*

$$\chi_{do}^j(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) = \frac{[\tau_{do}^j C_o^j(\tilde{\theta}) t(\theta, \tilde{\theta})]^{-\zeta} \cdot d\Theta_o^j(\tilde{\theta})}{\sum_{o'} [\tau_{do'}^j \Lambda_{o'}^j(\theta)]^{-\zeta}}, \quad (\text{A.5})$$

where $d\Theta_o^j(\tilde{\theta})$ denotes the measure of firms in (o, j) with technology $\tilde{\theta}$ and

$$\Lambda_o^j(\theta) \equiv \left(\int [C_o^j(\tilde{\theta}) t(\theta, \tilde{\theta})]^{-\zeta} d\Theta_o^j(\tilde{\theta}) \right)^{-1/\zeta}. \quad (\text{A.6})$$

2. *The expenditure share of the firm's input j produced by firms in country o (across all $\tilde{\theta}$) is*

$$\chi_{do}^j(\theta) = \int \chi_{do}^j(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) = \frac{[\tau_{do}^j \Lambda_o^j(\theta)]^{-\zeta}}{\sum_{o'} [\tau_{do'}^j \Lambda_{o'}^j(\theta)]^{-\zeta}}. \quad (\text{A.7})$$

Proof. See Appendix A.2. □

Final-good producers pay $\frac{\eta}{\eta-1} \tau_{do}^{Uj} p_o^j(\omega)$ per unit purchased from an intermediate-good firm ω , where the factory-gate price $p_o^j(\omega)$ follows the distribution in (A.3). Their sourcing decision is characterized as follows.

Corollary A.2. *For the final-good producer in country d , when purchasing goods from sector j :*

1. *The expenditure share allocated to goods produced by firms in country o with technology $\tilde{\theta}$ is*

$$\pi_{do}^j(\tilde{\theta}) d\Theta_o^j(\tilde{\theta}) = \frac{[\tau_{do}^{Uj} C_o^j(\tilde{\theta})]^{1-\eta}}{\sum_{o'} [\tau_{do'}^{Uj} \bar{\Lambda}_{o'}^j]^{1-\eta}} d\Theta_o^j(\tilde{\theta}), \quad (\text{A.8})$$

$$\text{where } \bar{\Lambda}_o^j \equiv \left(\int [C_o^j(\tilde{\theta})]^{1-\eta} d\Theta_o^j(\tilde{\theta}) \right)^{1/(1-\eta)}. \quad (\text{A.9})$$

2. *The expenditure share allocated to goods produced by firms in country o is*

$$\pi_{do}^j = \int \pi_{do}^j(\tilde{\theta}) d\Theta_o^j(\tilde{\theta}) = \frac{[\tau_{do}^{Uj} \bar{\Lambda}_o^j]^{1-\eta}}{\sum_{o'} [\tau_{do'}^{Uj} \bar{\Lambda}_{o'}^j]^{1-\eta}}. \quad (\text{A.10})$$

Proof. See Appendix A.2. □

Here, $\bar{\Lambda}_o^j$ reflects the overall competitiveness of intermediate-good firms in (o, j) when serving final-good producers; because final-good producers do not consider technology compatibility, $t(\theta, \tilde{\theta})$ does not appear in $\bar{\Lambda}_o^j$. Tractability stems from the Poisson assumptions on $A(v, r)$ and $z(v, \omega)$: the assumption on $A(v, r)$ ensures $p(v, r)$ in (A.1) is Weibull if all $c^j(v, r)$ are Weibull; likewise, the assumption on $z(v, \omega)$ ensures $\tilde{c}^j(v, \omega)$ in (A.2) is Weibull if $p(\omega)$ is Weibull. The Weibull price distributions can be summarized by a small set of parameters (e.g., $C_d^i(\theta)$ in (A.3)) and deliver the shares above, which coincide with the CES shares of the main text under $\zeta = \xi - 1$.

A.2 Proof of Proposition A.1 and Corollaries A.1 and A.2

We restate Proposition A.1 below:

Proposition A.2. *Under Assumption 3, $p_d^i(\theta)$, the factory-gate price of a firm in (d, i) with technology location θ , follows a Weibull (inverse Fréchet) distribution with c.d.f.*

$$F_d^i(p; \theta) = 1 - \exp(-[p/C_d^i(\theta)]^\lambda), \quad (\text{A.11})$$

The location parameter $C_d^i(\theta)$ is determined as the fixed point of

$$C_d^i(\theta) = \frac{\Xi^i}{A_d^i} [w_d]^{\gamma^{iL}} \prod_j \left[\sum_o \int [\tau_{do}^j C_o^j(\tilde{\theta}) t(\theta, \tilde{\theta})]^{-\zeta} d\Theta_o^j(\tilde{\theta}) \right]^{-\frac{\gamma^{ij}}{\zeta}}, \quad (\text{A.12})$$

where Ξ^i is a sector-specific constant, and Θ_o^j is the probability measure that describes the distribution of chosen technologies across firms in (o, j) .

We prove Proposition A.1 by guess and verification. We start with introducing a lemma. The lemma employs the Poisson assumption in the draws of suppliers to characterize the unconditional distribution of supplier efficiency.

Lemma A.1. *Let $\{X_z\}_z$ be a collection of random variables indexed by $z \in (0, \infty)$ such that*

$$F(x, z) \equiv \Pr(X_z \leq x \mid z)$$

is jointly continuous in (x, z) . Let $\mathcal{X}$ be a random set of uniformly drawn X_z such that for any $z_1 < z_2 \in (0, +\infty)$, $|\{X_z \in \mathcal{X} : z_1 < z \leq z_2\}|$ follows an independent Poisson distribution with mean $H(z_1) - H(z_2)$, where $|\cdot|$ denotes the number of elements in a set, and $H(z)$ is a decreasing function on $(0, +\infty)$. Then, $|\{X_z \in \mathcal{X} : X_z \leq x\}|$ follows a Poisson distribution with mean

$$\int_0^\infty F(x, z) d(-H(z)),$$

provided that the integral exists.

Proof. Since X_z is uniformly drawn, $|\{X_z \in \mathcal{X} : X_z \leq x, z_1 < z \leq z_2\}|$ follows a Poisson distribution with mean $G(x, z_1, z_2)$ that satisfies

$$\inf_{\bar{z} \in (z_1, z_2]} F(x, \bar{z})[H(z_1) - H(z_2)] \leq G(x, z_1, z_2) \leq \sup_{\bar{z} \in (z_1, z_2]} F(x, \bar{z})[H(z_1) - H(z_2)].$$

Taking any monotonically increasing sequence $\{z_i\}_{i=1}^\infty$ with $z_1 = 0$ and $\lim_{i \rightarrow \infty} z_i = \infty$, we have: (since the number draws in any interval follow independent Poisson, the sum of draws

across these intervals follow a Poisson)

$$\sum_{i=1}^{\infty} \inf_{\bar{z} \in (z_i, z_{i+1}]} F(x, \bar{z}) [H(z_i) - H(z_{i+1})] \leq \lim_{\bar{z} \rightarrow \infty} G(x, 0, \bar{z}) \leq \sum_{i=1}^{\infty} \sup_{\bar{z} \in (z_i, z_{i+1}]} F(x, \bar{z}) [H(z_i) - H(z_{i+1})].$$

Therefore,

$$\lim_{\bar{z} \rightarrow \infty} G(x, 0, \bar{z}) = \int_0^{\infty} F(x, z) d(-H(z)),$$

provided that the integral exists (in the sense of Riemann integration). $\square$

Proof of Proposition A.1. We now prove Proposition A.1.

Proof. Suppose $p_d^i(\theta)$, the factory-gate price of a firm in (d, i) with technology location θ , follows a Weibull distribution with c.d.f. specified in (A.3). We verify that this is consistent with firms' behaviors in the equilibrium.

Consider a firm v in (d, i) with technology location $\theta = \theta(v)$. This firm chooses among all techniques one that minimizes its production cost. For each technique r , the firm chooses one supplier in each sector j to source input.

We first characterize the distribution of production cost for different techniques, which we denote by $c^j(v, r)$. Recall that

$$c^j(v, r) = \min_o \min_{\omega \in \Omega_o^j(v, r)} \tilde{c}^j(v, \omega),$$

where $\tilde{c}^j(v, \omega)$, the (compatibility-adjusted) cost of for input j from supplier $\omega \in \Omega_o^j(v, r)$ is

$$\tilde{c}^j(v, \omega) = \tau_{do}^j \cdot p(\omega) \cdot \frac{1}{z(v, \omega)} \cdot t(\theta(v), \theta(\omega)),$$

Assumption 3 specifies the distribution of the match-specific efficiency of potential suppliers, $z(v, \omega)$. Following Lemma A.1, from any origin country-sector pair (o, j) , the number of suppliers such that each supplier ω has a technology location $\theta(\omega)$ around $\tilde{\theta}$ and an effective marginal cost $\tilde{c}^j(v, \omega)$ less or equal to $c > 0$ follows a Poisson distribution with mean

$$\begin{aligned} & \int_0^{\infty} F_o^j \left[\frac{z \cdot c}{\tau_{do}^j \cdot t(\theta, \tilde{\theta})}; \tilde{\theta} \right] \zeta z^{-\zeta-1} dz \cdot d\Theta_o^j(\tilde{\theta}) \\ &= \int_0^{\infty} F_o^j(\kappa; \tilde{\theta}) \zeta \kappa^{-\zeta-1} \left[\frac{c}{\tau_{do}^j \cdot t(\theta, \tilde{\theta})} \right]^{\zeta} d\kappa \cdot d\Theta_o^j(\tilde{\theta}) \\ &= \int_0^{\infty} [t(\theta, \tilde{\theta}) \kappa]^{-\zeta} dF_o^j(\kappa; \tilde{\theta}) \cdot \left(\frac{c}{\tau_{do}^j} \right)^{\zeta} \cdot d\Theta_o^j(\tilde{\theta}) \\ &= \Gamma(1 - \zeta/\lambda) \cdot [t(\theta, \tilde{\theta}) C_o^j(\tilde{\theta})]^{-\zeta} \cdot \left(\frac{c}{\tau_{do}^j} \right)^{\zeta} \cdot d\Theta_o^j(\tilde{\theta}). \end{aligned} \tag{A.13}$$

Integrating over $\tilde{\theta}$, the number of suppliers with effective marginal cost $\tilde{c}^j(v, \omega)$ less or equal to any level $c > 0$ follows a Poisson distribution with mean

$$\Gamma(1 - \zeta/\lambda) \cdot \left(\frac{c}{\tau_{do}^j \cdot \Lambda_o^j(\theta)} \right)^{\zeta},$$

where $\Lambda_o^j(\cdot)$ is defined in (A.6),

$$\Lambda_o^j(\theta) \equiv \left(\int [C_o^j(\tilde{\theta})t(\theta, \tilde{\theta})]^{-\zeta} d\Theta_o^j(\tilde{\theta}) \right)^{-1/\zeta}.$$

Therefore, for each (o, j) , the probability that no such supplier arrives is

$$\Pr \left[\min_{\omega \in \Omega_o^j(\nu, r)} \tilde{c}^j(\nu, \omega) > c \right] = \exp \left[-\Gamma(1 - \zeta/\lambda) \cdot \left(\frac{c}{\tau_{do}^j \cdot \Lambda_o^j(\theta)} \right)^\zeta \right].$$

The cost distribution for each j when choosing technique r , $c^j(\nu, r)$, which is the minimum over all countries, is hence characterized by

$$\begin{aligned} \Pr[c^j(\nu, r) > c] &= \exp[-\Gamma(1 - \zeta/\lambda) \cdot \sum_o \left(\frac{c}{\tau_{do}^j \cdot \Lambda_o^j(\theta)} \right)^\zeta] \\ &= \exp[-\tilde{\Lambda}_d^j(\theta) c^\zeta], \end{aligned} \quad (\text{A.14})$$

where $\tilde{\Lambda}_d^j(\theta) \equiv \Gamma(1 - \zeta/\lambda) \sum_o (\tau_{do}^j \Lambda_o^j(\theta))^{-\zeta}$.

Among all available techniques, the firm chooses one to minimize its factory-gate price:

$$p(\nu) = \min_{r \in R(\nu)} \frac{1}{A(\nu, r)} \cdot [w_d]^{\gamma^{iL}} \cdot \prod_j [c^j(\nu, r)]^{\gamma^{ij}}.$$

Assumption 3 specifies the distribution of production efficiency $A(\nu, r)$. By applying Lemma A.1 again, the number of techniques such that the factory-gate price is weakly less than p follows a Poisson distribution with mean

$$\begin{aligned} & \int_0^\infty \int_0^\infty \dots \int_0^\infty \mathbb{I} \left[\frac{1}{a} [w_d]^{\gamma^{iL}} \prod_j [c^j]^{\gamma^{ij}} \leq p \right] \\ & \quad \cdot \prod_j \zeta [c^j]^{\zeta-1} \tilde{\Lambda}_d^j(\theta) \exp[-\underbrace{\tilde{\Lambda}_d^j(\theta) [c^j]^\zeta}_{\equiv m^j}] \lambda [A_d^i]^\lambda a^{-\lambda-1} da^1 \dots da^S da \\ &= \int_0^\infty \int_0^\infty \dots \int_0^\infty \mathbb{I} \left[\frac{1}{a} [w_d]^{\gamma^{iL}} \prod_j \left(\frac{m^j}{\tilde{\Lambda}_d^j(\theta)} \right)^{\frac{\gamma^{ij}}{\zeta}} \leq p \right] \prod_j \exp(-m^j) \lambda [A_d^i]^\lambda a^{-\lambda-1} dm^1 \dots dm^S da \\ &= \int_0^\infty \int_0^\infty \dots \int_0^\infty \mathbb{I} \left[\underbrace{\prod_j [m^j]^{\frac{\gamma^{ij}}{\zeta}}}_{\equiv \kappa} \leq ap [w_d]^{-\gamma^{iL}} \prod_j \tilde{\Lambda}_d^j(\theta)^{\frac{\gamma^{ij}}{\zeta}} \right] \prod_j \exp(-m^j) \lambda [A_d^i]^\lambda a^{-\lambda-1} dm^1 \dots dm^S da \\ &= \int_0^\infty \int_0^\infty \dots \int_0^\infty \mathbb{I} \left[\prod_j [m^j]^{\frac{\gamma^{ij}}{\zeta}} \leq \kappa \right] \prod_j \exp(-m^j) [A_d^i]^\lambda (p [w_d]^{-\gamma^{iL}} \prod_j \tilde{\Lambda}_d^j(\theta)^{\frac{\gamma^{ij}}{\zeta}})^\lambda \lambda \kappa^{-\lambda-1} dm^1 \dots dm^S d\kappa \\ &= \int_0^\infty \int_0^\infty \dots \int_0^\infty \mathbb{I} \left[\prod_j [m^j]^{\frac{\gamma^{ij}}{\zeta}} \leq \kappa \right] \prod_j [\Gamma(1 - \zeta/\lambda)]^{\frac{\gamma^{ij}}{\zeta}} \exp(-m^j) \lambda \kappa^{-\lambda-1} dm^1 \dots dm^S d\kappa \\ & \quad \cdot [A_d^i]^\lambda (p [w_d]^{-\gamma^{iL}} \prod_j [\sum_o (\tau_{do}^j \Lambda_o^j(\theta))^{-\zeta}]^{\frac{\gamma^{ij}}{\zeta}})^\lambda \\ &= [\Xi^i]^{-\lambda} [A_d^i]^\lambda ([w_d]^{-\gamma^{iL}} \prod_j [\sum_o (\tau_{do}^j \Lambda_o^j(\theta))^{-\zeta}]^{\frac{\gamma^{ij}}{\zeta}})^\lambda p^\lambda, \end{aligned}$$

where

$$\Xi^i \equiv \left(\int_0^\infty \int_0^\infty \dots \int_0^\infty \mathbb{I}[\prod_j [m^j]^{\frac{\gamma^{ij}}{\zeta}} \leq \kappa] \prod_j [\Gamma(1 - \zeta/\lambda)]^{\frac{\gamma^{ij}}{\zeta}} \exp(-m^j) \lambda \kappa^{-\lambda-1} dm^1 \dots dm^S d\kappa \right)^{-1/\lambda} \quad (\text{A.15})$$

is a sector-specific constant that depends on the technology parameters only.

The probability that no such technique arrives is

$$\Pr[p(\nu) > p] = \exp(-[p/C_d^i(\theta)]^\lambda),$$

where

$$C_d^i(\theta) = \frac{\Xi_i}{A_d^i} [w_d]^{\gamma^{iL}} \prod_j (\sum_o [\tau_{do}^j \Lambda_o^j(\theta)]^{-\zeta})^{-\frac{\gamma^{ij}}{\zeta}} = \frac{\Xi_i}{A_d^i} [w_d]^{\gamma^{iL}} \prod_j (\sum_o \int [\tau_{do}^j C_o^j(\tilde{\theta}) t(\theta, \tilde{\theta})]^{-\zeta} d\Theta_o^j(\tilde{\theta}))^{-\frac{\gamma^{ij}}{\zeta}}.$$

Therefore, for each destination country-sector pair (d, i) , the factory-gate price of a firm with technology location θ is characterized by

$$F_d^i(p; \theta) = 1 - \exp(-[p/C_d^i(\theta)]^\lambda).$$

This completes the proof of Proposition A.1. $\square$

Proof of Corollary A.1. As is shown in (A.13), for any firm in (d, i) with targeted technology location θ , in any sourcing country o , the number of suppliers from sector j with technology location $\tilde{\theta}$ and effective marginal cost less or equal to any level $c > 0$ follows a Poisson distribution with mean

$$\Gamma(1 - \zeta/\lambda) \cdot [t(\theta, \tilde{\theta}) C_o^j(\tilde{\theta})]^{-\zeta} \cdot \left(\frac{c}{\tau_{do}^j}\right)^\zeta d\Theta_o^j(\tilde{\theta}).$$

This implies that the effective cost of sourcing sector- j input from firms with technology location around $\tilde{\theta}$ from o , denoted $\tilde{c}_{do}^j(\theta, \tilde{\theta})$, is distributed with c.d.f.

$$\tilde{G}_{do}^j(c; \theta, \tilde{\theta}) = 1 - \exp[-\Gamma(1 - \zeta/\lambda) \cdot [\tau_{do}^j t(\theta, \tilde{\theta}) C_o^j(\tilde{\theta})]^{-\zeta} \cdot c^\zeta d\Theta_o^j(\tilde{\theta})].$$

Therefore, the probability of sourcing sector- j inputs from firms in country o with technology location around $\tilde{\theta}$ is

$$\begin{aligned} & \Pr[(o, \tilde{\theta}) = \arg \min_{o', \tilde{\theta}'} \tilde{c}_{do'}^j(\theta, \tilde{\theta}')] \\ &= \int_0^{+\infty} \Pr[\tilde{c}_{do}^j(\theta, \tilde{\theta}) = c \cap \tilde{c}_{do'}^j(\theta, \tilde{\theta}) > c, \forall (o', \tilde{\theta}') \neq (o, \tilde{\theta})] dc \\ &= \int_0^{+\infty} \prod_{(o', \tilde{\theta}') \neq (o, \tilde{\theta})} [1 - \tilde{G}_{do'}^j(c; \theta, \tilde{\theta}')] d\tilde{G}_{do}^j(c; \theta, \tilde{\theta}) \\ &= \int_0^{+\infty} \left\{ \exp[-\Gamma(1 - \zeta/\lambda) \cdot \sum_{o'} \int [\tau_{do'}^j t(\theta, \tilde{\theta}) C_{o'}^j(\tilde{\theta})]^{-\zeta} d\Theta_{o'}^j(\tilde{\theta}) \cdot c^\zeta] \right. \\ & \quad \cdot \Gamma(1 - \zeta/\lambda) \cdot [\tau_{do}^j t(\theta, \tilde{\theta}) C_o^j(\tilde{\theta})]^{-\zeta} d\Theta_o^j(\tilde{\theta}) \cdot \zeta c^{\zeta-1} \Big\} dc \\ &= \frac{[\tau_{do}^j t(\theta, \tilde{\theta}) C_o^j(\tilde{\theta})]^{-\zeta}}{\sum_{o'} [\tau_{do'}^j \Lambda_{o'}^j(\theta)]^{-\zeta}} d\Theta_o^j(\tilde{\theta}) \\ &\equiv \chi_{do}^j(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}). \end{aligned}$$

Integrating over $\tilde{\theta}$, the probability of sourcing from firms in country o is

$$\chi_{do}^j(\theta) = \int \chi_{do}^j(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) = \frac{[\tau_{do}^j \Lambda_o^j(\theta)]^{-\zeta}}{\sum_{o'} [\tau_{do'}^j \Lambda_{o'}^j(\theta)]^{-\zeta}}.$$

This completes the proof of Corollary A.1.

Proof of Corollary A.2. Since firms engage in monopolistic competition when selling to final-good producers, they charge a monopolistic markup $\eta/(\eta - 1)$. Final-good producers maximize their profits

$$P_d Q_d - \sum_j \sum_o \int_0^1 \left[\frac{\eta}{\eta - 1} \tau_{do}^{Uj} p_{do}^j(\omega) \right] q_{do}^j(\omega) d\omega$$

subject to (1), where the factory-gate price $p_{do}^j(\omega)$ follows the distribution characterized by $F_d^i(p; \theta) = 1 - \exp(-[p/C_d^i(\theta)]^\lambda)$, and the ideal price index for the final good in d ,

$$P_d = \prod_j \left(\frac{P_d^j}{\rho_d^j} \right)^{\rho_d^j}, \quad \text{with } P_d^j \equiv \left(\sum_o \int_0^1 \left[\frac{\eta}{\eta - 1} \tau_{do}^{Uj} p_{do}^j(\omega) \right]^{1-\eta} d\omega \right)^{\frac{1}{1-\eta}}$$

Optimization of the final-good production implies that in country d , the market share of any good ω from country o in sector j over all goods in the sector is given by

$$\frac{\frac{\eta}{\eta-1} \tau_{do}^{Uj} p_{do}^j(\omega) x_{do}^j(\omega)}{\rho_d^j P_d Q_d} = \frac{\left[\frac{\eta}{\eta-1} \tau_{do}^{Uj} p_{do}^j(\omega) \right]^{1-\eta}}{(P_d^j)^{1-\eta}}.$$

Taking expectation overs firms with technology location around $\tilde{\theta}$, we have the expected expenditure share allocated to these firms:

$$\begin{aligned} \pi_{do}^j(\tilde{\theta}) d\Theta_o^j(\tilde{\theta}) &\equiv \frac{\mathbb{E} \left[\left(\frac{\eta}{\eta-1} \tau_{do}^{Uj} p_o^j(\tilde{\theta}) \right)^{1-\eta} \right]}{(P_d^j)^{1-\eta}} \cdot d\Theta_o^j(\tilde{\theta}) \\ &= \frac{\left[\frac{\eta}{\eta-1} \tau_{do}^{Uj} \right]^{1-\eta} \mathbb{E} [p_{do'}^j(\tilde{\theta})^{1-\eta}]}{\sum_{o'} \left[\frac{\eta}{\eta-1} \tau_{do'}^{Uj} \right]^{1-\eta} \int \mathbb{E} [p_{do'}^j(\tilde{\theta})^{1-\eta}] d\Theta_{o'}^j(\tilde{\theta})} \cdot d\Theta_o^j(\tilde{\theta}) \\ &= \frac{\Gamma(1 + \frac{1-\eta}{\lambda}) \cdot \left[\frac{\eta}{\eta-1} \tau_{do}^{Uj} \right]^{1-\eta} \cdot C_o^j(\tilde{\theta})^{1-\eta}}{\sum_{o'} \Gamma(1 + \frac{1-\eta}{\lambda}) \cdot \left[\frac{\eta}{\eta-1} \tau_{do'}^{Uj} \right]^{1-\eta} \cdot (\bar{\Lambda}_{o'}^j)^{1-\eta}} \cdot d\Theta_o^j(\tilde{\theta}), \end{aligned}$$

where the last equation follows the property of Weibull distribution,⁴ and

$$\bar{\Lambda}_o^j \equiv \left(\int [C_o^j(\tilde{\theta})]^{1-\eta} d\Theta_o^j(\tilde{\theta}) \right)^{1/(1-\eta)}.$$

Integrating over $\tilde{\theta}$, the total expenditure share allocated to goods produced by firms in

⁴Suppose random variable X follows a Weibull distribution with c.d.f. $F(x) = 1 - \exp[-(x/C)^\lambda]$, where $C > 0$ and $\lambda > 1$ are parameters. Then, for any $\varepsilon > 0$,

$$\mathbb{E}[X^\varepsilon] = \int_0^{+\infty} x^\varepsilon \cdot \exp[-(x/C)^\lambda] \cdot C^{-\lambda} \lambda x^{\lambda-1} dx = C^\varepsilon \cdot \int_0^{+\infty} \kappa^{\varepsilon/\lambda} \exp(-\kappa) d\kappa = C^\varepsilon \cdot \Gamma(1 + \frac{\varepsilon}{\lambda}),$$

where $\Gamma(\cdot)$ denotes the Gamma function.

country o is

$$\pi_{do}^j = \int \pi_{do}^j(\tilde{\theta}) d\Theta_o^j(\tilde{\theta}) = \frac{[\tau_{do}^{Uj} \bar{\Lambda}_o^j]^{1-\eta}}{\sum_{o'} [\tau_{do'}^{Uj} \bar{\Lambda}_{o'}^j]^{1-\eta}}.$$

This completes the proof of Corollary A.2.

A.3 Proof of Proposition 1

We restate Proposition 1 below:

Proposition A.3. *Suppose wages $\{w_d\}$ are given.*

1. *Assume $\{\bar{\Theta}_d^i\}$ have bounded support that is contained in $[-M, M]$ for some $M > 0$ and have associated density functions $\{\bar{\varsigma}_d^i\}$. If $(\xi - 1)\bar{t} < 1/M^2$, then there exists an equilibrium with firms' technology choice $\{g_d^i\}$ being continuously differentiable functions. Moreover, in this equilibrium, the choice of firms from (d, i) with endowment technology $\bar{\theta}$ is characterized by the following first-order condition with a unique solution.*

$$g_d^i(\bar{\theta}) = (1 - \omega^i)\bar{\theta} + \omega^i \sum_{j,o} \frac{\gamma^{ij}}{1 - \gamma^{iL}} \int \chi_{do}^j(g_d^i(\bar{\theta}), \bar{\theta}) \bar{\theta} d\Theta_o^j(\bar{\theta}), \quad \forall \bar{\theta} \in [-M, M] \quad (\text{A.16})$$

$$\text{where } \omega^i \equiv \frac{(\eta-1)(1-\gamma^{iL})\bar{t}}{(\eta-1)(1-\gamma^{iL})\bar{t} + \bar{\phi}} < 1.$$

2. *If, in addition, $\bar{t} < \frac{1}{2M}$ and $\bar{\phi} > \underline{\phi}$, where $\underline{\phi} > 0$ is a constant determined by parameters $(\xi - 1, \bar{t}, \eta, M, \gamma^{iL})$ as detailed in the proof, then such an equilibrium is unique.*

We outline the proof and explain the intuition behind each step, with technical details deferred to subsequent lemmas. To simplify notation, we begin by introducing the following constants.

Definition A.1. *Given wages w_d and parameters $M, \eta, \gamma^{iL}, \xi, \bar{t}, A_d^i, \tau_{do}^j$ of the model, define constants*

- $\underline{\gamma}^L \equiv \min_i \gamma^{iL}$.
- $\omega^i \equiv \frac{(\eta-1)(1-\gamma^{iL})\bar{t}}{(\eta-1)(1-\gamma^{iL})\bar{t} + \bar{\phi}}, \bar{\omega} \equiv \max_i \omega^i$.
- $\varkappa_d^i \equiv (w_d)^{\gamma^{iL}} / A_d^i$.
- $\bar{M}' \equiv \max_i \{1 - \omega^i(1 - (\xi - 1)\bar{t}M^2)\}, \underline{M}' \equiv 1 - \bar{\omega}$.
- $\bar{M}'' \equiv \frac{3\bar{\omega}(\xi-1)\bar{t}M^3}{1 - \bar{\omega}(\xi-1)\bar{t}M^2}$.
- $M^C \equiv \max_{d,i,j} \left| \frac{1}{\gamma^{iL}} \left[\ln \varkappa_d^i + (1 - \gamma^{iL}) \ln \{\sum_o [\tau_{do}^j]^{1-\xi}\}^{-\frac{1}{\xi-1}} \right] \right|$.

We first characterize the optimality conditions for technology choices. Under Assumption 1, with the objective function specified in (15), the first-order condition with respect to

technology choice θ is

$$\bar{\phi}(\bar{\theta} - \theta) = (\eta - 1) \frac{d}{d\theta} \ln c_d^i(\theta) = (\eta - 1) \bar{t} \cdot \sum_{j,o} \gamma^{ij} \int \chi_{do}^j(\theta, \tilde{\theta})(\theta - \tilde{\theta}) d\Theta_o^j(\tilde{\theta}),$$

which can be rearranged into

$$\theta = \omega^i \sum_{j,o} \frac{\gamma^{ij}}{1 - \gamma^{iL}} \int \chi_{do}^j(\theta, \tilde{\theta}) \tilde{\theta} d\Theta_o^j(\tilde{\theta}) + (1 - \omega^i) \bar{\theta}, \quad (\text{A.17})$$

with $\omega^i < 1$. The second-order condition requires that $-\bar{\phi} - (\eta - 1) \frac{d^2}{d\theta^2} \ln c_d^i(\theta) < 0$.

To explicitly work with the policy function, we rewrite (A.17) into

$$g_d^i(\bar{\theta}) = \omega^i \sum_{j,o} \frac{\gamma^{ij}}{1 - \gamma^{iL}} \int \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}') g_o^j(\tilde{\theta}) d\tilde{\theta} + (1 - \omega^i) \bar{\theta}, \quad (\text{A.18})$$

where, to slightly abuse notations and make them dependent on $\mathbf{g}$ and $\mathbf{g}'$ explicitly,⁵

$$\chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}') \equiv \frac{[\tau_{do}^j]^{1-\xi} \exp\left(-(\xi - 1)C_o^j(\bar{\theta}; \mathbf{g}, \mathbf{g}')\right) \exp\left(-\frac{1}{2}(\xi - 1)\bar{t}(g_d^i(\bar{\theta}) - g_o^j(\tilde{\theta}))^2\right) \bar{\zeta}_o^j(\tilde{\theta})}{\sum_m \int [\tau_{dm}^j]^{1-\xi} \exp\left(-(\xi - 1)C_m^j(\bar{\theta}; \mathbf{g}, \mathbf{g}')\right) \exp\left(-\frac{1}{2}(\xi - 1)\bar{t}(g_d^i(\bar{\theta}) - g_m^j(\tilde{\theta}))^2\right) \bar{\zeta}_m^j(\tilde{\theta}) d\tilde{\theta}}, \quad (\text{A.19})$$

$$C_d^i(\bar{\theta}; \mathbf{g}, \mathbf{g}') \equiv \ln \mathcal{C}_d^i - \sum_j \frac{\gamma^{ij}}{\xi - 1} \ln \left(\sum_o \int [\tau_{do}^j]^{1-\xi} \exp\left(-(\xi - 1)C_o^j(\bar{\theta}; \mathbf{g}, \mathbf{g}')\right) \exp\left(-\frac{1}{2}(\xi - 1)\bar{t}(g_d^i(\bar{\theta}) - g_o^j(\tilde{\theta}))^2\right) \bar{\zeta}_o^j(\tilde{\theta}) d\tilde{\theta} \right). \quad (\text{A.20})$$

Note that the integrals are over *endowment* technologies: because Θ_o^j is the pushforward of $\bar{\Theta}_o^j$ under g_o^j (equation (16)), integrals against $d\Theta_o^j$ rewrite as integrals against the ex-ante density, $\int f(\tilde{\theta}') d\Theta_o^j(\tilde{\theta}') = \int f(g_o^j(\tilde{\theta})) \bar{\zeta}_o^j(\tilde{\theta}) d\tilde{\theta}$, with no Jacobian term. In particular, with $C_d^i(\bar{\theta}; \mathbf{g}, \mathbf{g}') = \ln c_d^i(g_d^i(\bar{\theta}))$, equation (A.20) is the log of the cost fixed point (5) evaluated along the policy functions. Our goal is to prove the existence and uniqueness of an equilibrium with the policy function characterized by (A.18).

To show existence, we formulate a fixed-point problem for the policy function $\mathbf{g}$. We work on the space of function defined over $[-M, M]$ with uniformly bounded value and Lipschitz continuous first derivative. Define

$$\mathcal{G} = \{\mathbf{g} : [-M, M] \rightarrow \mathbb{R}^{N \times S}, \mathbf{g} \text{ is differentiable, } \|\mathbf{g}\|_\infty \leq M;$$

$$[g_d^i]'(\bar{\theta}) \in [\underline{M}', \overline{M}'] \text{ and } \mathbf{g}' \text{ is Lipschitz continuous with a Lipschitz constant } \overline{M}''\}.$$

Define an operator $\mathcal{T}$ on $\mathcal{G}$ as below (viewing $\mathbf{g}'$ as an operator mapping $\mathbf{g}$ to its first derivative):

$$[\mathcal{T}\mathbf{g}]_d^i(\bar{\theta}) = \omega^i \sum_{j,o} \frac{\gamma^{ij}}{1 - \gamma^{iL}} \int \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}') g_o^j(\tilde{\theta}) d\tilde{\theta} + (1 - \omega^i) \bar{\theta}, \quad (\text{A.21})$$

with $\omega^i < 1$. The fixed point of operator $\mathcal{T}$, if exists, solves the first-order condition of the technology adaptation problem, i.e. equation (A.18).

Intuitively, if χ_{do}^{ij} does not vary across $\bar{\theta}$ or change by $\mathbf{g}$, then we have already established the unique existence of $\mathbf{g}$ by the contraction mapping. To see this, denote $\tilde{\chi}_{do}^{ij}(\tilde{\theta}) = \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta})$

⁵Note that $\chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}')$ takes the *endowed* technologies as the first two arguments, which is different from $\chi_{do}^j(\cdot, \cdot)$ in (A.16) that takes the chosen technologies as the arguments. Also, $C_d^i(\bar{\theta}; \mathbf{g}, \mathbf{g}') = \ln c_d^i(g_d^i(\bar{\theta}))$.

under the premise. Then the equation above is reduced to

$$[\mathcal{T}\mathbf{g}]_d^i(\bar{\theta}) = \omega^i \sum_{j,o} \frac{\gamma^{ij}}{1 - \gamma^{iL}} \int \tilde{\chi}_{do}^{ij}(\bar{\theta}) g_o^j(\bar{\theta}) d\bar{\theta} + (1 - \omega^i) \bar{\theta},$$

with $\mathcal{T}$ trivially satisfying Blackwell's sufficiency conditions for contraction with a modulus $\bar{\omega} \equiv \max_i \omega^i$.

Following this intuition, Part 1 of the proposition says that if $(\xi - 1)\bar{t}$ is not too large relative to the variation in $\bar{\theta}$, then the derivative of $\chi_{do}^{ij}(\bar{\theta}, \bar{\theta}; \mathbf{g}, \mathbf{g}')$ in $\bar{\theta}$ is uniformly bounded. The first and second derivative of $\mathbf{g}$ thus exist and are uniformly bounded. The first derivative of $\mathbf{g}$ being bounded also implies that the value of $\mathbf{g}$ is bounded on the domain $[-M, M]$. This is sufficient to establish compactness of the space of $\mathbf{g}$ and hence ensures existence.

The formal proof of existence consists of three steps. Firstly, in Lemma A.2, we establish the existence and uniqueness of $\mathcal{C}$ given $\mathbf{g}, \mathbf{g}'$ by formulating (A.20) as a fixed point problem for $\mathcal{C}$. It also establishes that $\mathcal{C}$ is continuous in $\mathbf{g}, \mathbf{g}'$ and is differentiable in $\bar{\theta}$ with bounded derivatives for any given $\mathbf{g}, \mathbf{g}'$. Since $[\mathcal{T}\mathbf{g}]'$ involves $\mathbf{g}'$ through the chain rule, we equip $\mathcal{G}$ with the C^1 norm: $\|\mathbf{g}\|_{\mathcal{G}} = \|\mathbf{g}\|_{\infty} + \|\mathbf{g}'\|_{\infty}$. It follows that χ is also continuous in $\mathbf{g}, \mathbf{g}'$, and that $\mathcal{T}$ is continuous in $\mathbf{g}$ with the norm $\|\cdot\|_{\mathcal{G}}$.

Secondly, Lemma A.3 further characterizes the uniform bounds for $\mathcal{T}\mathbf{g}, [\mathcal{T}\mathbf{g}]'$ and the Lipschitz continuity of $[\mathcal{T}\mathbf{g}]'$. This implies that $\mathcal{T}\mathbf{g} \in \mathcal{G}$.

Lastly, in Proposition A.4, we show that since $\mathbf{g} \in \mathcal{G}$ is closed and has uniformly bounded values and Lipschitz-continuous first derivatives, by the Arzelà-Ascoli theorem, $\mathcal{G}$ is compact under the norm $\|\cdot\|_{\mathcal{G}}$. That $\mathcal{T}\mathbf{g} \in \mathcal{G}$ and $\mathcal{T}$ is continuous under the norm $\|\cdot\|_{\mathcal{G}}$ thus ensures the existence of a fixed point of $\mathcal{T}$ in $\mathcal{G}$ by the Schauder fixed-point theorem.

For uniqueness, Part 2 of the proposition says that if further $\bar{\phi}$ is large enough, then the variation of $\chi_{do}^{ij}(\bar{\theta}, \bar{\theta}; \mathbf{g}, \mathbf{g}')$ in $(\mathbf{g}, \mathbf{g}')$ is bounded uniformly to ensure that $\mathcal{T}$ is a contraction. To show this, we treat (A.18) and (A.20) as a joint fixed point problem for $(\mathbf{g}, \mathcal{C})$. We show that with $\bar{t}$ small enough and $\bar{\phi}$ large enough, the joint operator defined by the left-hand side of these equations forms a contraction mapping under the C^1 norm of the space of $\mathbf{g}$ combined with the C^0 norm of the space of $\mathcal{C}$. This is achieved by proving that the induced matrix norm—specifically, the maximum absolute row sum norm—of the Jacobian matrix containing the Fréchet derivatives of the operator with respect to $(\mathbf{g}, \mathcal{C})$ is uniformly bounded by a constant less than one, provided that $\bar{t}$ is sufficiently small and $\bar{\phi}$ is sufficiently large. The bounds on the Jacobian entries are detailed in Lemma A.4, and the formal proof of uniqueness is presented in Proposition A.5.

Lemma A.2. *For $\mathbf{g} \in \mathcal{G}$, there uniquely exists a $\mathcal{C}(\bar{\theta}; \mathbf{g}, \mathbf{g}')$ defined by (A.20) that is bounded and continuous in $(\bar{\theta}, \mathbf{g}, \mathbf{g}')$. Further, $\mathcal{C}$ satisfies*

- (1) $\|\mathcal{C}\|_{\infty} \leq M^{\mathcal{C}}$, in which $M^{\mathcal{C}}$ is defined in Definition A.1.
- (2) $\forall (\mathbf{g}, \mathbf{g}'), \mathcal{C}(\bar{\theta}; \mathbf{g}, \mathbf{g}')$ is differentiable in $\bar{\theta}$ and $\|\mathcal{C}'\|_{\infty} \leq (1 - \gamma^L) 2\bar{t} M \bar{M}'$.

Proof. Denote $\tilde{\mathcal{G}} = \{\mathbf{g} \in C^0([-M, M] \rightarrow \mathbb{R}^{N \times S}) : \|\mathbf{g}\|_{\infty} \leq M\}$, $\tilde{\mathcal{G}}' = \{\mathbf{g}' \in C^0([-M, M] \rightarrow \mathbb{R}^{N \times S}) : \underline{M}' \leq \mathbf{g}' \leq \bar{M}'\}$, and $\mathcal{M} = [-M, M]$, where $\underline{M}'$ and $\bar{M}'$ are the constants defined in Definition A.1.

Define $\mathbf{C} = \{\mathcal{C} : \mathcal{M} \times \mathcal{G} \times \tilde{\mathcal{G}} \rightarrow \mathbb{R}^{N \times S}, \|\mathcal{C}\|_\infty \leq M^C, \mathcal{C} \text{ is continuous}\}$, for M^C defined in the proposition. It can be shown that $\mathbf{C}$ is complete under the infinity norm.

We now prove operator $\mathcal{T}^C$ mapping from $\mathbf{C}$ defined below has an image contained in $\mathbf{C}$:

$$[\mathcal{T}^C \mathcal{C}]_d^i(\bar{\theta}; \mathbf{g}, \mathbf{g}') = \ln \mathcal{K}_d^i - \sum_j \frac{\gamma^{ij}}{\xi - 1} \ln \left(\sum_o \int [\tau_{do}^j]^{1-\xi} \exp \left(-(\xi - 1) \mathcal{C}_o^j(\bar{\theta}; \mathbf{g}, \mathbf{g}') \right) \exp \left(-\frac{1}{2}(\xi - 1) \bar{f}(g_d^i(\bar{\theta}) - g_o^j(\bar{\theta}))^2 \right) \bar{c}_o^j(\bar{\theta}) d\bar{\theta} \right).$$

Since $\mathcal{C} \in \mathbf{C}$ is continuous, $\mathcal{T}^C \mathcal{C}$ is also continuous. Since

$$\exp \left(-(\xi - 1) \mathcal{C}_o^j(\bar{\theta}; \mathbf{g}, \mathbf{g}') \right) \in [\exp(-(\xi - 1)M^C), \exp((\xi - 1)M^C)],$$

we have

$$\begin{aligned} \|[\mathcal{T}^C \mathcal{C}]_d^i\|_\infty &\leq \ln \mathcal{K}_d^i + (1 - \gamma^{iL}) \ln \left\{ \sum_o [\tau_{do}^j]^{1-\xi} \right\}^{-\frac{1}{\xi-1}} + (1 - \gamma^{iL}) M^C \\ \Rightarrow \|\mathcal{T}^C \mathcal{C}\|_\infty &\leq \max_{d,i,j} \left| \left[\ln \mathcal{K}_d^i + (1 - \gamma^{iL}) \ln \left\{ \sum_o [\tau_{do}^j]^{1-\xi} \right\}^{-\frac{1}{\xi-1}} \right] \right| + (1 - \gamma^{iL}) M^C \leq M^C, \end{aligned}$$

in which the last inequality applies the definition of M^C .

We now verify that $\mathcal{T}^C$ satisfies Blackwell's sufficiency conditions for contraction. For any $\mathcal{C}, \hat{\mathcal{C}} \in \mathbf{C}$, such that $\mathcal{C} \leq \hat{\mathcal{C}}$ point-wisely, it trivially holds that $[\mathcal{T}^C \mathcal{C}]_d^i \leq [\mathcal{T}^C \hat{\mathcal{C}}]_d^i$. And it holds that $\mathcal{T}^C[\mathcal{C}(\bar{\theta}; \mathbf{g}, \mathbf{g}') + c]_d^i = [\mathcal{T}^C \mathcal{C}]_d^i(\bar{\theta}; \mathbf{g}, \mathbf{g}') + (1 - \gamma^{iL})c \leq [\mathcal{T}^C \mathcal{C}]_d^i(\bar{\theta}; \mathbf{g}, \mathbf{g}') + (1 - \underline{\gamma}^L)c$ for $c \geq 0$. $\mathcal{T}^C$ is thus a contraction mapping with a modulus $1 - \underline{\gamma}^L$. $\mathcal{T}^C$ thus has a unique fixed point that is contained in $\mathbf{C}$ by the contraction mapping theorem.

Next, consider

$$\begin{aligned} [\mathcal{T}^C \mathcal{C}]_d^{i'}(\bar{\theta}; \mathbf{g}, \mathbf{g}') &= \sum_j \gamma^{ij} \sum_o \int \chi_{do}^{ij}(\bar{\theta}, \bar{\theta}; \mathbf{g}, \mathbf{g}') \left[\bar{f}(g_d^i(\bar{\theta}) - g_o^j(\bar{\theta})) \right] [g_d^i]'(\bar{\theta}) d\bar{\theta} \\ \Rightarrow \left| [\mathcal{T}^C \mathcal{C}]_d^{i'}(\bar{\theta}; \mathbf{g}, \mathbf{g}') \right| &\leq (1 - \underline{\gamma}^L) 2\bar{f} M \bar{M}', \end{aligned}$$

where the last line applies that $\sum_o \int \chi_{do}^{ij}(\bar{\theta}, \bar{\theta}; \mathbf{g}, \mathbf{g}') d\bar{\theta} = 1$ and $|g_d^i(\bar{\theta})| \leq M$. Note that the derivation for $[\mathcal{T}^C \mathcal{C}]_d^{i'}$ does not rely on the starting point $\mathcal{C}$ being differentiable in $\bar{\theta}$. Thus, $\mathcal{C}$ is differentiable in $\bar{\theta}$ and $\|\mathcal{C}'\|_\infty \leq (1 - \underline{\gamma}^L) 2\bar{f} M \bar{M}'$.

□

Lemma A.3. For $\mathbf{g} \in \mathcal{G}$ and $\mathcal{T}\mathbf{g}$ defined by (A.18), restated below

$$[\mathcal{T}\mathbf{g}]_d^i(\bar{\theta}) = \omega^i \sum_{j,o} \frac{\gamma^{ij}}{1 - \gamma^{iL}} \int \chi_{do}^{ij}(\bar{\theta}, \bar{\theta}; \mathbf{g}, \mathbf{g}') g_o^j(\bar{\theta}) d\bar{\theta} + (1 - \omega^i) \bar{\theta},$$

- (1) $\mathcal{T}\mathbf{g}$ is continuous in $\mathbf{g}$ under the C^1 norm, $\|\mathbf{g}\|_\infty + \|\mathbf{g}'\|_\infty$.
- (2) $\|\mathcal{T}\mathbf{g}\|_\infty \leq M$.
- (3) $\forall \mathbf{g} \in \mathcal{G}$, $\mathcal{T}\mathbf{g}(\bar{\theta})$ is twice differentiable in $\bar{\theta}$ with $[\mathcal{T}\mathbf{g}]_d^{i'}(\bar{\theta}) \in [\underline{M}', \bar{M}']$, for $\underline{M}'$ and $\bar{M}'$ the constants defined in Definition A.1; $[\mathcal{T}\mathbf{g}]_d^{i'}(\bar{\theta})$ is Lipschitz continuous with a Lipschitz constant $\omega^i(\xi - 1)\bar{f}M^2\bar{M}'' + 3\omega^i(\xi - 1)\bar{f}M^3 \leq \bar{M}''$.

Proof. From Lemma A.2, $\mathcal{C}$ is continuous in $\mathbf{g}, \mathbf{g}'$ under the infinity norm. By the definition of χ in (A.19), χ is continuous in $(\mathbf{g}, \mathbf{g}')$ and so is $\mathcal{T}\mathbf{g}$. $\mathcal{T}\mathbf{g}$ is thus continuous in $\mathbf{g}$ under the C^1 norm. This proves part (1).

Next, since $g_o^j(\bar{\theta}) \in [-M, M], \forall \bar{\theta}$, we have that $[\mathcal{T}\mathbf{g}]_d^i(\bar{\theta}) - \bar{\theta} > 0$ for $\bar{\theta} = -M$ and $[\mathcal{T}\mathbf{g}]_d^i(\bar{\theta}) - \bar{\theta} < 0$ for $\bar{\theta} = M$. By the intermediate value theorem, $\exists \bar{\theta}^* \in [-M, M], [\mathcal{T}\mathbf{g}]_d^i(\bar{\theta}^*) = \bar{\theta}^*$. Now consider $\forall \bar{\theta} \in (\bar{\theta}^*, M]$. Since $\bar{M}' < 1$, we have that

$$[\mathcal{T}\mathbf{g}]_d^i(\bar{\theta}) \leq \bar{\theta}^* + \bar{M}'(\bar{\theta} - \bar{\theta}^*) \leq \bar{\theta}^* + (\bar{\theta} - \bar{\theta}^*) \leq \bar{\theta} \leq M,$$

Similarly, $\forall \bar{\theta} \in [-M, \bar{\theta}^*)$

$$[\mathcal{T}\mathbf{g}]_d^i(\bar{\theta}) \geq \bar{\theta}^* - \bar{M}'(\bar{\theta}^* - \bar{\theta}) \geq \bar{\theta}^* - (\bar{\theta}^* - \bar{\theta}) \geq \bar{\theta} \geq -M.$$

We thus have $[\mathcal{T}\mathbf{g}]_d^i(\bar{\theta}) \in [-M, M], \forall \bar{\theta}$. This proves part (2).

Lastly, consider

$$\begin{aligned} [\mathcal{T}\mathbf{g}]_d^{i'}(\bar{\theta}) &= \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} \sum_o \int \partial_{\bar{\theta}} \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) g_o^j(\tilde{\theta}) d\tilde{\theta} + (1 - \omega^i) \\ &= \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} (\zeta - 1) \bar{t} [g_d^i]'(\bar{\theta}) cov_d^{ij}[g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})] + (1 - \omega^i), \end{aligned}$$

where $cov_d^{ij}[\cdot, \cdot]$ is the covariance between the two arguments across $(o, \tilde{\theta})$ under the distribution characterized by $\chi_{do}^{ij}(\bar{\theta}, \tilde{\theta})$ and the second line applies Lemma A.5 stated below. Since $0 < [g_d^i]'(\bar{\theta}) < 1$ and by Popoviciu's inequality on variances $cov_d^{ij}[g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})] \leq M^2$, we have that

$$1 - \omega^i \leq [\mathcal{T}\mathbf{g}]_d^{i'}(\bar{\theta}; \mathbf{g}, \mathbf{g}') \leq 1 - \omega^i(1 - (\zeta - 1)\bar{t}M^2).$$

Further, observe that $\forall \bar{\theta}, \hat{\theta} \in [-M, M]$,

$$\begin{aligned} & \left| [\mathcal{T}\mathbf{g}]_d^{i'}(\bar{\theta}) - [\mathcal{T}\mathbf{g}]_d^{i'}(\hat{\theta}) \right| \\ &= \left| \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} (\zeta - 1) \bar{t} \left\{ [g_d^i]'(\bar{\theta}) cov_{d,\bar{\theta}}^{ij}[g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})] - [g_d^i]'(\hat{\theta}) cov_{d,\hat{\theta}}^{ij}[g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})] \right\} \right| \\ &= \left| \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} (\zeta - 1) \bar{t} \left\{ ([g_d^i]'(\bar{\theta}) - [g_d^i]'(\hat{\theta})) cov_{d,\bar{\theta}}^{ij}[g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})] + [g_d^i]'(\hat{\theta}) [cov_{d,\bar{\theta}}^{ij}[g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})] - cov_{d,\hat{\theta}}^{ij}[g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})]] \right\} \right| \\ &\leq \omega^i (\zeta - 1) \bar{t} \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} \left(\mathcal{L}([g_d^i]') |\bar{\theta} - \hat{\theta}| |cov_{d,\bar{\theta}}^{ij}[g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})]| \right. \\ &\quad \left. + [g_d^i]'(\bar{\theta}) |cov_{d,\hat{\theta}}^{ij}([g_o^j(\tilde{\theta})]^2, g_o^j(\tilde{\theta})) - 2cov_{d,\hat{\theta}}^{ij}[g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})](\sum_o \int \chi_{do}^{ij}(\hat{\theta}, \tilde{\theta}) g_o^j(\tilde{\theta}) d\tilde{\theta})| |\bar{\theta} - \hat{\theta}| \right) \\ &\leq [\omega^i (\zeta - 1) \bar{t} M^2 \bar{M}'' + 3\omega^i (\zeta - 1) \bar{t} M^3] |\bar{\theta} - \hat{\theta}|, \end{aligned}$$

where $\mathcal{L}(\cdot)$ denotes the Lipschitz constant of a Lipschitz continuous function $(\cdot)$, $\hat{\theta}$ is between $\bar{\theta}$ and $\hat{\theta}$, and the third inequality applies the definition of Lipschitz continuity of $\mathbf{g}'$ and the mean value theorem. Therefore, $[\mathcal{T}\mathbf{g}]_d^{i'}(\bar{\theta})$ is Lipschitz continuous with a Lipschitz constant $\omega^i (\zeta - 1) \bar{t} M^2 \bar{M}'' + 3\omega^i (\zeta - 1) \bar{t} M^3$ which satisfies $\omega^i (\zeta - 1) \bar{t} M^2 \bar{M}'' + 3\omega^i (\zeta - 1) \bar{t} M^3 \leq \bar{M}''$ by the definition of $\bar{M}''$. This proves part (3).

□

Proposition A.4. *Given wages, assume $\{\tilde{\Theta}_o^j\}$ have bounded support that is contained in $[-M, M]$ for a positive constant $M > 0$ and have associated density functions $\tilde{\zeta}_o^j$. If $\bar{\phi} > 0$ and $(\xi - 1)\bar{t} < 1/M^2$, then an equilibrium exists which satisfies that the policy function $\mathbf{g}$ is twice differentiable; $\|\mathbf{g}\|_\infty \leq M$; $[g_d^i]'(\bar{\theta}) \in [\underline{M}', \overline{M}']$, $\forall (\bar{\theta}, d, i)$; $\mathbf{g}'$ is Lipschitz continuous with a Lipschitz constant $\overline{M}''$; for constants $\underline{M}', \overline{M}', \overline{M}''$ defined in Definition A.1. Moreover, under such an equilibrium, the first order condition of the technology direction choice problem has a unique solution that characterizes the optimal decision.*

Proof. Define $\mathcal{G} = \{\mathbf{g} : [-M, M] \rightarrow \mathbb{R}^{N \times S}, \mathbf{g} \text{ is differentiable, } \|\mathbf{g}\|_\infty \leq M; [g_d^i]'(\bar{\theta}) \in [\underline{M}', \overline{M}'], \mathbf{g}' \text{ is Lipschitz continuous with a Lipschitz constant } \overline{M}''\}$, for $\underline{M}', \overline{M}', \overline{M}''$ defined in the proposition. Equip $\mathcal{G}$ with the C^1 norm: $\|\mathbf{g}\|_{\mathcal{G}} = \|\mathbf{g}\|_\infty + \|\mathbf{g}'\|_\infty$. It can be shown that $\mathcal{G}$ is closed. By the Arzelà-Ascoli theorem, $\mathcal{G}$ is compact under the norm $\|\cdot\|_{\mathcal{G}}$.

From Lemma A.3, $\mathcal{T}$ defined by (A.18) is continuous under the C^1 norm of $\mathcal{G}$ and $\mathcal{T}\mathbf{g} \in \mathcal{G}$. By the Schauder fixed-point theorem, $\mathcal{G}$ contains a fixed point of $\mathcal{T}$.

For a fixed point $\mathbf{g}$, i.e. $\mathcal{T}\mathbf{g} = \mathbf{g}$, it solves the first-order condition of the technology adaptation problem. We now verify that the second-order optimality condition holds for the technology direction choice problem. With the objective function specified in (15), the second-order derivative of the objective with respect to choice θ is

$$-\bar{\phi} - (\eta - 1) \frac{d^2}{d\theta^2} \ln c_d^i(\theta)$$

where

$$\begin{aligned} \frac{d^2}{d\theta^2} \ln c_d^i(\theta) &= \bar{t} \cdot \sum_{j,o} \gamma^{ij} \int \partial_\theta \chi_{do}^j(\theta, \tilde{\theta})(\theta - \tilde{\theta}) + \chi_{do}^j(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) \\ &= \bar{t} \cdot \sum_{j,o} \gamma^{ij} \left[\int \partial_\theta \chi_{do}^j(\theta, \tilde{\theta})(\theta - \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) + 1 \right] \\ &= -(\xi - 1)\bar{t}^2 \sum_j \gamma^{ij} \text{var}_d^j(\tilde{\theta}) + \bar{t}(1 - \gamma^{iL}) \\ &\in [\bar{t}(1 - \gamma^{iL})(1 - (\xi - 1)\bar{t}M^2), \bar{t}(1 - \gamma^{iL})]. \end{aligned}$$

Here, $\text{var}_d^j(\tilde{\theta})$ in the third line is the variance taken under distribution $\chi_{do}^j(\theta, \tilde{\theta})$ across $(o, \tilde{\theta})$, and the line follows Lemma A.6. The last line applies that $\|\mathbf{g}\|_\infty \leq M$ and the Popoviciu's inequality. Therefore, the second-order condition of the technology choice problem

$$-\bar{\phi} - (\eta - 1) \frac{d^2}{d\theta^2} \ln c_d^i(\theta) < 0$$

holds globally, provided that $1 - (\xi - 1)\bar{t}M^2 \geq 0$. □

Lemma A.4. Denote $\tilde{\mathcal{G}} = \{\mathbf{g} \in C^0([-M, M] \rightarrow \mathbb{R}^{N \times S}) : \|\mathbf{g}\|_\infty \leq M\}$, $\tilde{\mathcal{G}}' = \{\mathbf{g}' \in C^0([-M, M] \rightarrow \mathbb{R}^{N \times S}) : \underline{M}' \leq \mathbf{g}' \leq \overline{M}'\}$, and $\mathbf{C} = \{\mathbf{C} \in C^0([-M, M] \rightarrow \mathbb{R}^{N \times S}) : \|\mathbf{C}\|_\infty \leq M^C\}$, for $\underline{M}', \overline{M}'$,

and M^C defined in Definition A.1. Define $\mathcal{T}^g, \mathcal{T}^{g'}, \mathcal{T}^C$ mapping from $\tilde{\mathcal{G}} \times \tilde{\mathcal{G}}' \times \mathbb{C}$, given below

$$\begin{aligned} [\mathcal{T}^g(\mathbf{g}, \mathbf{g}', \mathbf{C})]_d^i(\bar{\theta}) &= \omega^i \sum_{j,o} \frac{\gamma^{ij}}{1 - \gamma^{iL}} \int \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}', \mathbf{C}) g_o^j(\tilde{\theta}) d\tilde{\theta} + (1 - \omega^i) \bar{\theta}, \\ [\mathcal{T}^{g'}(\mathbf{g}, \mathbf{g}', \mathbf{C})]_d^i(\bar{\theta}) &= \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} \sum_o \int \partial_{\tilde{\theta}} \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}', \mathbf{C}) g_o^j(\tilde{\theta}) d\tilde{\theta} + (1 - \omega^i), \\ [\mathcal{T}^C(\mathbf{g}, \mathbf{g}', \mathbf{C})]_d^i(\bar{\theta}) &= \ln \varkappa_d^i - \sum_j \frac{\gamma^{ij}}{\xi - 1} \ln \left(\sum_o \int [\tau_{do}^j]^{1-\xi} \exp \left(-(\xi - 1) C_o^j(\tilde{\theta}) \right) \exp \left(-\frac{1}{2} (\xi - 1) \bar{I} (g_d^i(\bar{\theta}) - g_o^j(\tilde{\theta}))^2 \right) \bar{\zeta}_o^j(\tilde{\theta}) d\tilde{\theta} \right) \end{aligned}$$

where to slightly abuse notations, χ is the one defined in (A.19), but also highlighting the dependence on $\mathbf{C}$:

$$\chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}', \mathbf{C}) \equiv \frac{[\tau_{do}^j]^{1-\xi} \exp \left(-(\xi - 1) C_o^j(\tilde{\theta}) \right) \exp \left(-\frac{1}{2} (\xi - 1) \bar{I} (g_d^i(\bar{\theta}) - g_o^j(\tilde{\theta}))^2 \right) \bar{\zeta}_o^j(\tilde{\theta})}{\sum_m \int [\tau_{dm}^j]^{1-\xi} \exp \left(-(\xi - 1) C_m^j(\tilde{\theta}) \right) \exp \left(-\frac{1}{2} (\xi - 1) \bar{I} (g_d^i(\bar{\theta}) - g_m^j(\tilde{\theta}))^2 \right) \bar{\zeta}_m^j(\tilde{\theta}) d\tilde{\theta}}. \quad (\text{A.22})$$

Then we have

- (1) $\mathcal{T}^g(\mathbf{g}, \mathbf{g}', \mathbf{C}) \in \tilde{\mathcal{G}}, \mathcal{T}^{g'}(\mathbf{g}, \mathbf{g}', \mathbf{C}) \in \tilde{\mathcal{G}}', \mathcal{T}^C(\mathbf{g}, \mathbf{g}', \mathbf{C}) \in \mathbb{C}$.
- (2) $\sum_{m,k} \|\partial_{g,m}^k \mathcal{T}^g\| \leq \bar{\omega} [3(\xi - 1) \bar{I} M^2 + 1]$. $\sum_{m,k} \|\partial_{g',m}^k \mathcal{T}^g\| = 0$. $\sum_{m,k} \|\partial_{C,m}^k \mathcal{T}^g\| \leq \bar{\omega} (\xi - 1) M$.
- (3) $\sum_{m,k} \|\partial_{g,m}^k \mathcal{T}^{g'}\| \leq 9\bar{\omega} (\xi - 1)^2 \bar{I}^2 M^3$. $\sum_{m,k} \|\partial_{g',m}^k \mathcal{T}^{g'}\| \leq \bar{\omega} (\xi - 1) \bar{I} M^2$. $\sum_{m,k} \|\partial_{C,m}^k \mathcal{T}^{g'}\| \leq 3\bar{\omega} (\xi - 1)^2 \bar{I} M^2$.
- (4) $\sum_{m,k} \|\partial_{g,m}^k \mathcal{T}^C\| \leq (1 - \underline{\gamma}^L) 2\bar{I} M$. $\sum_{m,k} \|\partial_{g',m}^k \mathcal{T}^C\| = 0$. $\sum_{m,k} \|\partial_{C,m}^k \mathcal{T}^C\| \leq 1 - \underline{\gamma}^L$.

Proof. Combining the proofs for part (2) and (3) of Lemma A.3, and the proof for part (1) of Lemma A.2 we have proved part (1).

For part (2), suppressing the argument $(\mathbf{g}, \mathbf{g}', \mathbf{C})$ to simplify notations, consider

$$[\partial_{g,m}^k \mathcal{T}^g]_d^i(\bar{\theta}) = \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} \left\{ \sum_o \int \partial_{g,m}^k \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) g_o^j(\tilde{\theta}) d\tilde{\theta} + \sum_o \int \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) \partial_{g,m}^k g_o^j(\tilde{\theta}) d\tilde{\theta} \right\}.$$

Apply part (1) of Lemma A.7 stated below:

$$\begin{aligned} \sum_o \int \partial_{g,m}^k \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) g_o^j(\tilde{\theta}) d\tilde{\theta} &= -(\xi - 1) \left\{ -\bar{I} g_d^i(\bar{\theta}) \text{cov}_d^{ij}(\partial_{g,m}^k g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})) \right. \\ &\quad \left. - \bar{I} [\partial_{g,m}^k g_d^i(\bar{\theta})] \text{cov}_d^{ij}(g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})) + \bar{I} \cdot \text{cov}_d^{ij}(g_o^j(\tilde{\theta}) \partial_{g,m}^k g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})) \right\}. \end{aligned}$$

By Popoviciu's inequality on variances and note that $\partial_{g,m}^k g_d^i = 1$ for $(m, k) = (d, i)$ and zero otherwise, we have

$$\sum_{m,k} \left| \sum_o \int \partial_{g,m}^k \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) g_o^j(\tilde{\theta}) d\tilde{\theta} \right| \leq (\xi - 1) \left\{ \bar{I} M^2 + \bar{I} M^2 + \bar{I} M^2 \right\} = 3(\xi - 1) \bar{I} M^2.$$

Therefore,

$$\sum_{m,k} \left| [\partial_{g,m}^k \mathcal{T}^g]_d^i(\bar{\theta}) \right| \leq \omega^i [3(\xi - 1) \bar{I} M^2 + 1].$$

Because χ_{do}^{ij} in (A.22) does not involve $\mathbf{g}'$,

$$[\partial_{g',m}^k \mathcal{T}^g]_d^i(\bar{\theta}) = 0.$$

Similarly,

$$\begin{aligned} [\partial_{\mathcal{C},m}^k \mathcal{T}^g]_d^i(\bar{\theta}) &= \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} \sum_o \int \partial_{\mathcal{C},m}^k \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) g_o^j(\tilde{\theta}) d\tilde{\theta} \\ &= -\omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} (\xi - 1) \text{cov}_d^{ij} \left(\partial_{\mathcal{C},m}^k C_o^j(\tilde{\theta}), g_o^j(\tilde{\theta}) \right), \end{aligned}$$

which implies

$$\sum_{m,k} \left| [\partial_{\mathcal{C},m}^k \mathcal{T}^g]_d^i(\bar{\theta}) \right| \leq \omega^i (\xi - 1) M.$$

For part (3), expand

$$[\mathcal{T}^{g'}]_d^i(\bar{\theta}) = \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} (\xi - 1) \bar{t} [g_d^i]'(\bar{\theta}) \left[\sum_o \int \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) [g_o^j(\tilde{\theta})]^2 d\tilde{\theta} - \left(\sum_o \int \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) g_o^j(\tilde{\theta}) d\tilde{\theta} \right)^2 \right] + (1 - \omega^i)$$

Apply part (1) of Lemma A.7:

$$\begin{aligned} [\partial_{g,m}^k \mathcal{T}^{g'}]_d^i(\bar{\theta}) &= \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} (\xi - 1) \bar{t} [g_d^i]'(\bar{\theta}) \left[-(\xi - 1) \left\{ -\bar{t} g_d^i(\bar{\theta}) \text{cov}_d^{ij}(\partial_{g,m}^k g_o^j(\tilde{\theta}), [g_o^j(\tilde{\theta})]^2) \right. \right. \\ &\quad \left. \left. - \bar{t} \partial_{g,m}^k g_d^i(\bar{\theta}) \text{cov}_d^{ij}(g_o^j(\tilde{\theta}), [g_o^j(\tilde{\theta})]^2) + \bar{t} \cdot \text{cov}_d^{ij}(g_o^j(\tilde{\theta}) \partial_{g,m}^k g_o^j(\tilde{\theta}), [g_o^j(\tilde{\theta})]^2) \right\} \right. \\ &\quad \left. + 2(\xi - 1) \left(\sum_o \int \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}') g_o^j(\tilde{\theta}) d\tilde{\theta} \right) \left\{ -\bar{t} g_d^i(\bar{\theta}) \text{cov}_d^{ij}(\partial_{g,m}^k g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})) \right. \right. \\ &\quad \left. \left. - \bar{t} \partial_{g,m}^k g_d^i(\bar{\theta}) \text{cov}_d^{ij}(g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})) + \bar{t} \cdot \text{cov}_d^{ij}(g_o^j(\tilde{\theta}) \partial_{g,m}^k g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})) \right\} \right] \\ \Rightarrow \sum_{m,k} \left\| [\partial_{g,m}^k \mathcal{T}^{g'}]_d^i \right\|_\infty &\leq \omega^i (\xi - 1)^2 \bar{t}^2 \left[\left\{ M^3 + M^3 + M^3 \right\} + 2M \left\{ M^2 + M^2 + M^2 \right\} \right] = 9\omega^i (\xi - 1)^2 \bar{t}^2 M^3 \end{aligned}$$

Because χ_{do}^{ij} does not involve $\mathbf{g}'$, the only dependence of $[\mathcal{T}^{g'}]_d^i$ on $\mathbf{g}'$ is through the factor $[g_d^i]'(\bar{\theta})$, with $\partial_{g',m}^k [g_d^i]' = 1$ for $(m, k) = (d, i)$ and zero otherwise:

$$\begin{aligned} [\partial_{g',m}^k \mathcal{T}^{g'}]_d^i(\bar{\theta}) &= \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} (\xi - 1) \bar{t} \partial_{g',m}^k [g_d^i]'(\bar{\theta}) \text{cov}_d^{ij} [g_o^j(\tilde{\theta}), g_o^j(\tilde{\theta})] \\ &\Rightarrow \sum_{m,k} \left\| [\partial_{g',m}^k \mathcal{T}^{g'}]_d^i \right\|_\infty \leq \omega^i (\xi - 1) \bar{t} M^2, \end{aligned}$$

where the bound applies Popoviciu's inequality on variances. Apply part (2) of Lemma A.7:

$$\begin{aligned}
[\partial_{\mathcal{C},m}^k \mathcal{T}^{\mathcal{S}'}]_d^i(\bar{\theta}) &= \omega^i \sum_j \frac{\gamma^{ij}}{1 - \gamma^{iL}} (\xi - 1) \bar{t} [g_d^i]'(\bar{\theta}) \left[-(\xi - 1) \left\{ cov_d^{ij}(\partial_{\mathcal{C},m}^k \mathcal{C}_o^j(\bar{\theta}), [g_o^j(\bar{\theta})]^2) \right\} \right. \\
&\quad \left. + 2(\xi - 1) \left(\sum_o \int \chi_{do}^{ij}(\bar{\theta}, \bar{\theta}) g_o^j(\bar{\theta}) d\bar{\theta} \right) \left\{ cov_d^{ij}(\partial_{\mathcal{C},m}^k \mathcal{C}_o^j(\bar{\theta}), g_o^j(\bar{\theta})) \right\} \right] \\
\Rightarrow \sum_{m,k} \|[\partial_{\mathcal{C},m}^k \mathcal{T}^{\mathcal{S}'}]_d^i\|_\infty &\leq \omega^i (\xi - 1) \bar{t} \left[(\xi - 1) \left\{ M^2 \right\} + 2(\xi - 1) M \left\{ M \right\} \right] = 3\omega^i (\xi - 1)^2 \bar{t} M^2
\end{aligned}$$

For part (4). Directly apply the chain rule

$$\begin{aligned}
\partial_{g,m}^k [\mathcal{T}^{\mathcal{C}}]_d^i(\bar{\theta}) &= \sum_j \gamma^{ij} \sum_o \int \chi_{do}^{ij}(\bar{\theta}, \bar{\theta}) \left[\bar{t} (g_d^i(\bar{\theta}) - g_o^j(\bar{\theta})) (\partial_{g,m}^k g_d^i(\bar{\theta}) - \partial_{g,m}^k g_o^j(\bar{\theta})) \right] d\bar{\theta} \\
\Rightarrow \sum_{m,k} \|\partial_{g,m}^k [\mathcal{T}^{\mathcal{C}}]_d^i\|_\infty &\leq (1 - \underline{\gamma}^L) 2\bar{t} M. \\
\partial_{g',m}^k [\mathcal{T}^{\mathcal{C}}]_d^i(\bar{\theta}) &= 0, \text{ as the integrand does not involve } g'. \\
\partial_{\mathcal{C},m}^k [\mathcal{T}^{\mathcal{C}}]_d^i(\bar{\theta}) &= \sum_j \gamma^{ij} \sum_o \int \chi_{do}^{ij}(\bar{\theta}, \bar{\theta}) \partial_{\mathcal{C},m}^k \mathcal{C}_o^j(\bar{\theta}) d\bar{\theta} \\
\Rightarrow \sum_{m,k} \|\partial_{\mathcal{C},m}^k [\mathcal{T}^{\mathcal{C}}]_d^i\|_\infty &\leq 1 - \underline{\gamma}^L.
\end{aligned}$$

□

Proposition A.5. *Given wages, $\{w_d\}$, assume $\{\bar{\Theta}_o^j\}$ have bounded support that is contained in $[-M, M]$ for some $M > 0$ and have associated density functions $\bar{\zeta}_o^j$. If $(\xi - 1)\bar{t} < 1/M^2$, $\bar{t} < \frac{1}{2M}$ and $\bar{\phi} > \phi$, where $\phi > 0$ is a constant determined by parameters $(\xi - 1, \bar{t}, \eta, M, \underline{\gamma}^L)$ detailed in the proof, then an equilibrium uniquely exists and satisfies all properties stated in Proposition A.4.*

Proof. Denote $\mathcal{G} = \{\mathbf{g} : [-M, M] \rightarrow \mathbb{R}^{N \times S}; \mathbf{g} \text{ is differentiable}; \|\mathbf{g}\|_\infty \leq M; [g_d^i]'(\bar{\theta}) \in [\underline{M}', \bar{M}'], \forall d, i, \bar{\theta}\}$. Denote $\mathbf{C} = \{\mathbf{C} \in C^0([-M, M] \rightarrow \mathbb{R}^{N \times S}) : \|\mathbf{C}\|_\infty \leq M^{\mathcal{C}}\}$. Denote $\mathcal{X} = \mathcal{G} \times \mathbf{C}$. Endow $\mathcal{X}$ with the C^1 norm of $\mathbf{g}$ combined with the C^0 norm of $\mathbf{C}$: $\|(\mathbf{g}, \mathbf{C})\|_{\mathcal{X}} = \|\mathbf{g}\|_\infty + \|\mathbf{g}'\|_\infty + \|\mathbf{C}\|_\infty$. It can be verified that $\mathcal{X}$ is a complete metric space with the norm $\|\cdot\|_{\mathcal{X}}$.

Define $\tilde{\mathcal{T}} = (\tilde{\mathcal{T}}^{\mathcal{S}}, \tilde{\mathcal{T}}^{\mathcal{C}})$ mapping from $\mathcal{X}$ given by

$$\begin{aligned}
[\tilde{\mathcal{T}}^{\mathcal{S}}(\mathbf{g}, \mathbf{C})]_d^i(\bar{\theta}) &= \omega^i \sum_{j,o} \frac{\gamma^{ij}}{1 - \gamma^{iL}} \int \chi_{do}^{ij}(\bar{\theta}, \bar{\theta}; \mathbf{g}, \mathbf{C}) g_o^j(\bar{\theta}) d\bar{\theta} + (1 - \omega^i) \bar{\theta} \\
[\tilde{\mathcal{T}}^{\mathcal{C}}(\mathbf{g}, \mathbf{C})]_d^i(\bar{\theta}) &= \ln \kappa_d^i - \sum_j \frac{\gamma^{ij}}{\xi - 1} \ln \left(\sum_o \int [\tau_{do}^j]^{1-\xi} \exp \left(-(\xi - 1) \mathcal{C}_o^j(\bar{\theta}) \right) \exp \left(-\frac{1}{2} (\xi - 1) \bar{t} (g_d^i(\bar{\theta}) - g_o^j(\bar{\theta}))^2 \right) \bar{\zeta}_o^j(\bar{\theta}) d\bar{\theta} \right),
\end{aligned}$$

where $\mathbf{g}'$ is viewed as an operator applied to $\mathbf{g}$, and to slightly abuse notations, χ is redefined below to highlight its dependence on $(\mathbf{g}, \mathbf{C})$:

$$\chi_{do}^{ij}(\bar{\theta}, \bar{\theta}; \mathbf{g}, \mathbf{C}) \equiv \frac{[\tau_{do}^j]^{1-\xi} \exp \left(-(\xi - 1) \mathcal{C}_o^j(\bar{\theta}) \right) \exp \left(-\frac{1}{2} (\xi - 1) \bar{t} (g_d^i(\bar{\theta}) - g_o^j(\bar{\theta}))^2 \right) \bar{\zeta}_o^j(\bar{\theta})}{\sum_m \int [\tau_{dm}^j]^{1-\xi} \exp \left(-(\xi - 1) \mathcal{C}_m^j(\bar{\theta}) \right) \exp \left(-\frac{1}{2} (\xi - 1) \bar{t} (g_d^i(\bar{\theta}) - g_m^j(\bar{\theta}))^2 \right) \bar{\zeta}_m^j(\bar{\theta}) d\bar{\theta}}.$$

Part (1) of Lemma A.4 shows that $\tilde{\mathcal{T}}(\mathbf{g}, \mathbf{C}) \in \mathcal{X}$.

Consider $\forall (\mathbf{g}, \mathbf{C}) \in \mathcal{X}, (\hat{\mathbf{g}}, \hat{\mathbf{C}}) \in \mathcal{X}$

$$\begin{aligned}
& \|\tilde{\mathcal{T}}(\mathbf{g}, \mathbf{C}) - \tilde{\mathcal{T}}(\hat{\mathbf{g}}, \hat{\mathbf{C}})\|_{\mathcal{X}} \\
&= \|\tilde{\mathcal{T}}^g(\mathbf{g}, \mathbf{C}) - \tilde{\mathcal{T}}^g(\hat{\mathbf{g}}, \hat{\mathbf{C}})\|_{\infty} + \|[\tilde{\mathcal{T}}^g(\mathbf{g}, \mathbf{C})]' - [\tilde{\mathcal{T}}^g(\hat{\mathbf{g}}, \hat{\mathbf{C}})]'\|_{\infty} + \|\tilde{\mathcal{T}}^c(\mathbf{g}, \mathbf{C}) - \tilde{\mathcal{T}}^c(\hat{\mathbf{g}}, \hat{\mathbf{C}})\|_{\infty} \\
&\leq (\sum_{m,k} \|\partial_{g,m}^k \mathcal{T}^g\| + \sum_{m,k} \|\partial_{g,m}^k \mathcal{T}^{g'}\| + \sum_{m,k} \|\partial_{g,m}^k \mathcal{T}^c\|) \|\mathbf{g} - \hat{\mathbf{g}}\|_{\infty} \\
&\quad + (\sum_{m,k} \|\partial_{g',m}^k \mathcal{T}^g\| + \sum_{m,k} \|\partial_{g',m}^k \mathcal{T}^{g'}\| + \sum_{m,k} \|\partial_{g',m}^k \mathcal{T}^c\|) \|\mathbf{g}' - \hat{\mathbf{g}}'\|_{\infty} \\
&\quad + (\sum_{m,k} \|\partial_{c,m}^k \mathcal{T}^g\| + \sum_{m,k} \|\partial_{c,m}^k \mathcal{T}^{g'}\| + \sum_{m,k} \|\partial_{c,m}^k \mathcal{T}^c\|) \|\mathbf{C} - \hat{\mathbf{C}}\|_{\infty},
\end{aligned}$$

where the second line applies the definition of $\|\cdot\|_{\mathcal{X}}$, the third line applies the mean value theorem for Fréchet derivatives,⁶ and $\mathcal{T}^g, \mathcal{T}^{g'}, \mathcal{T}^c$ are the operators defined in Lemma A.4. From the estimates in part (2)-(4) of Lemma A.4

$$\|\tilde{\mathcal{T}}(\mathbf{g}, \mathbf{C}) - \tilde{\mathcal{T}}(\hat{\mathbf{g}}, \hat{\mathbf{C}})\|_{\mathcal{X}} \leq \Omega_g \|\mathbf{g} - \hat{\mathbf{g}}\|_{\infty} + \Omega_{g'} \|\mathbf{g}' - \hat{\mathbf{g}}'\|_{\infty} + \Omega_c \|\mathbf{C} - \hat{\mathbf{C}}\|_{\infty},$$

where $\Omega_g = \bar{\omega} [3(\xi - 1)\bar{t}M^2 + 1] + 9\bar{\omega}(\xi - 1)^2\bar{t}^2M^3 + (1 - \gamma^L)2\bar{t}M$, $\Omega_{g'} = \bar{\omega}(\xi - 1)\bar{t}M^2$, $\Omega_c = \bar{\omega}(\xi - 1)M + 3\bar{\omega}(\xi - 1)^2\bar{t}M^2 + 1 - \gamma^L$. Since it is assumed that $2\bar{t}M < 1$, $\bar{\Omega} \equiv \max\{\Omega_g, \Omega_{g'}, \Omega_c\}$ is thus increasing in $\bar{\omega}$ and $\bar{\Omega}|_{\bar{\omega}=0} = (1 - \gamma^L) < 1$. Choose any $\bar{\Omega}^* \in (1 - \gamma^L, 1)$. Since $\bar{\omega} = \max_i \frac{(\eta-1)(1-\gamma^{iL})\bar{t}}{(\eta-1)(1-\gamma^{iL})\bar{t} + \bar{\phi}}$ which is decreasing in $\bar{\phi}$, $\exists \underline{\phi} > 0$ such that for all $\bar{\phi} > \underline{\phi}$, $\bar{\Omega} < \bar{\Omega}^*$. We thus have found $\underline{\phi}$ such that for all $\bar{\phi} > \underline{\phi}$, $\tilde{\mathcal{T}}$ is a contraction mapping with the norm $\|\cdot\|_{\mathcal{X}}$ with a modulus $\bar{\Omega}^*$. The existence and uniqueness of the equilibrium then follows from the contraction mapping theorem.

Since the conditions stated in Proposition A.4 also hold, we have that the unique fixed point also satisfies the properties stated in Proposition A.4. $\square$

Lemma A.5. For χ defined in (A.19) and any function $f_o(\tilde{\theta})$ we have

$$\sum_o \int \partial_{\tilde{\theta}} \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}') f_o(\tilde{\theta}) d\tilde{\theta} = (\xi - 1)\bar{t}[g_d^i]'(\tilde{\theta}) cov_d^{ij}(g_o^j(\tilde{\theta}), f_o(\tilde{\theta})),$$

where cov_d^{ij} is the covariance taken under the distribution $\chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}; \mathbf{g}, \mathbf{g}')$ across $(o, \tilde{\theta})$.

⁶This can be shown as a corollary of the Hahn-Banach Theorem, see e.g., Theorem 1.8 of Ambrosetti and Prodi (1995).

Proof. For simplicity we omit $\mathbf{g}, \mathbf{g}'$ in the arguments.

$$\begin{aligned}
& \sum_o \int \partial_{\tilde{\theta}} \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} \\
&= \sum_o \int \partial_{\tilde{\theta}} \ln \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}) \cdot \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} \\
&= \sum_o \int \left\{ -(\xi - 1) \bar{t} (g_d^i(\tilde{\theta}) - g_o^j(\tilde{\theta})) + \sum_{\tilde{o}} \int \chi_{d\tilde{o}}^{ij}(\tilde{\theta}, \tilde{\theta}) \left[(\xi - 1) \bar{t} (g_d^i(\tilde{\theta}) - g_{\tilde{o}}^j(\tilde{\theta})) \right] d\tilde{\theta} \right\} \cdot [g_d^i]'(\tilde{\theta}) \cdot \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} \\
&= -(\xi - 1) \bar{t} [g_d^i]'(\tilde{\theta}) \left\{ \sum_o \int (g_d^i(\tilde{\theta}) - g_o^j(\tilde{\theta})) f_o(\tilde{\theta}) \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}) d\tilde{\theta} \right. \\
&\quad \left. - \left(\sum_o \int (g_d^i(\tilde{\theta}) - g_o^j(\tilde{\theta})) \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}) d\tilde{\theta} \right) \left(\sum_o \int f_o(\tilde{\theta}) \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}) d\tilde{\theta} \right) \right\} \\
&= -(\xi - 1) \bar{t} [g_d^i]'(\tilde{\theta}) \text{cov}_d^{ij} [g_d^i(\tilde{\theta}) - g_o^j(\tilde{\theta}), f_o(\tilde{\theta})] \\
&= (\xi - 1) \bar{t} [g_d^i]'(\tilde{\theta}) \text{cov}_d^{ij} [g_o^j(\tilde{\theta}), f_o(\tilde{\theta})].
\end{aligned}$$

□

Lemma A.6. For $\chi_{do}^j(\theta, \tilde{\theta})$ defined in equation (6) and any function $f_o(\theta, \tilde{\theta})$ we have

$$\sum_o \int \partial_{\theta} \chi_{do}^j(\theta, \tilde{\theta}) f_o(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) = -(\xi - 1) \bar{t} \cdot \text{cov}_d^j(\theta - \tilde{\theta}, f_o(\theta, \tilde{\theta})),$$

where cov_d^j is the covariance under the distribution $\chi_{do}^j(\theta, \tilde{\theta})$ across $(o, \tilde{\theta})$.

Proof. Consider

$$\begin{aligned}
& \sum_o \int \partial_{\theta} \chi_{do}^j(\theta, \tilde{\theta}) f_o(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) \\
&= \sum_o \int \partial_{\theta} \ln \chi_{do}^j(\theta, \tilde{\theta}) \cdot \chi_{do}^j(\theta, \tilde{\theta}) f_o(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) \\
&= \sum_o \int \left\{ -(\xi - 1) \bar{t} (\theta - \tilde{\theta}) + \sum_{\tilde{o}} \int \chi_{d\tilde{o}}^j(\theta, \tilde{\theta}) \left[(\xi - 1) \bar{t} (\theta - \tilde{\theta}) \right] d\Theta_{\tilde{o}}^j(\tilde{\theta}) \right\} \cdot \chi_{do}^j(\theta, \tilde{\theta}) f_o(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) \\
&= -(\xi - 1) \bar{t} \left\{ \sum_o \int (\theta - \tilde{\theta}) f_o(\theta, \tilde{\theta}) \chi_{do}^j(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) \right. \\
&\quad \left. - \left(\sum_{\tilde{o}} \int (\theta - \tilde{\theta}) \chi_{d\tilde{o}}^j(\theta, \tilde{\theta}) d\Theta_{\tilde{o}}^j(\tilde{\theta}) \right) \left(\sum_o \int f_o(\theta, \tilde{\theta}) \chi_{do}^j(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}) \right) \right\} \\
&= -(\xi - 1) \bar{t} \cdot \text{cov}_d^j(\theta - \tilde{\theta}, f_o(\theta, \tilde{\theta}))
\end{aligned}$$

□

Lemma A.7. For χ defined in (A.22) and any function $f_o(\tilde{\theta})$ we have

$$\begin{aligned}
(1) \quad & \sum_o \int \partial_{g,m}^k \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} = -(\xi - 1) \left\{ -\bar{t} g_d^i(\tilde{\theta}) \text{cov}_d^{ij}(\partial_{g,m}^k g_o^j(\tilde{\theta}), f_o(\tilde{\theta})) \right. \\
& \quad \left. - \bar{t} \partial_{g,m}^k g_d^i(\tilde{\theta}) \text{cov}_d^{ij}(g_o^j(\tilde{\theta}), f_o(\tilde{\theta})) + \bar{t} \cdot \text{cov}_d^{ij}(g_o^j(\tilde{\theta}) \partial_{g,m}^k g_o^j(\tilde{\theta}), f_o(\tilde{\theta})) \right\}. \\
(2) \quad & \sum_o \int \partial_{\mathcal{C},m}^k \chi_{do}^{ij}(\tilde{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} = -(\xi - 1) \text{cov}_d^{ij}(\partial_{\mathcal{C},m}^k c_o^j(\tilde{\theta}), f_o(\tilde{\theta})).
\end{aligned}$$

Proof. For part (1),

$$\begin{aligned}
& \sum_o \int \partial_{g,m}^k \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} \\
&= \sum_o \int \partial_{g,m}^k \ln \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) \cdot \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} \\
&= -(\xi - 1) \sum_o \int \left\{ \bar{t}(g_d^i(\bar{\theta}) - g_o^j(\tilde{\theta}))(\partial_{g,m}^k g_d^i(\bar{\theta}) - \partial_{g,m}^k g_o^j(\tilde{\theta})) \right. \\
&\quad \left. - \sum_{\tilde{o}} \int \chi_{d\tilde{o}}^{ij}(\bar{\theta}, \tilde{\theta}) \left[\bar{t}(g_d^i(\bar{\theta}) - g_{\tilde{o}}^j(\tilde{\theta}))(\partial_{g,m}^k g_d^i(\bar{\theta}) - \partial_{g,m}^k g_{\tilde{o}}^j(\tilde{\theta})) \right] d\tilde{\theta} \right\} \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} \\
&= -(\xi - 1) cov_d^{ij} \left(\bar{t}(g_d^i(\bar{\theta}) - g_o^j(\tilde{\theta}))(\partial_{g,m}^k g_d^i(\bar{\theta}) - \partial_{g,m}^k g_o^j(\tilde{\theta})), f_o(\tilde{\theta}) \right) \\
&= -(\xi - 1) \left\{ -\bar{t} g_d^i(\bar{\theta}) cov_d^{ij}(\partial_{g,m}^k g_o^j(\tilde{\theta}), f_o(\tilde{\theta})) \right. \\
&\quad \left. - \bar{t} \partial_{g,m}^k g_d^i(\bar{\theta}) cov_d^{ij}(g_o^j(\tilde{\theta}), f_o(\tilde{\theta})) + \bar{t} \cdot cov_d^{ij}(g_o^j(\tilde{\theta}) \partial_{g,m}^k g_o^j(\tilde{\theta}), f_o(\tilde{\theta})) \right\}.
\end{aligned}$$

For part (2),

$$\begin{aligned}
& \sum_o \int \partial_{c,m}^k \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} \\
&= \sum_o \int \partial_{c,m}^k \ln \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) \cdot \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} \\
&= -(\xi - 1) \sum_o \int \left\{ \partial_{c,m}^k C_o^j(\tilde{\theta}) - \sum_{\tilde{o}} \int \chi_{d\tilde{o}}^{ij}(\bar{\theta}, \tilde{\theta}) \left[\partial_{c,m}^k C_{\tilde{o}}^j(\tilde{\theta}) \right] d\tilde{\theta} \right\} \chi_{do}^{ij}(\bar{\theta}, \tilde{\theta}) f_o(\tilde{\theta}) d\tilde{\theta} \\
&= -(\xi - 1) cov_d^{ij} \left(\partial_{c,m}^k C_o^j(\tilde{\theta}), f_o(\tilde{\theta}) \right).
\end{aligned}$$

□

A.4 Proof of Proposition 2

Proof. Slightly abusing notation, we denote the change in the trade cost as $d \ln \tau_{do}^{ij}$. Differentiating the first-order condition (23), with $\tilde{\omega}^i \equiv \omega^i / (1 - \gamma^{iL})$ as in the statement of the

proposition,

$$\begin{aligned}
d\theta_d^i &= \tilde{\omega}^i \sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} (d\theta_{o'}^{j'} + \theta_{o'}^{j'} d \ln \tilde{\chi}_{do'}^{ij'}) \\
&= \underbrace{\tilde{\omega}^i \sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} d\theta_{o'}^{j'}}_{=0 \text{ noting the assumption implies } \tilde{\chi}_{dd}^{ii}=0} \\
&\quad - (\xi - 1) \tilde{\omega}^i \sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \theta_{o'}^{j'} [\underbrace{d \ln \tilde{c}_{o'}^{j'}}_{=0} + d \ln \tau_{do'}^{ij'} + \bar{t}(\theta_d^i - \theta_{o'}^{j'}) d\theta_d^i - \underbrace{\bar{t}(\theta_d^i - \theta_{o'}^{j'}) d\theta_{o'}^{j'}}_{=0}] \\
&\quad + (\xi - 1) \tilde{\omega}^i \sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \theta_{o'}^{j'} \sum_m \tilde{\chi}_{dm}^{ij'} [\underbrace{d \ln \tilde{c}_m^{j'}}_{=0} + d \ln \tau_{dm}^{ij'} + \bar{t}(\theta_d^i - \theta_m^{j'}) d\theta_d^i - \underbrace{\bar{t}(\theta_d^i - \theta_m^{j'}) d\theta_m^{j'}}_{=0}] \\
&= -(\xi - 1) \tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} \theta_o^j d \ln \tau_{do}^{ij} - (\xi - 1) \bar{t} \tilde{\omega}^i [\sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \theta_{o'}^{j'} (\theta_d^i - \theta_{o'}^{j'})] d\theta_d^i \\
&\quad + (\xi - 1) \tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} (\sum_{o'} \tilde{\chi}_{do'}^{ij} \theta_{o'}^j) d \ln \tau_{do}^{ij} + (\xi - 1) \bar{t} \tilde{\omega}^i [\sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \theta_{o'}^{j'} (\theta_d^i - \sum_m \tilde{\chi}_{dm}^{ij'} \theta_m^{j'})] d\theta_d^i \\
&= -(\xi - 1) \tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} [\theta_o^j - \sum_m \tilde{\chi}_{dm}^{ij} \theta_m^j] d \ln \tau_{do}^{ij} + (\xi - 1) \bar{t} \tilde{\omega}^i \underbrace{\sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \theta_{o'}^{j'} [\theta_o^j - \sum_m \tilde{\chi}_{dm}^{ij'} \theta_m^{j'}]}_{\text{effect due to the change in } \theta_d^i \text{ through trade shares}} d\theta_d^i
\end{aligned}$$

Hence,

$$\begin{aligned}
d\theta_d^i &= \frac{-(\xi - 1) \tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} [\theta_o^j - \sum_m \tilde{\chi}_{dm}^{ij} \theta_m^j] d \ln \tau_{do}^{ij}}{1 - (\xi - 1) \bar{t} \tilde{\omega}^i \sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \theta_{o'}^{j'} [\theta_o^j - \sum_m \tilde{\chi}_{dm}^{ij'} \theta_m^{j'}]} \\
&= - \frac{(\xi - 1) \tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} \|\theta_o^j - \theta_d^{ij}\|}{1 - \bar{t}(\xi - 1) \tilde{\omega}^i \sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \|\theta_{o'}^{j'} - \theta_d^{ij'}\|} \times \frac{d \ln \tau_{do}^{ij}}{\theta_o^j - \theta_d^{ij}}.
\end{aligned}$$

The second equality multiplies and divides by $\theta_o^j - \theta_d^{ij}$ (nonzero by assumption) and uses

$$\begin{aligned}
&\sum_{j',o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \sum_m \tilde{\chi}_{dm}^{ij'} \theta_m^{j'} [\theta_o^j - \sum_m \tilde{\chi}_{dm}^{ij'} \theta_m^{j'}] \\
&= \sum_{j'} \gamma^{ij'} \sum_{o'} \tilde{\chi}_{do'}^{ij'} \theta_{o'}^{j'} \sum_m \tilde{\chi}_{dm}^{ij'} \theta_m^{j'} - \sum_{j'} \gamma^{ij'} \sum_{o'} \tilde{\chi}_{do'}^{ij'} \left[\sum_m \tilde{\chi}_{dm}^{ij'} \theta_m^{j'} \right]^2 \\
&= 0.
\end{aligned}$$

Since $\|\theta_d^i - \theta_{o'}^j\| \equiv (\theta_d^i - \theta_{o'}^j)^2$ and $d\theta_{o'}^j = 0$ by (d, i) being small, $\Delta \|\theta_d^i - \theta_{o'}^j\| = 2(\theta_d^i -$

$\theta_{o'}^j) d\theta_{d'}^j$. Setting $d \ln \tau_{do}^{ij} = x$, we have $\forall o', o$

$$\begin{aligned}\Delta \|\theta_d^i - \theta_o^j\| &= -\frac{2(\xi - 1)\tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} \|\theta_o^j - \vartheta_d^{ij}\|}{1 - \bar{\ell}(\xi - 1)\tilde{\omega}^i \sum_{j', o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \|\theta_{o'}^{j'} - \vartheta_d^{ij'}\|} \times \frac{\theta_o^j - \vartheta_o^{ij}}{\theta_o^j - \vartheta_d^{ij}} \times x, \\ \Delta \|\theta_d^i - \theta_{o'}^j\| &= -\frac{2(\xi - 1)\tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} \|\theta_o^j - \vartheta_d^{ij}\|}{1 - \bar{\ell}(\xi - 1)\tilde{\omega}^i \sum_{j', o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \|\theta_{o'}^{j'} - \vartheta_d^{ij'}\|} \times \frac{\theta_o^j - \vartheta_{o'}^{ij}}{\theta_o^j - \vartheta_d^{ij}} \times x.\end{aligned}$$

It follows that

$$\begin{aligned}\Delta \|\theta_d^i - \theta_o^j\| &= \sum_{o'} \tilde{\chi}_{do'}^{ij} \Delta \|\theta_d^i - \theta_{o'}^j\| \\ &= -\frac{2(\xi - 1)\tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} \|\theta_o^j - \vartheta_d^{ij}\|}{1 - \bar{\ell}(\xi - 1)\tilde{\omega}^i \sum_{j', o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \|\theta_{o'}^{j'} - \vartheta_d^{ij'}\|} \times \frac{x}{\theta_o^j - \vartheta_d^{ij}} [(\theta_d^i - \theta_o^j) - \sum_{o'} \tilde{\chi}_{do'}^{ij} (\theta_d^i - \theta_{o'}^j)] \\ &= -\frac{2(\xi - 1)\tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} \|\theta_o^j - \vartheta_d^{ij}\|}{1 - \bar{\ell}(\xi - 1)\tilde{\omega}^i \sum_{j', o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \|\theta_{o'}^{j'} - \vartheta_d^{ij'}\|} \times \frac{x}{\theta_o^j - \vartheta_d^{ij}} \times (\vartheta_d^{ij} - \theta_o^j) \\ &= \frac{2(\xi - 1)\tilde{\omega}^i \gamma^{ij} \tilde{\chi}_{do}^{ij} \|\theta_o^j - \vartheta_d^{ij}\|}{1 - \bar{\ell}(\xi - 1)\tilde{\omega}^i \sum_{j', o'} \gamma^{ij'} \tilde{\chi}_{do'}^{ij'} \|\theta_{o'}^{j'} - \vartheta_d^{ij'}\|} \times x,\end{aligned}$$

where the denominator is positive by the second-order condition of θ_d^i . $\square$

A.5 Proof of Proposition 3

Proof. From the first-order condition,

$$\begin{aligned}\theta_d^i(\nu) &= \frac{(\eta - 1)\bar{\ell}}{(\eta - 1)(1 - \gamma^{iL})\bar{\ell} + \bar{\phi}} \sum_{j, o} \gamma^{ij} \tilde{\chi}_{do}^{ij}(\nu) \theta_o^j + \frac{\bar{\phi}}{(\eta - 1)(1 - \gamma^{iL})\bar{\ell} + \bar{\phi}} \bar{\theta}(\nu), \\ \text{with } \tilde{\chi}_{do}^{ij}(\nu) &= \frac{[\tau_{do}^j \bar{c}_o^j \exp(\frac{1}{2}\bar{\ell}(\theta_d^i(\nu) - \theta_o^j)^2)]^{1-\xi}}{\sum_{o'} [\tau_{do'}^j \bar{c}_{o'}^j \exp(\frac{1}{2}\bar{\ell}(\theta_d^i(\nu) - \theta_{o'}^j)^2)]^{1-\xi}}.\end{aligned}$$

Totally differentiate $\theta_d^i(\nu)$ and $\tilde{\chi}_{do}^{ij}(\nu)$ w.r.t $\bar{\theta}(\nu)$ around θ_d^i . Since only one firm is deviating, all aggregate outcomes will not change. Using $d \ln \tilde{\chi}_{do}^{ij}(\nu) = (\xi - 1)\bar{\ell}[\theta_o^j - \vartheta_d^{ij}] d\theta_d^i(\nu)$, where $\vartheta_d^{ij} \equiv \sum_m \tilde{\chi}_{dm}^{ij} \theta_m^j$,

$$\begin{aligned}d\theta_d^i(\nu) &= \frac{(\eta - 1)\bar{\ell}}{(\eta - 1)(1 - \gamma^{iL})\bar{\ell} + \bar{\phi}} (\xi - 1)\bar{\ell} \sum_{j, o} \gamma^{ij} \theta_o^j \cdot \tilde{\chi}_{do}^{ij} [\theta_o^j - \vartheta_d^{ij}] \cdot d\theta_d^i(\nu) \\ &\quad + \frac{\bar{\phi}}{(\eta - 1)(1 - \gamma^{iL})\bar{\ell} + \bar{\phi}} d\bar{\theta}(\nu) \\ \Rightarrow d\theta_d^i(\nu) &= \left(\frac{\frac{\bar{\phi}}{(\eta - 1)(1 - \gamma^{iL})\bar{\ell} + \bar{\phi}}}{1 - \frac{(\eta - 1)\bar{\ell}}{(\eta - 1)(1 - \gamma^{iL})\bar{\ell} + \bar{\phi}} (\xi - 1)\bar{\ell} \sum_{j, o} \gamma^{ij} \tilde{\chi}_{do}^{ij} \|\theta_o^j - \vartheta_d^{ij}\|} \right) d\bar{\theta}(\nu),\end{aligned}$$

where the last line uses $\sum_o \tilde{\chi}_{do}^{ij} [\theta_o^j - \vartheta_d^{ij}] = 0$ to write $\sum_o \tilde{\chi}_{do}^{ij} \theta_o^j [\theta_o^j - \vartheta_d^{ij}] = \sum_o \tilde{\chi}_{do}^{ij} \|\theta_o^j - \vartheta_d^{ij}\|$, and the denominator is positive by the second order condition of firms' optimal $\theta_d^i(\nu)$.

Therefore,

$$\begin{aligned} d[\|\theta_d^i(\nu) - \theta_o^j\| - \|\theta_d^i(\nu) - \theta_{o'}^j\|] \\ = -2d\theta_d^i(\nu)[\theta_o^j - \theta_{o'}^j], \end{aligned}$$

whose sign agrees with $-2d\bar{\theta}(\nu)[\theta_o^j - \theta_{o'}^j] = d[\|\bar{\theta}(\nu) - \theta_o^j\| - \|\bar{\theta}(\nu) - \theta_{o'}^j\|]$.

The rest of the proposition follows by taking the log ratio of the expenditure shares above:

$$\ln \frac{\bar{\chi}_{do}^{ij}(\nu)}{\bar{\chi}_{do'}^{ij}(\nu)} = (1 - \xi) \ln \frac{\tau_{do}^j \bar{c}_o^j}{\tau_{do'}^j \bar{c}_{o'}^j} - \frac{1}{2}(\xi - 1)\bar{t}[\|\theta_d^i(\nu) - \theta_o^j\| - \|\theta_d^i(\nu) - \theta_{o'}^j\|].$$

□

A.6 Proof of Proposition 4

Proof. Drop country subscripts; profit accrues from final sales only, $\Pi^i = \frac{1}{\eta}X^i$. Write $a^i \equiv 1 - \phi(\theta^i, \bar{\theta}^i) = \exp(-\frac{1}{2}\bar{\phi}(\theta^i - \bar{\theta}^i)^2)$ for the retained fraction of profit. By (20)–(21), final output splits into consumption and adaptation, $PQ = I + \Phi$ with $\Phi = \sum_k \phi^k \Pi^k$, household income is $I = wL + \sum_k a^k \Pi^k$, and welfare is $U = I/P$.

The unit cost is determined by the fixed point equation $\ln c^i = \sum_j \gamma^{ij} \ln c^j + \frac{1}{2}\bar{t} \sum_j \gamma^{ij} (\theta^i - \theta^j)^2$. Linearity in $\ln c$ gives $\ln c^i = \frac{1}{2}\bar{t} \sum_m \Omega^{im} [\sum_j \gamma^{mj} (\theta^m - \theta^j)^2]$, where $\Omega = (\mathbb{I}_{S \times S} - \Gamma)^{-1}$. Hence, with $\lambda^m \equiv \sum_n \rho^n \Omega^{nm}$,

$$\ln P = \text{const} + \sum_i \rho^i \ln c^i = \text{const} + \frac{1}{2}\bar{t} \sum_m \lambda^m \sum_j \gamma^{mj} (\theta^m - \theta^j)^2,$$

so θ^i enters as the buyer of row i and as supplier $j = i$ of rows $m \neq i$:

$$\frac{\partial \ln P}{\partial \theta^i} = \bar{t} \lambda^i \sum_j \gamma^{ij} (\theta^i - \theta^j) + \bar{t} \sum_{m \neq i} \lambda^m \gamma^{mi} (\theta^i - \theta^m).$$

Final demand is Cobb–Douglas across sectors, so $X^i = \rho^i PQ$ and $\Pi^i = \frac{1}{\eta} \rho^i PQ$. Substituting into $PQ = I + \Phi$ and $I = wL + \sum_k a^k \Pi^k$ gives $PQ = \frac{\eta}{\eta-1} wL$, independent of θ ; labor-market clearing (22) follows from summing $e^i = \frac{\eta-1}{\eta} X^i + M^i$ over i and using $M^i = \sum_k \gamma^{ki} e^k$, which gives $\sum_k \gamma^{kL} e^k = \frac{\eta-1}{\eta} PQ = wL$. Hence each $\Pi^k = \frac{1}{\eta} \rho^k PQ$ is independent of θ , which enters income only through the retained fractions a^k . Thus $I = wL + \frac{1}{\eta} PQ \sum_k \rho^k a^k = \frac{\eta-1+\sum_k \rho^k a^k}{\eta-1} wL$, and using $\partial_{\theta^i} a^i = a^i \bar{\phi}(\bar{\theta}^i - \theta^i)$,

$$\frac{\partial \ln I}{\partial \theta^i} = \frac{a^i \Pi^i}{I} \bar{\phi}(\bar{\theta}^i - \theta^i) = \frac{\rho^i a^i}{\eta - 1 + \sum_k \rho^k a^k} \bar{\phi}(\bar{\theta}^i - \theta^i).$$

Since $\partial \ln U / \partial \theta^i = \partial \ln I / \partial \theta^i - \partial \ln P / \partial \theta^i$, this establishes (25), proving Part 1.

For Part 2, under the symmetric structure $\Gamma \mathbf{1} = (1 - \gamma^{iL}) \mathbf{1}$, hence $\Omega \mathbf{1} = (\gamma^{iL})^{-1} \mathbf{1}$: every row of Ω sums to $1/\gamma^{iL}$, and by symmetry so does every column, so $\lambda^i = \sum_n \rho^n \Omega^{ni} = \rho^i / \gamma^{iL}$. Because $\gamma^{mi} = \gamma^{im}$, the downstream price sum equals the own-sector sum, and the two price

terms combine:

$$\frac{\partial \ln U}{\partial \theta^i} = \frac{\rho^i a^i}{\eta - 1 + \sum_k \rho^k a^k} \bar{\phi}(\bar{\theta}^i - \theta^i) - 2\bar{t} \frac{\rho^i}{\gamma^{iL}} \sum_j \gamma^{ij} (\theta^i - \theta^j).$$

At the firm's own choice the private condition (26) gives $\bar{t} \sum_j \gamma^{ij} (\theta^i - \theta^j) = \frac{1}{\eta-1} \bar{\phi}(\bar{\theta}^i - \theta^i)$; substituting,

$$\left. \frac{\partial \ln U}{\partial \theta^i} \right|_{\text{dec}} = \rho^i \bar{\phi}(\bar{\theta}^i - \theta^i) \left[\frac{a^i}{\eta - 1 + \sum_k \rho^k a^k} - \frac{2}{\gamma^{iL}(\eta - 1)} \right].$$

Since $a^i \leq 1$ and $\sum_k \rho^k a^k > 0$, the first bracket term is strictly below $1/(\eta - 1)$, itself below $2/(\gamma^{iL}(\eta - 1))$ as $\gamma^{iL} < 1$; the bracket is strictly negative. Thus $\text{sgn}(\partial \ln U / \partial \theta^i)|_{\text{dec}} = \text{sgn}(\theta^i - \bar{\theta}^i)$: firms under-invest in adaptation. $\square$

A.7 Proof of Proposition 5

Proof. Drop the sector superscript and normalize the common productivity $A = 1$. We construct the equilibrium with equal wages, $w_1 = w_2 \equiv w$, normalized to one, and verify below that it clears every market for any (θ_1, θ_2) , so the unilateral perturbation of θ_2 leaves relative wages unchanged. Symmetry ($\bar{\theta}_1 = -\bar{\theta}_2 > 0$) implies a symmetric decentralized equilibrium, $\theta_2 = -\theta_1 < 0$, with retained-profit fraction $a_d \equiv \exp(-\frac{1}{2}\bar{\phi}(\theta_d - \bar{\theta}_d)^2)$, $a_1 = a_2 \equiv a$. By (20)–(21), $P_d Q_d = I_d + \Phi_d$, $I_d = wL + a_d \Pi_d$, and $U_d = I_d / P_d$.

Given the common wage, $c_1 = c_2$ for any (θ_1, θ_2) : writing $u_d \equiv c_d^{1-\xi}$ and $B \equiv (\tau \exp(\frac{1}{2}\bar{t}(\theta_1 - \theta_2)^2))^{1-\xi}$, the cost fixed point (5) reads $\ln u_d = (1 - \gamma^L) \ln(u_d + B u_{-d})$, so $s \equiv u_1 / u_2$ satisfies $\ln s = (1 - \gamma^L) \ln \frac{s+B}{1+B s}$; since $|\ln \frac{s+B}{1+B s}| \leq |\ln s|$ for all $s, B > 0$, $|\ln s| \leq (1 - \gamma^L) |\ln s|$, hence $s = 1$. The fixed point (5) then gives $\ln c_d = \ln w + \frac{1 - \gamma^L}{\gamma^L} \frac{1}{1 - \xi} \ln[1 + (\tau \exp(\frac{1}{2}\bar{t}(\theta_1 - \theta_2)^2))^{1-\xi}]$, the same in both countries, with the foreign intermediate-input expenditure share $\bar{\chi}$ of (27). The final-good price index differs from c_d only by θ -independent constants, so $\partial \ln P_d = \partial \ln c_d$ and

$$\frac{\partial \ln P_d}{\partial \theta_2} = -\bar{t} \frac{1 - \gamma^L}{\gamma^L} \bar{\chi} (\theta_1 - \theta_2) < 0, \quad d = 1, 2.$$

Let $E_d \equiv P_d Q_d$ denote gross final expenditure, so $E_d = I_d + \Phi_d = wL + \Pi_d$ for any (a_1, a_2) . Profit accrues from final sales only, $\Pi_d = \frac{1}{\eta} X_d$, where $X_d = \sum_{d'} \pi_{d'd} E_{d'}$ by (19). Because $c_1 = c_2$ and the final-use trade cost τ^U is common to both directions, the final-expenditure share matrix $[\pi_{d'd}]$ of (10) is symmetric; its rows sum to one, hence so do its columns. Substituting,

$$E_d = wL + \frac{1}{\eta} \sum_{d'} \pi_{d'd} E_{d'}, \quad d = 1, 2,$$

and since $\frac{1}{\eta} \sum_{d'} \pi_{d'd} = \frac{1}{\eta} < 1$, the unique solution is

$$E_1 = E_2 = \frac{\eta}{\eta - 1} wL, \quad X_d = E_d, \quad \Pi_1 = \Pi_2 = \frac{wL}{\eta - 1}.$$

Labor markets also clear at the common wage: $e_d = \frac{\eta-1}{\eta} X_d + M_d$ with $M_d = (1 - \gamma^L)[(1 -$

$\bar{\chi})e_d + \bar{\chi}e_{-d}]$ gives $e_1 = e_2$ and $\gamma^L e_d = \frac{\eta-1}{\eta} E_d = wL$, matching (22). Hence the equal-wage allocation clears every market for any (θ_1, θ_2) , including at the initial decentralized equilibrium, and E_d , Π_d , and w remain constant as θ_2 varies.

Since $\Pi_d = \frac{wL}{\eta-1}$ is constant, θ_2 enters $I_d = wL + a_d \Pi_d = wL \frac{\eta-1+a_d}{\eta-1}$ only through a_d . Using $\partial_{\theta_2} a_2 = a_2 \bar{\phi}(\bar{\theta}_2 - \theta_2)$,

$$\frac{\partial \ln I_2}{\partial \theta_2} = \frac{a \Pi}{I} \bar{\phi}(\bar{\theta}_2 - \theta_2) = \frac{a}{\eta-1+a} \bar{\phi}(\bar{\theta}_2 - \theta_2), \quad \frac{\partial \ln I_1}{\partial \theta_2} = 0,$$

the second because a_1 does not depend on θ_2 .

With $U_d = I_d / P_d$, combining the income and price effects gives (27):

$$\frac{\partial \ln U_2}{\partial \theta_2} = \frac{a}{\eta-1+a} \bar{\phi}(\bar{\theta}_2 - \theta_2) + \bar{t} \frac{1-\gamma^L}{\gamma^L} \bar{\chi}(\theta_1 - \theta_2), \quad \frac{\partial \ln U_1}{\partial \theta_2} = \bar{t} \frac{1-\gamma^L}{\gamma^L} \bar{\chi}(\theta_1 - \theta_2).$$

Take $\theta_1 > \theta_2$, a finite trade cost (so $\bar{\chi} > 0$), $\eta > 1$, and $\gamma^L \in (0, 1)$. Then $\partial \ln U_1 / \partial \theta_2 > 0$. For country 2, a firm maximizes $a(\theta) \Pi(\theta)$ taking its suppliers' costs as given, so $\partial_{\theta} \ln \Pi = -(\eta-1) \partial_{\theta} \ln c = -(\eta-1) \bar{t} (1-\gamma^L) \bar{\chi}(\theta_2 - \theta_1)$, and the first-order condition (the one-sector, two-country specialization of (17)) is

$$\frac{1}{\eta-1} \bar{\phi}(\theta_2 - \bar{\theta}_2) = \bar{t} (1-\gamma^L) \bar{\chi}(\theta_1 - \theta_2).$$

Substituting $\bar{\phi}(\bar{\theta}_2 - \theta_2) = -(\eta-1) \bar{t} (1-\gamma^L) \bar{\chi}(\theta_1 - \theta_2)$,

$$\left. \frac{\partial \ln U_2}{\partial \theta_2} \right|_{\text{dec}} = \bar{t} (1-\gamma^L) \bar{\chi}(\theta_1 - \theta_2) \left[\frac{1}{\gamma^L} - \frac{a(\eta-1)}{\eta-1+a} \right] > 0,$$

since $\frac{a(\eta-1)}{\eta-1+a} < 1 < \frac{1}{\gamma^L}$. With equal incomes, world welfare also rises. $\square$

A.8 Proof of Proposition 6

Proof. The bilateral change preserves the symmetry of the two countries, so the equilibrium remains symmetric: $\theta_2 = -\theta_1$, a common retained fraction $a = \exp(-\frac{1}{2} \bar{\phi}(\theta_1 - \bar{\theta}_1)^2)$, and a common wage, normalized to one. As shown in the proof of Proposition 5, for any $(\theta_1, \theta_2, \tau, \tau^U)$ we have $E_d = \frac{\eta}{\eta-1} wL$, $X_d = E_d$, and $\Pi_d = \frac{wL}{\eta-1}$, so these remain constant along the perturbation. Write $\Delta \equiv \theta_1 - \theta_2$ and $t = \exp(\frac{1}{2} \bar{t} \Delta^2)$, so $d \ln t = \bar{t} \Delta d\Delta$ and, by symmetry, $d\theta_1 = -d\theta_2 = \frac{1}{2} d\Delta$. We differentiate the full equilibrium in $\ln \tau$, valid by the implicit function theorem whenever the denominator of (29) is nonzero.

From the proof of Proposition 5, $\ln c = \ln w + \frac{1-\gamma^L}{\gamma^L} \frac{1}{1-\bar{\xi}} \ln[1 + (\tau t)^{1-\bar{\xi}}]$, so

$$\frac{d \ln c}{d \ln \tau} = \frac{1-\gamma^L}{\gamma^L} \bar{\chi} \left(1 + \frac{d \ln t}{d \ln \tau} \right),$$

and differentiating $\ln P_d = \ln \mu_X + \ln c + \frac{1}{1-\eta} \ln[1 + (\tau^U)^{1-\eta}]$ with $d \ln \tau^U = d \ln \tau$ gives

$$\frac{d \ln P_d}{d \ln \tau} = \bar{\pi} + \frac{1-\gamma^L}{\gamma^L} \bar{\chi} \left(1 + \frac{d \ln t}{d \ln \tau} \right), \quad d = 1, 2.$$

With Π_d constant, $I_d = wL \frac{\eta-1+a}{\eta-1}$ moves only through a , and $d \ln a = \bar{\phi}(\bar{\theta}_1 - \theta_1) d\theta_1 =$

$\bar{\phi}(\bar{\theta}_1 - \theta_1) d \ln t / (2\bar{t}\Delta)$:

$$\frac{d \ln I_d}{d \ln \tau} = \frac{a}{\eta - 1 + a} \frac{\bar{\phi}(\bar{\theta}_1 - \theta_1)}{2\bar{t}\Delta} \frac{d \ln t}{d \ln \tau}, \quad d = 1, 2.$$

Then $d \ln U_d = d \ln I_d - d \ln P_d$ gives (28) with $\varrho = \frac{a}{\eta-1+a} \frac{\bar{\phi}(\bar{\theta}_1 - \theta_1)}{2\bar{t}\Delta} - \frac{1-\gamma^L}{\gamma^L} \bar{\chi}$.

The first-order condition from the proof of Proposition 5, written for country 1, is $\frac{1}{\eta-1} \bar{\phi}(\bar{\theta}_1 - \theta_1) = \bar{t}(1 - \gamma^L) \bar{\chi} \Delta$. Substituting into ϱ ,

$$\varrho = (1 - \gamma^L) \bar{\chi} \left[\frac{a(\eta - 1)}{2(\eta - 1 + a)} - \frac{1}{\gamma^L} \right] = -\frac{1 - \gamma^L}{\gamma^L} \bar{\chi} \left[1 - \frac{\gamma^L}{2} \frac{a(\eta - 1)}{\eta - 1 + a} \right] < 0,$$

since $a(\eta - 1)/(\eta - 1 + a) < 1$ and $\gamma^L < 1$.

For (29), take logs of the first-order condition, $\ln \bar{\phi} + \ln(\bar{\theta}_1 - \theta_1) = \ln[(\eta - 1)(1 - \gamma^L)\bar{t}] + \ln \bar{\chi} + \ln \Delta$, and differentiate totally in $\ln \tau$. Using $d \ln \bar{\chi} = -(\xi - 1)(1 - \bar{\chi})(d \ln \tau + d \ln t)$, $d \ln \Delta = d \ln t / (\bar{t}\Delta^2)$, and $d \ln(\bar{\theta}_1 - \theta_1) = -\frac{1}{2} d\Delta / (\bar{\theta}_1 - \theta_1)$ with $d\Delta = d \ln t / (\bar{t}\Delta)$:

$$-\frac{1}{2\bar{t}\Delta(\bar{\theta}_1 - \theta_1)} \frac{d \ln t}{d \ln \tau} = -(\xi - 1)(1 - \bar{\chi}) \left(1 + \frac{d \ln t}{d \ln \tau} \right) + \frac{1}{\bar{t}\Delta^2} \frac{d \ln t}{d \ln \tau};$$

collecting the $d \ln t / d \ln \tau$ terms gives (29). Positivity of its denominator is implied the local-stability condition of the proposition. Then $d \ln t / d \ln \tau > 0$ and, since $\varrho < 0$, the technology response reinforces the direct effect. $\square$

A.9 Efficiency Result for Models with Intermediate Sales Markups

Setup. As stated in Section 3.3, a supplier charges $\mu_M c_o^j(\tilde{\theta})$ to intermediate buyers, so the cost in (3) becomes $\tilde{c}_{do}^j(v, \omega) = \mu_M c_o^j(\theta(\omega)) \tau_{do}^j t(\theta(v), \theta(\omega))$, and the cost fixed point (5) becomes,

$$c_d^i(\theta) = \frac{\mu_M^{1-\gamma^{iL}}}{A_d^i} w_d^{\gamma^{iL}} \prod_{j=1}^S \left[\sum_o \int [c_o^j(\tilde{\theta}) \tau_{do}^j t(\theta, \tilde{\theta})]^{1-\xi} d\Theta_o^j(\tilde{\theta}) \right]^{\frac{\gamma^{ij}}{1-\xi}}. \quad (\text{A.23})$$

As noted in the main text, the trade shares (6) and (8) are unchanged. Total cost and profit now carry the margin $1/\xi$ on firm sales,

$$e_d^i(\theta) = \frac{\xi - 1}{\xi} M_d^i(\theta) + \frac{\eta - 1}{\eta} X_d^i(\theta), \quad \Pi_d^i(\theta) = \frac{1}{\xi} M_d^i(\theta) + \frac{1}{\eta} X_d^i(\theta), \quad (\text{A.24})$$

with $e_d^i(\theta)$ replacing the production cost $M_d^i(\theta) + (1 - \frac{1}{\eta})X_d^i(\theta)$ in the market clearing (18) and labor clearing (22), and $\Pi_d^i(\theta)$ replacing $\frac{1}{\eta} X_d^i(\theta)$ in the adaptation cost (13). Firms choose θ to maximize $[1 - \phi(\theta, \bar{\theta})] \Pi_d^i(\theta)$, replacing (14); slightly abusing notation, the one-argument $\Pi_d^i(\theta)$ denotes variable profit before adaptation costs, and net profit $[1 - \phi] \Pi_d^i(\theta)$ enters the income condition (21).

Technology choice. With the markup, firm sales $M_d^i(\theta)$ contribute to profit and respond to θ both through the firm's own cost (its appeal to buyers as a supplier) and through the technologies of the buyers it serves. Define the supplier- and buyer-side weighted-average tech-

nologies for a firm choosing technology θ as:

$$\vartheta_S(\theta) \equiv \sum_{j,o} \frac{\gamma^{ij}}{1 - \gamma^{iL}} \int \chi_{do}^j(\theta, \tilde{\theta}) \tilde{\theta} d\Theta_o^j(\tilde{\theta}), \quad \vartheta_B(\theta) \equiv \sum_{d',i'} \int s_{d'd}^{i'i}(\tilde{\theta}, \theta) \tilde{\theta} d\Theta_{d'}^{i'}(\tilde{\theta}), \quad (\text{A.25})$$

where the buyer-side share $s_{d'd}^{i'i}(\tilde{\theta}, \theta) d\Theta_{d'}^{i'}(\tilde{\theta}) \equiv \gamma^{i'i} e_{d'}^{i'}(\tilde{\theta}) \chi_{d'd}^i(\tilde{\theta}, \theta) d\Theta_{d'}^{i'}(\tilde{\theta}) / M_d^i(\theta)$ integrates to one, $e_{d'}^{i'}(\tilde{\theta})$ is the buyer's total variable cost defined with (A.24), and we suppress the firm's (d, i) indices on ϑ_S and ϑ_B .

Under Assumption 1, at an interior optimum the technology $\theta_d^i(\bar{\theta})$ chosen by a firm in (d, i) with endowment technology $\bar{\theta}$ can be written as a weighted average of its endowment technology and the average technologies of its suppliers and buyers:

$$\theta_d^i(\bar{\theta}) = \omega_E^i \bar{\theta} + \omega_S^i \vartheta_S(\theta_d^i(\bar{\theta})) + \omega_B^i \vartheta_B(\theta_d^i(\bar{\theta})), \quad \omega_E^i + \omega_S^i + \omega_B^i = 1, \quad (\text{A.26})$$

with

$$\omega_E^i = \frac{\bar{\phi}}{D^i}, \quad \omega_S^i = \frac{\bar{t}(1 - \gamma^{iL})[(\eta - 1)\sigma_X^i + (\xi - 1)\sigma_M^i]}{D^i}, \quad \omega_B^i = \frac{\bar{t}(\xi - 1)\sigma_M^i}{D^i},$$

$$D^i \equiv \bar{\phi} + \bar{t}(1 - \gamma^{iL})[(\eta - 1)\sigma_X^i + (\xi - 1)\sigma_M^i] + \bar{t}(\xi - 1)\sigma_M^i,$$

where $\sigma_X^i \equiv \frac{X_d^i/\eta}{\Pi_d^i}$ and $\sigma_M^i \equiv \frac{M_d^i/\xi}{\Pi_d^i}$, evaluated at $\theta_d^i(\bar{\theta})$, are the shares of the firm's operating profit earned on final-use sales and on sales to other firms, respectively, with $\sigma_X^i + \sigma_M^i = 1$. When $\sigma_M^i = 0$, $\omega_B^i = 0$, $\omega_S^i = \omega^i$, and $\omega_E^i = 1 - \omega^i$: (A.26) collapses to the baseline condition (17).

To derive (A.26), note that an interior optimum satisfies

$$\partial_\theta \ln(1 - \phi) + \partial_\theta \ln \Pi_d^i = 0.$$

Under Assumption 1, $\partial_\theta \ln(1 - \phi) = -\bar{\phi}(\theta - \bar{\theta})$, and from (A.24), $\partial_\theta \ln \Pi_d^i = \sigma_M^i \partial_\theta \ln M_d^i + \sigma_X^i \partial_\theta \ln X_d^i$. Three derivatives are needed.

Cost. Taking logs of (A.23), only $t(\theta, \tilde{\theta})^{1-\xi} = \exp(\frac{1}{2}(1 - \xi)\bar{t}(\theta - \tilde{\theta})^2)$ depends on θ ; with $\sum_j \gamma^{ij} = 1 - \gamma^{iL}$ and (A.25),

$$\partial_\theta \ln c_d^i(\theta) = \bar{t}(1 - \gamma^{iL})(\theta - \vartheta_S(\theta)). \quad (\text{A.27})$$

Final sales. Since $X_d^i(\theta) \propto c_d^i(\theta)^{1-\eta}$ by (10) and (19), $\partial_\theta \ln X_d^i = -(\eta - 1)\bar{t}(1 - \gamma^{iL})(\theta - \vartheta_S(\theta))$.

Firm sales. The firm is atomistic, so its θ moves only the numerator of each buyer's sourcing share (6)—not buyers' price indices or expenditures—giving $\partial_\theta \ln \chi_{d'd}^i(\tilde{\theta}, \theta) = -(\xi - 1)[\partial_\theta \ln c_d^i(\theta) + \bar{t}(\theta - \tilde{\theta})]$. Differentiating $M_d^i(\theta)$ in (18) and using (A.27) and (A.25),

$$\partial_\theta \ln M_d^i(\theta) = -(\xi - 1)\bar{t}(1 - \gamma^{iL})(\theta - \vartheta_S(\theta)) - (\xi - 1)\bar{t}(\theta - \vartheta_B(\theta)).$$

Substituting the X- and M-derivatives above into $\bar{\phi}(\theta - \bar{\theta}) = \sigma_M^i \partial_\theta \ln M_d^i + \sigma_X^i \partial_\theta \ln X_d^i$ and collecting the coefficients of θ , $\bar{\theta}$, ϑ_S , and ϑ_B gives (A.26) with the stated weights.

Closed economy. With markups, dropping country subscripts, the marginal effect of θ^i on real income becomes

$$\frac{\partial \ln U}{\partial \theta^i} = \frac{a^i \Pi^i}{I} \bar{\phi}(\bar{\theta}^i - \theta^i) - \bar{t} \lambda^i \sum_j \gamma^{ij}(\theta^i - \theta^j) - \bar{t} \sum_{m \neq i} \lambda^m \gamma^{mi}(\theta^i - \theta^m), \quad (\text{A.28})$$

with $\Pi^i = \frac{1}{\xi} M^i + \frac{1}{\eta} X^i$, $a^i = \exp(-\frac{1}{2} \bar{\phi}(\theta^i - \bar{\theta}^i)^2)$, and λ^i the Domar weight defined in Proposition 4: the two price terms are exactly those of (25), and the markup enters only through the profit base Π^i .

Contrasting this with the firm's trade-off, the closed-economy specialization of (A.26),

$$\underbrace{\bar{\phi}(\bar{\theta}^i - \theta^i)}_{\text{adaptation cost}} = \underbrace{\bar{t}[(\eta - 1)\sigma_X^i + (\xi - 1)\sigma_M^i] \sum_j \gamma^{ij}(\theta^i - \theta^j)}_{\text{production cost (supplier side)}} + \underbrace{\bar{t}(\xi - 1)\sigma_M^i \sum_m s^{mi}(\theta^i - \theta^m)}_{\text{downstream stake (buyer side)}}, \quad (\text{A.29})$$

where $s^{mi} \equiv \gamma^{mi} e^m / M^i$, the closed-economy counterpart of the buyer share in (A.25), is sector m 's share in sector i 's sales to firms (the $m = i$ summand vanishes since $\theta^i - \theta^m = 0$), we can see that the firm now has a stake in its downstream users. The buyer-side term is a private counterpart of the downstream price term in (A.28), but it is weighted by the firm's sales shares s^{mi} and the margin $(\xi - 1)\sigma_M^i$ rather than by the social weights $\lambda^m \gamma^{mi}$. Setting $\sigma_M^i = 0$ and dividing by $\eta - 1$ recovers the baseline condition (26), up to the ρ^i factor.

Symmetric case. Under the symmetric structure of Proposition 4, substituting the firm's condition (A.29) into the gradient (A.28) shows that the gradient has the sign of $(\theta^i - \bar{\theta}^i)$ if and only if the ratio of the private to the social marginal incentive for adaptation,

$$\mathcal{R}^i = \frac{\gamma^{iL} a^i}{2\rho^i I} \left[\frac{\eta - 1}{\eta} X^i + \frac{\xi - 1}{\xi} M^i \frac{2 - \gamma^{iL}}{1 - \gamma^{iL}} \right], \quad (\text{A.30})$$

is below one. The no-markup case has only the final-sales term $\frac{\eta-1}{\eta} X^i$ in the bracket; the markup adds the margin on sales to other firms, which responds to the firm's technology through both its supplier appeal and its buyer-side stake, and income I now includes those margins. The market-clearing accounting nevertheless bounds $\mathcal{R}^i$ below $1 - \frac{1}{2\xi}$ for every technology configuration. The gradient thus has the sign of $(\theta^i - \bar{\theta}^i)$ at the decentralized equilibrium: firms under-invest, and without the markup $\mathcal{R}^i < 1$ collapses to the bracket of Proposition 4.

Two countries. In the one-sector, two-country case, the convergence elasticity (29) is exactly as stated in Proposition 6, and the coefficient converting the technology response into welfare becomes

$$\varrho = \frac{1 - \gamma^L}{\gamma^L} \bar{\chi} (\mathcal{R} - 1), \quad \mathcal{R} = \frac{\gamma^L a (\eta - 1)(2\xi - 1)}{2[\gamma^L \xi (\eta - 1) + a(\eta(1 - \gamma^L) + \gamma^L \xi)]} \leq a \left(1 - \frac{1}{2\xi} \right) < 1,$$

where $\mathcal{R}$ is the one-sector value of (A.30), with the market-clearing accounting substituted in. In the benchmark of Proposition 6, ϱ takes the same form with $\mathcal{R} = \frac{\gamma^L}{2} \frac{(\eta-1)a}{\eta-1+a}$. The markup raises $\mathcal{R}$ for all parameter values: firms internalize part of the downstream compatibility benefit, shrinking the externality. Because $\mathcal{R}$ remains strictly below one, $\varrho < 0$: the amplification result remains robust.

Unilateral adaptation. The same ratio governs whether a unilateral move benefits both countries under markup. A marginal increase in θ_2 toward θ_1 , holding θ_1 fixed, always benefits country 1, whose gain materializes through cheaper inputs as before. Country 2 gains if and only if $2\mathcal{R} < 1$: twice the threshold that governs symmetric adaptation. Since $\mathcal{R} < 1$ always holds while $2\mathcal{R}$ may exceed one, a unilateral move can lower the mover's welfare even though symmetric convergence raises welfare in both. Without the markup, $2\mathcal{R} = \gamma^L \frac{(\eta-1)a}{\eta-1+a} < 1$ for all parameters, and country 2 always gains, recovering Proposition 5.

Appendix B Reduced-Form Evidence

We discuss the details for our reduced-form evidence in this section. Section B.1 introduces the data sources, dataset construction, and cleaning procedures. Section B.2 provides validation exercises for our text-based measure on technology similarity. Section B.3 describes the construction of the technology competition score. Section B.4 introduces details in constructing the measure of compatibility intensity for each industry. Section B.5 presents firm-level evidence on the relationship between technology similarity and importing, which underlies one of our calibration targets. Finally, Section B.6 presents additional robustness checks for the analysis in Section 4.2.

B.1 Data

Our dataset is built from four main sources, the World Input-Output Tables, patent data from the PATSTAT Global, Chinese firm-level data, and tariff data from the TRAINS database. We explain in detail where we source the data and how we construct the final datasets.

World Input-Output Tables (WIOTs). Our multi-country analyses center around countries in the WIOTs (Timmer et al., 2015) sourced from the World Input-Output Database (2016 Release). To facilitate analysis, we classify all countries in the WIOTs into 29 regions based on their geographic and political proximity. Table B.1 lists the regions.

Patent Data. Our patent data are sourced from the PATSTAT Global (the 2023 Fall version), a comprehensive database comprising bibliographical information on over 100 million patent documents. This database encompasses patent records from 90 patent-issuing authorities, including all major national (e.g. USPTO), regional (e.g. EPO), and global (e.g. the Patent Cooperative Treaty) patent offices. To focus on production technologies, we narrow our focus to invention patents and utility models while excluding design patents from our analysis.

Throughout our analysis, we define a patent as a *patent family*, which typically protects one invention. An invention can be a new product, a new process to produce, or a new technical solution. A patent family can involve multiple patent *applications* when this technology seeks patent protection from different authorities. For each patent, we denote its *year of invention* as the year in which the first application is filed. We keep those patents with at least one application granted.

Each patent record contains details about the inventors of the patent (the individuals who invent the patent, though not necessarily the applicant or the owner of the patent) and their countries of residence. We first map the countries into the geo-political regions and take all these regions as the *regions of invention* of that patent, assigning weights to each region based on the number of inventors from that specific location. For example, if a patent involves five inventors—two from China and three from the US—we consider that China holds 2/5 of the patent, while the US holds 3/5. In case where the inventor information is missing, we designate the region of its first application as the region of invention. This assumption is grounded in the idea that a patent would typically be filed domestically before seeking international protection.

Chinese Firm-Level Data. Our firm-level analysis focuses on Chinese manufacturing firms in the Annual Survey of Industrial Enterprise maintained by the National Bureau of Statistics of China (NBSC). The dataset offers a yearly census of all state-owned manufac-

Table B.1: Geo-Political Regions

Region Code	Region	ISO3 in WIOTs
AUS	Australia	AUS
AUT	Austria	AUT
BLK	Balkans	BGR, HRV, GRC
BLT	Baltic States	EST, LVA, LTU
BNE	Benelux	BEL, LUX, NLD
BRA	Brazil	BRA
CAN	Canada	CAN
CHE	Switzerland	CHE
CHN	China (Mainland)	CHN
CNE	Central Europe	CZE, HUN, POL, SVK, SVN
DEU	Germany	DEU
ESP	Spain	ESP
FRA	France	FRA
GBR	United Kingdom	GBR
IDN	Indonesia	IDN
IND	India	IND
IRL	Ireland	IRL
ITA	Italy	ITA
JPN	Japan	JPN
KOR	South Korea	KOR
MEX	Mexico	MEX
NRD	Nordic Countries	DNK, FIN, NOR, SWE
PRT	Portugal	PRT
ROU	Romania	ROU
ROW	Rest of the World	CYP, MLT, ROW
RUS	Russia	RUS
TUR	Turkey	TUR
TWN	Taiwan	TWN
USA	United States	USA

turing firms and all non-state manufacturing firms with sales greater than RMB 5 million (approx. US\$600,000) over 1998–2014, including plant-level information on industry, location, sales, employment, etc.

Crucial to our analysis are a firm’s identity (ω) and the prime industry (i) it belongs to. We link each firm consistently over time using information on the NBS ID, firm name, the name of legal person representative, phone number, address, name of main products, founding year, etc. To investigate the input-output linkage, we manually map the industry codes to the 3-digit industry classification used in the 2007 Chinese Input-Output Table.

To link Chinese patents to their patent portfolios, we use as a bridge patent application data provided by China’s State Intellectual Property Office (SIPO). For each firm in the SIPO data, we link it to firms in the NBSC database by matching the names (in Chinese). We then trace individual patents in SIPO to patents in PATSTAT using application numbers that can be linked between the two databases. With this link, for each Chinese firm, we observe its global portfolio of patents, and their (forward and backward) citations with other worldwide

Table B.2: Summary Statistics: Technology Similarity Measures

	Mean	Std. Dev.	Min	Max	Obs.
Similarity _{ωot}	0.564	0.090	0.071	0.965	5,147,016
Similarity _{ωojt}	0.506	0.110	-0.011	0.978	13,781,232
Similarity _{$do,HS6,t$}	0.694	0.112	0.171	0.977	5,720,642

Notes: Similarity _{ωot} is the cosine similarity between the text embeddings of patents in period t from a Chinese firm ω and those from a foreign region o . Similarity _{ωojt} and Similarity _{$do,HS6,t$} are defined analogously. Similarity _{ωojt} compares firm ω with a region-sector pair (o, j) ; Similarity _{$do,HS6,t$} compares country d with patents in a specific HS-6 category in region o . Regions are classified as in Table B.1. Sectors are classified at the two-digit level of the International Standard Industrial Classification.

patents.⁷

Finally, we obtain information on firms' imports from China's General Administration of Customs, which provides detailed records on the universe of all Chinese trade transactions by both importing and exporting firms at the HS eight-digit level for the years 2000-2014. We merge the import data with the NBSC manufacturing firm survey data. The matching procedure consists of three main steps: (1) match by company names (in Chinese); (2) match by phone number and zip code; (3) match by phone number and the name of contact person.

Our final firm-level dataset includes all manufacturing firms with patents, regardless of whether they import goods from abroad. The panel is unbalanced with the number of firms increasing from 57,465 in the period 2000-2002 to 102,153 in the period 2012-2014.

Tariff Data. We source tariff data from UN TRAINS, which are downloaded from <https://wits.worldbank.org> for each year between 2000-2014. The raw data include the effectively applied tariff rates and MFN tariff rates at the importer-exporter-product level, where importers and exporters are in three-letter ISO country code, and products are at the level of 6-digit HS code (2007).

When cleaning the data, we drop all the observations where either the importer or the exporter is unspecified. For both the applied and MFN tariffs, when the observations are missing, we impute them with the first non-missing preceding tariff. If no earlier observation is available, we leave them missing and drop these observations in our regressions. Finally, we aggregate the tariff rates from yearly figures into three-year periods by computing simple averages.

Table B.2 reports summary statistics for the key similarity measures used in the main text and in this appendix.

B.2 Technology Similarity Measure Based on Patent Texts

Case Studies. In Section 4.1, we construct a measure of technology similarity based on the closeness of patent texts. To validate that our patent-based semantic similarity captures the level of compatibility between related technologies, we examine several cases of production of final products that have similar functions but follow different technological paths, which

⁷For this bridge to work, our assumption is that when Chinese firms file a patent in any foreign office, they also file for a patent for the same invention in the SIPO. The reason we use PATSTAT, instead of the SIPO data only is that the former allows us to measure citations and similarity between Chinese firms and worldwide patents that are not covered by SIPO.

Table B.3: Similarity Scores Between Selected Technology Pairs

IPC4 of input using technology	IPC4 of input supplying technology	Similarity
B60L (Electric Vehicle)	H01M (Battery)	0.81
B60L (Electric Vehicle)	F02M (Fuel System)	0.78
B60K (Fuel Vehicle)	H01M (Battery)	0.74
B60K (Fuel Vehicle)	F02M (Fuel System)	0.87
G02F (Flat Panel Display)	H10K (Organic Display)	0.85
G02F (Flat Panel Display)	H05H (Plasma Tech)	0.67
H01J (CRT Display)	H10K (Organic Display)	0.78
H01J (CRT Display)	H05H (Plasma Tech)	0.92

Notes: This table presents similarity scores between embeddings of downstream technologies (first column) and their corresponding upstream technologies (second column). The table is divided into two panels demonstrating different technological paths. The upper panel contrasts electric vs. fuel vehicles in relation to their key input technologies (batteries vs. fuel systems). The lower panel contrasts display technologies (CRT vs. flat panel) and their associated manufacturing processes (plasma technology vs. organic electronic devices). Each similarity score represents the cosine similarity between the average embeddings (over 2000-2014) of the technology pairs.

thus require different upstream technologies to be compatible.

Table B.3 shows the similarity scores between the IPC subclass (IPC4) of patents used by the production of a final product and that of its intimate upstream industry, averaged over 2000-2014. For the first example, the automotive sector, the technology class associated with electronic vehicles (B60L) has a similarity score of 0.81 with that of battery technology (H01M), while it has a similarity score of 0.78 with that of combustion engines (F02M). In contrast, the technology class associated with traditional fuel vehicles (B60K) has a similarity score of 0.74 with that of battery technology (H01M), and a higher similarity score of 0.87 with that of combustion engines (F02M). This matches the natural technological fit between different energy-based vehicles and the fundamental technology for converting the corresponding energy to motion.

The other example involves the technology transition from Cathode Ray Tube (CRT)-based display screens to flat panel displays (LCD and OLED). CRT technology (H01J) has a high similarity score (0.92) with plasma physics (H05H), reflecting CRT’s core dependence on electron beams. In contrast, flat panel displays (G02F) show lower similarity with plasma physics (0.67) but higher similarity with organic electronic devices (H10K, 0.85). This aligns with the fundamental technological differences between CRT and flat panel displays: CRT relies on electron beams in vacuum tubes, while flat panel displays use solid-state electronic control. These examples demonstrate that our semantic similarity measure captures meaningful technological compatibility relationships along production chains.

To illustrate what information the sentence embedding method encodes in terms of abstract semantic relationships, and to demonstrate its advantages over traditional word-count based measures like TF-IDF (used by e.g., Kelly, Papanikolaou, Seru and Taddy, 2021) for capturing degree of compatibility between technologies,⁸ we construct an example of technology descriptions of electric vehicles (EVs) and gasoline vehicles, along with their respective upstream technologies: batteries and combustion engines. Table B.4 reports the example

⁸TF-IDF (Term Frequency-Inverse Document Frequency) is a numerical statistic that reflects how important a word is to a document. It combines how frequently a term appears in a document (TF) with how unique that term is across all documents (IDF). Words that appear frequently in one document but rarely in others receive higher TF-IDF scores.

Table B.4: Examples of Technology Description and Embedding-based Similarity

Text1 (Input using)	Text2 (Input supplying)	Sentence Embedding Similarity	TF-IDF Similarity
Electric vehicles need batteries, not gas engines	Batteries power electric cars, not gas cars	0.77	0.21
Electric vehicles need batteries, not gas engines	Gas engines power gas cars, not electric cars	0.76	0.21
Gas cars need engines, not batteries	Batteries power electric cars, not gas cars	0.70	0.21
Gas cars need engines, not batteries	Gas engines power gas cars, not electric cars	0.75	0.21

Notes: This table presents examples of technology descriptions and their similarity scores calculated using two different methods: sentence embeddings and TF-IDF. The descriptions represent technologies related to electric and gasoline vehicles, along with their upstream components (batteries and engines). The last two columns report the computed similarity measures for different technology pairs under the two methods.

Table B.5: Average Similarity Across Different IPC Levels

Classification Level	Average Similarity
Within same IPC subclass (IPC4)	0.40
Within same IPC class (IPC2)	0.33
Within same IPC section (IPC1)	0.31
All sample	0.29

Notes: This table is based on a sample of 10 patent abstracts from each IPC4 subclass, drawn from U.S. patents in 2014, totaling 5,616 patents. Bilateral cosine similarity scores were calculated for each pair of patents (excluding self-similarity), and the average similarity was computed for patents within different IPC levels.

descriptions, and the similarity scores calculated for different technology pairs under different methods.

The example descriptions intentionally include both positive and negative statements, which are common in patent abstracts to delineate a technology’s applicability. The results show that sentence embeddings successfully capture the technological compatibility relationships: they assign higher similarity scores to naturally compatible pairs (EVs with batteries, gasoline vehicles with engines). In contrast, the TF-IDF method assigns identical similarity scores across all technology pairs as it relies solely on word overlap rather than semantic meaning.

These results show that sentence embeddings are particularly effective for understanding technological compatibility, as they can process and interpret complete descriptions rather than just individual terms, while word-count based methods remain valuable for other analytical purposes, such as identifying emerging technical terminology.

Statistical Validation. We now further validate this patent-text-based similarity measure by correlating it with other factors that are likely associated with technology proximity.

First, we show in Table B.5 that patent abstracts within the same technology class tend to have higher average similarity, with finer classification levels showing greater similarity.

Second, we show that sectors with stronger input-output linkages exhibit greater simi-

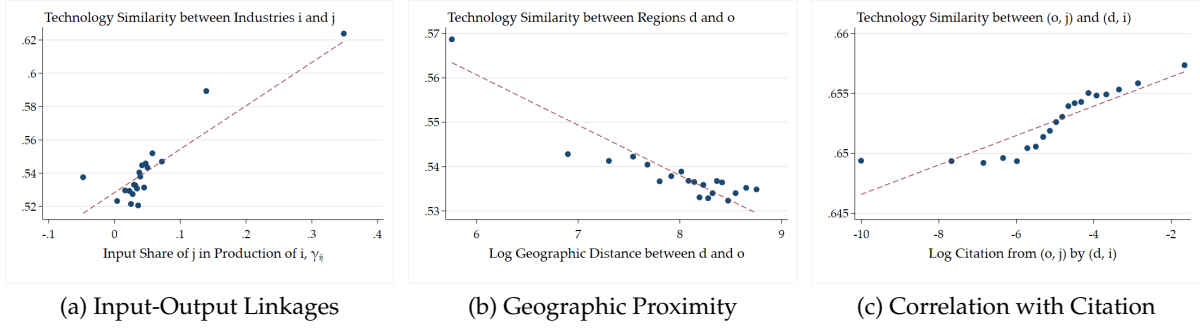


Figure B.1: Validation Checks for Technology Similarity

Notes: All figures are binned scatter plots. Figure B.1a plots the average similarity against the input-output weights between 2-digit industries i and j , controlling for i and j fixed effects. Figure B.1b plots the average similarity against the log of geographic distance between countries d and o , controlling for d and o fixed effects. Figure B.1c plots the average similarity against the log of citation flows between (d, i) and (o, j) , controlling for $d-i$, $o-j$, $d-o$ and $i-j$ fixed effects.

ilarity than those with weaker connections. Figure B.1a shows a binned scatter plot where we regress the average similarity between industries i and j against γ_{ij}^i , the input weight from i in the production of j , controlling for the fixed effects of i and j . We observe a positive correlation that is statistically significant.

Third, we confirm that countries in close geographical proximity tend to have more similar patents. We show this with a binned scatter plot in Figure B.1b, where the average similarity between countries d and o is regressed on the logged geographic distance between the two countries, controlling for the fixed effects of d and o . There is a significantly negative correlation.

Last, Figure B.1c demonstrates a strong correlation between our text-based similarity measure and citation flows between country-industry pairs (d, i) and (o, j) , even conditional on $d-i$, $o-j$, $d-o$ and $i-j$ fixed effects. This also motivates our robustness checks using citation as an alternative measure.

B.3 Measuring Competing Technology

Semantic similarity between patent texts may reflect technological relations other than compatibility. The leading alternative is competition: two patents that propose rival solutions to the same problem also read alike, yet such technologies are substitutes for each other in production, whereas compatibility is about complementarity. This appendix describes how we construct a measure of technological competition between patents and, by aggregation, between firms, countries, and sectors. We control for this measure in all empirical analyses, so that the residual variation in similarity isolates compatibility.⁹

A Bi-Encoder Model of Competition. Formally, an embedding model is a map f_{θ} from text to $\mathbb{R}^{768}$ with trainable parameters θ : a tokenizer first converts the text into a sequence of integer tokens, and a transformer network then converts the token sequence into a vec-

⁹In principle, one could instead finetune a model that scores compatibility directly. In our experiments, however, large language models agree with one another much more consistently when judging whether two technologies *compete* than whether they are *compatible*. We therefore score competition, the concept that can be measured reliably, and use it as a control.

tor. We normalize the output, letting $z_i = f_\theta(x_i) / \|f_\theta(x_i)\|$ denote the embedding of patent i 's abstract x_i . Our similarity measure in Section 4.1 uses such a model ('gte-base-en-v1.5') trained so that the inner product $z_i^\top z_j$ tracks the semantic similarity of two texts.¹⁰

We now finetune this same model so that the inner product scores competition instead. For a patent pair (i, j) , let the binary label y_{ij} equal one if the two patents represent competing technologies. The model predicts competition with probability $\sigma(s_{ij})$, where σ is the logistic function and

$$s_{ij} = \alpha z_i^\top z_j + \beta$$

is the inner product after an affine transformation, with α and β two trainable scalars whose only role is to rescale the inner product into probability units. Training solves

$$\min_{\vartheta, \alpha, \beta} \sum_{(i,j)} \ell(\sigma(s_{ij}), y_{ij}),$$

where ℓ is the binary cross-entropy loss. We initialize ϑ at the base model, so that finetuning refines the semantic information already encoded in the embeddings to capture competition, rather than learning it from scratch. The objects carried to the empirical analysis are the fine-tuned embeddings and their inner products. Because the score is bilinear in the embeddings, it aggregates exactly as our similarity measure does: the average pairwise score between two groups of patents equals the score between the groups' average embeddings.

Sampling and Labeling Patent Pairs. The training data consist of patent pairs whose competitive relation is assessed by a large language model. We draw 20,000 anchor patents from the same universe underlying our similarity measure. For each anchor, we sample the candidate partners for which raw similarity is least informative about the nature of the technological relation: the two nearest neighbors (by base-model embedding) within the anchor's IPC4 class, where competing technologies are most likely to be found; the two nearest neighbors outside the class, which mix competing, compatible, and more loosely related technologies across fields; one patent drawn at random outside the class, which is almost surely unrelated and anchors the low end of the scale; and up to four patents linked to the anchor by citation, which are related by construction and enrich the sample with compatible pairs at more moderate similarity levels. This procedure yields 126,124 unique pairs.

Each pair is classified by a large language model (DeepSeek v4-flash, at temperature zero with structured output) into one of five categories: compatible/complementary, substitute/competing, related without a clear relation of either kind, unrelated, or insufficient information. Figure B.2 reproduces the instruction prompt. The instruction directs the model to compare the main technical contributions of the two patents and, in particular, to distinguish technologies that occupy the same functional slot (competing) from technologies that play different, interacting roles (compatible), exactly the distinction that raw semantic similarity conflates. We drop pairs labeled as having insufficient information and retain the 56,397 pairs whose labels carry high confidence.

Table B.6 summarizes the labeled sample. Two patterns stand out. First, among nearest-neighbor pairs, whether within or outside the anchor's patent class, competing and compat-

¹⁰This architecture is called a 'bi-encoder': each text is mapped to an embedding separately, and the relation between two texts is computed from their embeddings. A 'cross-encoder,' in contrast, takes a pair of texts as a single input and returns a score directly. The bi-encoder architecture is what makes our measurement scalable: embeddings are computed once per patent, and any pairwise or group-level score is an inner product.

Table B.6: LLM-Labeled Patent Pairs: Counts and Raw Similarity

	Within IPC4 (nearest neighbors)	Outside IPC4 (nearest neighbors)	Citation- linked	Outside IPC4 (random)	All
<i>Panel A: Number of pairs</i>					
Competing	9,462	4,816	5,661	1	19,940
Compatible	2,527	2,404	1,999	7	6,937
Related	1,139	2,358	1,096	63	4,656
Unrelated	143	2,823	2,614	19,284	24,864
<i>Panel B: Mean similarity under the base model</i>					
Competing	0.691	0.678	0.640	0.270	0.673
Compatible	0.665	0.656	0.560	0.445	0.632
Related	0.664	0.641	0.531	0.402	0.617
Unrelated	0.611	0.596	0.369	0.289	0.334

Notes: The sample consists of the 56,397 high-confidence LLM-labeled patent pairs used to finetune the competition model. Columns correspond to how the pair was sampled around its anchor patent; rows to the label assigned by the LLM. Similarity is the cosine similarity of the two patents’ base-model (‘gte-base-en-v1.5’) embeddings.

ible pairs have similarly high average raw similarity (0.691 versus 0.665 within class, 0.678 versus 0.656 outside). Raw similarity identifies these pairs as technologically close, but is by itself less informative about whether the relation is one of competition or compatibility; the competition score is designed to add this information. Second, the easy cases behave as expected: randomly drawn pairs are almost all unrelated, with far lower similarity (0.29 on average).

Training and Evaluation. We split the pairs by patent family into training, validation, and test sets, comprising roughly 80%, 10%, and 10% of the sample. We train with the AdamW optimizer (learning rates 10^{-5} for the encoder and 10^{-3} for the two scalars, weight decay 0.01) in batches, keeping the model from the epoch with the highest validation accuracy, which is reached at epoch 9;¹¹ accuracy mildly deteriorates afterward from overfitting. We measure accuracy by the AUC.¹² On validation pairs from the nearest-neighbor bands, where raw similarity is uninformative about the relation (Table B.6), the AUC rises from 0.55 under the base model to 0.71 after finetuning. On the sharpest contrast, competing versus compatible pairs, it rises from 0.56 to 0.72. Test-set results are similar (0.54 to 0.69, and 0.55 to 0.71). In other words, the base model measures how close two technologies are, but closeness alone reveals little about whether the relation is one of rivalry or complementarity; the finetuned model adds this dimension while retaining the semantic information of the base model.

A Contrastive Example. Table B.7 illustrates with three patents on the thermal management of electric-vehicle battery packs (all three patents belong to IPC class B60L, electric-

¹¹An epoch is one complete pass of the training algorithm through the training sample. Training proceeds over multiple epochs, fitting the data gradually; choosing the model by validation accuracy, computed on pairs not used in training, guards against overfitting.

¹²The AUC (area under the receiver operating characteristic curve) is the probability that a randomly drawn competing pair receives a higher score than a randomly drawn non-competing pair: 0.5 corresponds to an uninformative classifier and 1 to perfect discrimination.

Table B.7: A Contrastive Example: Competition versus Compatibility

Anchor: battery-pack heating based on vehicle state. “The invention discloses a battery pack heating method, device and equipment [...]. The method comprises the steps that when a battery pack meets a heating condition, the vehicle state is determined; when the vehicle state is a driving state, determining a target temperature according to the historical operation data of the vehicle; and heating the battery pack to enable the temperature of the battery pack to reach the target temperature. The heating temperature needed by the battery pack is determined according to the actual running condition of the whole vehicle, unnecessary heat loss [...] is reduced, electricity consumption is reduced, and the running cost of the vehicle is reduced.”	Compatible: battery-pack temperature prediction. “The invention discloses a power battery pack temperature prediction method and system. The method comprises [...]: determining heat generation model parameters, heat exchange model parameters and temperature measurement and calculation model parameters; evaluating average heat generation power in a future period of time and heat exchange power between the battery pack and the environment [...]; and predicting the temperature after a period of time in the future [...]. The battery pack temperature change is predicted by considering the charging and discharging requirements of the future application condition of the vehicle [...], and a basis is provided for battery pack temperature management.”
Competing: heating by mutual charge and discharge. “The invention discloses a battery heating control method and device [...]: acquiring battery parameters of a battery module; judging whether the battery parameters [...] meet a preset heating condition or not; if yes, a power battery and a power supply battery in the battery module are controlled to charge and discharge mutually, so that the battery module is heated through mutual charge and discharge [...]; the problem of poor convenience caused by the fact that the battery needs to be heated in advance when the vehicle is used in the low-temperature environment [...] is solved [...]”	

	Similarity (base model)	Competition probability (finetuned)
Anchor vs. competing patent	0.903	0.99
Anchor vs. compatible patent	0.849	0.05

Notes: Abstracts are quoted from PATSTAT with light trimming, marked by [...]. All three patents are in IPC subclass B60L. The pairs are from the validation sample of the finetuning exercise (PATSTAT application identifiers 580446316, 583051629, and 541506938). ‘Similarity’ is the cosine similarity of base-model embeddings; ‘competition probability’ is the finetuned model’s predicted probability that the two technologies compete.

vehicle propulsion). The anchor patent heats the battery pack to a target temperature chosen from the vehicle’s operating history. The first partner patent heats the battery pack by letting two batteries within the module charge and discharge each other, an alternative solution to the same problem and hence a competing technology. The second partner patent predicts the battery pack’s future temperature from a heat-generation model, an input into deciding when and how much to heat and hence a compatible technology. To the base model, the two partners are almost equally similar to the anchor (0.90 and 0.85). The finetuned model, in contrast, assigns the competing pair a competition probability of 0.99 and the compatible pair a probability of 0.05.

From Patents to Controls. We compute finetuned embeddings for all patents in our sample and aggregate them into competition scores at the same levels as our similarity measures: firm–region, firm–region–sector, and importer–exporter–product pairs. In all regressions in the paper, we control for the competition score at the level of the identifying variation. The variation in semantic similarity that remains after this control thus better reflects genuine compatibility.

Figure B.2: The Labeling Instruction Given to the Large Language Model

You are classifying the technological relation between two patent abstracts.

Classify only from the technical content of the abstracts. Compare the main technical contribution of each patent, not incidental features. Do not rely on assignee, IPC class, market category, keyword overlap, or general semantic similarity.

Choose exactly one label:

- A. Compatible / complementary
- B. Substitute / competing
- C. Related but no clear compatibility or substitution
- D. Unrelated
- E. Insufficient information

Definitions:

A. Compatible / complementary
Choose A when the two technologies have different non-redundant technical roles and a concrete technical interface, so they can plausibly be used together in the same product, process, or system. This includes component-system, material-product, reagent-process, tool-process, control-device, monitoring-process, testing-product, upstream-downstream, or enabling-use relations.

B. Substitute / competing
Choose B when the two technologies occupy the same or highly overlapping functional slot for the same or practically interchangeable target, output, operation, or technical problem. They are alternative ways to perform the same function, make the same kind of product, treat the same kind of object, or serve the same role in a system or process.

C. Related but no clear compatibility or substitution
Choose C when the technologies are meaningfully related by field, target, material, mechanism, operation, or application, but the abstracts do not support either a concrete complementary interface or a clear substitute relationship.

D. Unrelated
Choose D when there is no meaningful technical relation beyond generic engineering or scientific concepts.

E. Insufficient information
Choose E only when one or both abstracts are too vague, incomplete, generic, or unintelligible to determine the relation.

Decision procedure:

1. Identify each invention's main role, target/output, and technical function.
2. If the abstracts are too vague to identify these, choose E.
3. If there is no meaningful shared technical field, target, material, mechanism, operation, or application, choose D.
4. If the main inventions serve the same functional slot for the same or practically interchangeable target/output/problem, choose B.
5. If the main inventions serve different interacting roles with a concrete technical interface, choose A.
6. If they are technically related but neither step 4 nor step 5 is satisfied, choose C.

Tie-break rules:

- A method and an apparatus are not automatically complementary. Choose B if both independently perform the same immediate operation on the same target to produce the same output. Choose A only if one is clearly a tool, input, component, control layer, or implementation vehicle for the other.
- A system and a component/input/control/testing method are usually A when the second can directly operate within, supply, measure, control, test, manufacture, maintain, or enable the first.
- Different improvements to the same type of product are not automatically A. Choose A only if the improvements occupy different interacting roles; choose B if they are alternative designs for the same functional slot; otherwise choose C.
- Similar function on materially different targets, species, products, materials, industries, or applications is usually C, not B, unless the abstracts imply practical interchangeability.
- General co-use in the same industry, plant, device type, or broad production chain is not enough for A.
- If both A and B remain plausible after applying these rules, choose C with low confidence unless the abstracts clearly favor one relation.

Confidence:

"high": the label follows directly from explicit or strongly implied abstract content.

"medium": the label is likely but requires some inference.

"low": the relation is ambiguous, weakly supported, or depends on interpretation.

Return valid JSON only, with this exact schema:

```
{
  "label": "A" | "B" | "C" | "D" | "E",
  "confidence": "high" | "medium" | "low",
  "reason": "one short sentence"
}
```

The reason must state the decisive role relationship, not merely repeat keywords. Do not include any other text.

Notes: The system instruction used to label all candidate patent pairs, reproduced verbatim. Each request supplies the two patent abstracts and returns a label, a confidence level, and a one-sentence reason.

B.4 Measuring Compatibility Intensity

Since the importance of input compatibility may vary across industries, we construct a novel measure of compatibility intensity for each industry based on the full text of U.S. patents from the Orbis Intellectual Property Database, which a link full list of patents to global firms. For each 3-digit ISIC industry, We count the number of patenting firms, and the number of patenting firms with patents containing keywords ‘compatibility/compatible’ or ‘interoperability.’

We use the share of patenting firms in an industry with such patents as a proxy for the likelihood that for a random firm in this industry, technology compatibility is important. Table B.8 lists the measure for all manufacturing industries at a 2-digit level. In the regressions, we further map this industry-specific measure of compatibility intensity from 3-digit ISIC to 3-digit CIC industries, denoted as $\text{Compatibility}_{i(\omega)}$.

A concern with this measure is that sometimes the two keywords can appear under a different meaning. For example, ‘compatibility’ can appear with other meanings such as ‘water compatibility’, ‘skin compatibility.’ As a robustness, we require at least one of the following words to appear in a patent containing ‘compatibility’ for it to be counted as about technological compatibility: technology, interface, system, software, hardware, standard, compliance, protocol, input, output, firmware, plug-and-play, backward, network, modular. This list of key words is generated by ChatGPT. Imposing this restriction does not materially alter either the intensity measure or the regression results based on the measure.

Table B.8: Compatibility Intensity by Industries

ISIC	Industry	Compatibility Intensity
21*	Manufacture of basic pharmaceutical products	0.52
60	Programming and broadcasting activities	0.48
12*	Manufacture of tobacco products	0.46
19*	Manufacture of coke and refined petroleum products	0.45
61	Telecommunications	0.42
20*	Manufacture of chemicals and chemical products	0.41
51	Air transport	0.36
63	Information service activities	0.35
26*	Manufacture of computer, electronic and optical products	0.35
58	Publishing activities	0.34
62	Computer programming, consultancy and related activities	0.33
53	Postal and courier activities	0.32
30*	Manufacture of other transport equipment	0.32
18*	Printing and reproduction of recorded media	0.29
32*	Other manufacturing	0.27
35	Electricity, gas, steam and air conditioning supply	0.26
52	Warehousing and support activities for transportation	0.25
29*	Manufacture of motor vehicles, trailers and semi-trailers	0.25
27*	Manufacture of electrical equipment	0.25
24*	Manufacture of basic metals	0.24
17*	Manufacture of paper and paper products	0.24
22*	Manufacture of rubber and plastic products	0.24
59	Motion picture, video and television programme production	0.23
11*	Manufacture of beverages	0.23
10*	Manufacture of food products	0.22
23*	Manufacture of other non-metallic mineral products	0.21
50	Water transport	0.21
13*	Manufacture of textiles	0.21
37	Sewerage	0.20
49	Land transport and transport via pipelines	0.20
33*	Repair and installation of machinery and equipment	0.20
46	Wholesale trade, except of motor vehicles and motorcycles	0.20
36	Water collection, treatment and supply	0.19
39	Remediation activities and other waste management services	0.19
56	Food and beverage service activities	0.19
42	Civil engineering	0.19
28*	Manufacture of machinery and equipment nec	0.18
47	Retail trade, except of motor vehicles and motorcycles	0.18
45	Wholesale and retail trade and repair of motor vehicles and motorcycles	0.18
16*	Manufacture of wood and of products of wood and cork, except furniture	0.16
41	Construction of buildings	0.16
43	Specialised construction activities	0.16
38	Waste collection, treatment and disposal activities; materials recovery	0.15
14*	Manufacture of wearing apparel	0.15
25*	Manufacture of fabricated metal products, except machinery and equipment	0.15
31*	Manufacture of furniture	0.12
55	Accommodation	0.10
15*	Manufacture of leather and related products	0.10

Notes: This table lists the compatibility intensity for two-digit industries in manufacturing and related services (Sections C-K). Compatibility intensity is defined as the fraction of patenting firms with patents containing keywords related to 'compatibility'. An ISIC code with a star suffix indicates a manufacturing industry.

B.5 Firm-Level Evidence: Technology Similarity and Importing

This section presents a firm-level analysis of the relationship between technology similarity and importing. Consistent with the model’s implication (Proposition 3), firms whose technologies are more similar to a foreign country’s are more likely to import from that country: we document this correlation using matched data on Chinese manufacturing firms, their patents, and their customs transactions. The correlation is stronger in sector pairs with tighter input-output linkages and in industries where compatibility is more important, and an ‘extended-gravity’ pattern lends further support to the compatibility interpretation. In Section 5, we use the baseline correlation to calibrate the compatibility parameter $\bar{t}$, along with other parameters. Below, we present the main analysis, followed by construction details and additional results.

Proposition 3 suggests that within a country-sector (d, i) , firms whose endowed technology is closer to a foreign country o tend to choose similar technologies to those of o and import from o . This leads to a *firm-level* correlation between imports and technology proximity.¹³

Data. We use data on manufacturing firms from the Annual Survey of Industrial Enterprises (ASIE) of China’s National Bureau of Statistics (NBS). This survey covers all state-owned firms and non-state firms with sales above 5 million RMB (approximately \$600,000) before year 2011 and with sales above 20 million RMB after 2011. We match these firms to *PATSTAT Global* to retrieve their patent families and then use the abstracts to construct $\text{Similarity}_{\omega ot}$ and $\text{Similarity}_{\omega jt}$. Our final sample contains all manufacturing firms with patents, forming an unbalanced panel that grows from 57,465 firms in the first period (2000–2002) to 102,153 in the last period (2012–2014). We match these firms to Chinese Customs Data (available to us over 2000–2014), with which we construct an indicator for whether firm ω imported from region o in period t .

Specification. We estimate the following regression:

$$\mathbb{I}[\text{Import}_{\omega ot} > 0] = \beta \text{Similarity}_{\omega ot} + \beta_2 X_{\omega ot} + FE_{\omega t} + FE_{\omega o} + FE_{ot} + \epsilon_{\omega ot}, \quad (\text{B.1})$$

where $\mathbb{I}[\text{Import}_{\omega ot} > 0]$ is an indicator for whether firm ω imports from o in period t and the key regressor is $\text{Similarity}_{\omega ot}$. We include the following controls: firm-time fixed effects $FE_{\omega t}$, which absorb time-varying firm characteristics that affect firms’ tendency to import or use foreign technologies (e.g., productivity, openness to foreign ideas); firm-region fixed effects $FE_{\omega o}$, which capture time-invariant linkages between a specific firm and a region. These fixed effects also alleviate the concern that our findings may reflect differences in firms’ English proficiency or the way they write patent applications. FE_{ot} represents region-time fixed effects, accounting for changes over time in region o . Lastly, $X_{\omega ot}$ includes additional controls for other determinants of import and technology choices. In all specifications, we also control for the competition score between ω and o (Section B.3), so that the coefficient on similarity reflects the compatibility, rather than the rivalry, of technologies.

Main Results. Table B.9 reports our findings. Column 1 shows that firms with technologies more similar to region o are more likely to import from o . One may be concerned that

¹³Our model is static, so firms’ endowed technologies are permanent. In the empirical specification, we control for firm-region fixed effects to absorb the permanent component of firm-country connections, exploiting only the variations from the over-time shifts in technology endowments. If any, the empirical correlation between technology proximity and imports becomes stronger if firm-region fixed effects are excluded.

Table B.9: Firm-Level Correlation between Trade and Technology Choice

	$\mathbb{I}[\text{Import}_{\omega ot} > 0]$			
	(1)	(2)	(3)	(4)
Similarity $_{\omega ot}$	0.019*** (0.005)	0.019*** (0.005)	0.022*** (0.006)	0.022*** (0.006)
Competition $_{\omega ot}$	-0.005 (0.006)	-0.005 (0.006)	-0.003 (0.007)	0.001 (0.007)
FE ω - t	Yes	Yes	Yes	Yes
FE ω - o	Yes	Yes	Yes	Yes
FE o - t	Yes	Yes	Yes	Yes
$X_{i(\omega)ot}$		Yes		
FE $p(\omega)$ - $i(\omega)$ - o - t			Yes	Yes
Exclude Foreign Firms				Yes
Observations	3,326,764	3,326,764	3,262,280	2,431,940
R^2	0.748	0.748	0.766	0.729

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered by firm. $i(\omega)$ and $p(\omega)$ represent the industry and province of firm ω , respectively. $X_{i(\omega)ot}$ includes the applied tariffs of industry i 's inputs and outputs. Industry i 's input tariff is defined by the average applied tariff on imports of all goods from world region o , weighted by each input's share in industry i 's production. Industries are classified at the three-digit level of China's Industry Classification.

the correlation is driven by tariffs, which may be correlated with both trade and other unobserved linkages between firms and foreign regions. However, controlling for input and output tariffs in the firm's industry (constructed in Section B.5.1) barely changes the coefficient (Column 2).

One may also be concerned that the correlation reflects differences among firms from different provinces or industries in China, which face different barriers when interacting with foreign suppliers due to either their geographic locations or historical connections with these countries. While the firm-country fixed effects absorb the time-invariant aspect of such connections, they do not capture the time-varying aspect. In Column 3, our preferred specification, we absorb all such variations by controlling for $i(\omega)$ - o - $p(\omega)$ - t fixed effects, in which $p(\omega)$ is the province of firm ω and $i(\omega)$ is the industry of ω . If any, the coefficient for Similarity $_{\omega ot}$ is larger. The estimate, 0.022, means that a one-standard-deviation rise in similarity with o is associated with a 0.20-percentage-point increase in the probability of importing from o , or 4% of the sample mean.

Finally, foreign-owned firms might drive this result if they import from their parent companies and adopt their parent's technology. In Column 4, we exclude foreign affiliates and joint ventures between foreign and Chinese firms. The sample shrinks by a third, but the estimate remains unchanged. Since our model applies to foreign affiliates as well as domestic firms, in the rest of this subsection, we use the full sample of firms. All results are robust if we exclude foreign firms and joint ventures. It is also worth noting that across all columns, the competition score itself enters insignificantly: what predicts importing from a country is the compatibility of technologies, not their rivalry.

Heterogeneity by Industry. The correlation documented in Table B.9 is consistent with our model's premise that firms choose technology and suppliers jointly for compatibility

Table B.10: Firm-Level Correlation: Mechanisms and Alternative Explanations

	$\mathbb{I}[\text{Import}_{\omega ojt} > 0]$			
	(1)	(2)	(3)	(4)
Similarity $_{\omega ojt}$	0.023*** (0.004)	0.020*** (0.004)	0.021*** (0.004)	0.018*** (0.004)
Competition $_{\omega ojt}$	-0.010*** (0.003)	-0.010*** (0.003)	-0.010*** (0.003)	-0.010*** (0.003)
Similarity $_{\omega ojt} \times \gamma_{i(\omega)j}$		0.054* (0.029)		0.057* (0.029)
Similarity $_{\omega ojt} \times \text{Compatibility}_{i(\omega)}$			0.082** (0.035)	0.083** (0.035)
FE ω - j - t	Yes	Yes	Yes	Yes
FE ω - o - j	Yes	Yes	Yes	Yes
FE o - j - t	Yes	Yes	Yes	Yes
Observations	8,656,925	8,570,998	8,572,704	8,570,998
R^2	0.750	0.750	0.750	0.750

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered by firm. $\gamma_{i(\omega)j}$ is the input share of sector j in firm ω 's sector $i(\omega)$. Compatibility $_{i(\omega)}$ measures the compatibility intensity of sector $i(\omega)$.

reasons. An alternative information-based explanation is that firms may discover foreign suppliers and technologies through correlated sources. For example, the firm owner may meet a contact from a region o , who bring the technologies as well as the suppliers of o . Although we include a rich set of fixed effects, we cannot fully rule out such mechanisms. To support our interpretation of the correlation, we provide auxiliary evidence by looking into two implications of the model on how the correlation varies across industries.

First, if the correlation is indeed driven by a desire for compatible inputs, it should be stronger in sector pairs with more intensive input-output linkages. To test this implication, we use Similarity $_{\omega ojt}$, the similarity between firm ω and region-sector (d, j) , as the explanatory variable. As the outcome variable, we use an indicator for whether a firm ω imports goods in sector j from country o , denoted by $\mathbb{I}[\text{Import}_{\omega ojt} > 0]$. Column 1 of Table B.10 reports a baseline specification where we simply regress this indicator on Similarity $_{\omega ojt}$. Given that the data have a j dimension, we are able to include as controls the interactions between the fixed effects in specification (B.1) and j . The estimate is 0.023, similar to those in Table B.9. Column 2 adds the interaction between Similarity $_{\omega ojt}$ and $\gamma_{i(\omega)j}$, the input share of sector j in firm ω 's sector $i(\omega)$. This interaction is sizable: when $\gamma_{i(\omega)j}$ increases by 10 percentage points, the correlation between similarity and importing increases by 0.0054—one quarter of the average correlation.

Second, the importance of input compatibility can vary across industries. For instance, electronics and telecommunications equipment often require stricter compatibility than apparel. If our estimates indeed capture technology compatibility, we should see a stronger correlation in industries where compatibility is more critical.

To test this hypothesis, we construct a measure of compatibility intensity using the full text of U.S. patents in the Orbis Intellectual Property Database, which links patents to firms

worldwide. For each three-digit industry, we compute the fraction of patenting firms with at least one patent containing in its text either the word ‘compatibility/compatible’ or the word ‘inter-operability.’ Treating patents with these key words as an indicator of technologies where compatibility is important, this fraction, which we denote as $\text{Compatibility}_{i(\omega)}$, captures how often firms in industry i require compatible technologies.¹⁴

Column 3 of Table B.10 adds the interaction between $\text{Compatibility}_{i(\omega)}$ and $\text{Similarity}_{\omega ojt}$. The coefficient on the interaction, 0.082, implies that a one-standard-deviation increase in industry compatibility intensity (0.11) raises the correlation by 0.9 percentage points—about 8.8% of the mean correlation. In Column 4, we further include the interaction with the input-output parameter, and the coefficients for both interactive terms remain similar. Together, these results offer support for our hypothesis that compatibility along the input–output network drives the correlation.

Extended Gravity. Another implication of our model is an ‘extended-gravity’-like effect (Morales et al., 2019) in the relationship between technology similarity and trade: if region o' has technology similar to o , then stronger similarity between firm ω and o' is associated with a higher likelihood of ω importing from o . Thus, if we regress whether ω imports from o on the technological similarity between ω and a region o' that shares similar technologies as o , the resulting coefficient should be positive. This relationship can hold even after controlling for $\text{Similarity}_{\omega ot}$ unless the similarity measure is perfectly accurate.

Section B.5.2 presents this test, together with checks against information-based alternative explanations. The similarity between firm ω and the technology neighbors of o is indeed strongly correlated with ω importing from o , further supporting our interpretation that technology closeness, rather than purely informational channels, drives the correlation between imports and technology similarity.

B.5.1 Input and Output Tariffs

In one of the specifications in Table B.9, we control for the input and output tariffs faced by an industry i in China when sourcing inputs from abroad. We construct these variables from the effectively applied tariffs.

In general, output tariffs $\tau_{doit}^{\text{Output}}$ are defined as the weighted average of applied tariffs that region d (a collection of countries) imposes on products in industry i from region o (another collection of countries) in period t :

$$\tau_{doit}^{\text{Output}} \equiv \sum_{HS6 \in i} \tau_{do,HS6,t}^{\text{Applied}} \times \frac{X_{d,HS6,\text{initial}}}{\sum_{HS6' \in i} X_{d,HS6',\text{initial}}},$$

where $\tau_{do,HS6,t}^{\text{Applied}}$ are aggregated from the country-pair level by weighing with each pair’s initial trade shares in the region

$$\tau_{do,HS6,t}^{\text{Applied}} \equiv \sum_{ISOd \in d} \sum_{ISOo \in o} \tau_{ISOd,ISOo,HS6,t}^{\text{Applied}} \times \frac{X_{ISOd,ISOo,HS6,\text{initial}}}{\sum_{ISOd' \in d} \sum_{ISOo' \in o} X_{ISOd',ISOo',HS6,\text{initial}}}.$$

For industry i , the input tariffs $\tau_{doit}^{\text{Input}}$ are defined as the averaged tariffs a firm in i faces when *sourcing inputs* from abroad. We construct this measure by weighing the import tariffs

¹⁴Table B.8 reports this compatibility intensity measure by industry. Our results remain robust if we refine the context in which these keywords appear in patent texts. See Section B.4 for details.

of products from other industries with their input-output weights and initial trade shares:

$$\tau_{doit}^{Input} \equiv \sum_{HS6 \notin i} \tau_{do,HS6,t}^{Applied} \times \gamma_{i,j(HS6)} \times \frac{X_{d,HS6,initial}}{\sum_{HS6' \in j(HS)} X_{d,HS6',initial}},$$

where $\gamma_{i,j(HS6)}$ denotes the industry-level input-output weight derived from the 2007 Chinese Input-Output Table. Throughout the firm-level regressions in this section, region d is restricted to China (CHN).

B.5.2 Extended Gravity

In this section, we provide empirical evidence for the ‘extended-gravity’-like implication on the relationship between technology similarity and importing decisions. Our theory predicts that if regions o' and o are close to each other in the space of technologies, then the technology similarity between a firm ω and o' would be positively correlated with the firm importing from o .

Table B.11: Extended Gravity

	$\mathbb{I}[\text{Import}_{\omega ot} > 0]$			
	(1)	(2)	(3)	(4)
Similarity $_{\omega ot}$	0.016*** (0.005)	0.014*** (0.005)	0.031*** (0.009)	0.031*** (0.009)
Similarity $_{\omega, \mathcal{R}_t^{tech}(o), t}$	0.083*** (0.031)	0.094*** (0.030)	0.154*** (0.046)	0.156*** (0.046)
Competition $_{\omega ot}$	-0.007 (0.006)	-0.006 (0.006)	-0.024** (0.011)	-0.024** (0.011)
Similarity $_{\omega, \mathcal{R}^{geo}(o), t}$		0.003 (0.009)		-0.005 (0.012)
Similarity $_{\omega, \mathcal{R}^{lang}(o), t}$			0.000 (0.010)	0.002 (0.011)
FE ω - t	Yes	Yes	Yes	Yes
FE ω - o	Yes	Yes	Yes	Yes
FE o - t	Yes	Yes	Yes	Yes
$X_{i(\omega)ot}$	Yes	Yes	Yes	Yes
$\mathbb{I}[\text{Import}_{\omega, \mathcal{R}_t^{tech}(o), t} > 0]$	Yes	Yes	Yes	Yes
Observations	3,326,764	3,207,951	2,019,821	2,019,821
R^2	0.751	0.749	0.758	0.758

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered by firm. $\mathcal{R}_t^{tech}(d)$, $\mathcal{R}^{geo}(o)$, and $\mathcal{R}^{lang}(o)$ represent the technological, geographical, and linguistic neighbors of region o , respectively. Competition $_{\omega ot}$ is the competition score between firm ω and region o , constructed in Appendix B.3. $X_{i(\omega)ot}$ includes the applied tariffs of industry i 's inputs and outputs. Industry i 's input tariff is defined by the average applied tariff on imports of all goods from world region o , weighted by each input's share in industry i 's production. Industries are classified at the three-digit level of China's Industry Classification.

To test this implication, we first define the set of technology neighbors of each region d , which are regions with technologies similar to those of region d . Specifically, we run the

following regression:

$$\text{Similarity}_{doi jt} = FE_{dij} + \varepsilon_{doi jt},$$

where $\text{Similarity}_{doi jt}$ denotes the technology similarity between region-industry pairs (d, i) and (o, j) in period t . We obtain the residuals from the regression, and calculate its mean over all industries i and j , which essentially capture the similarity of technologies in region o to d in period t , conditional on the fixed effects. Then, we define the ‘*technology neighbors*’ of region d , denoted with $\mathcal{R}_t^{tech}(d)$, as the regions o ’s with the averaged residual ranking at the top 10% for d during period t .

In column (1) of Table B.11, we extend our baseline specification (B.1) by including the similarity measure between firm ω and the technological neighbors of o . Because this similarity is likely correlated with firm ω ’s importing from these technological neighbors, and because importing from different regions can be substitutes of each other, we control for whether ω imports from these technological neighbors. The result shows that indeed, a firm ω ’s being technologically similar to the technology neighbors of a region o is strongly correlated with ω importing from o .

Still, it is possible that this result is driven by an information-based story: Firm ω meets a new contact, who introduces to ω both the suppliers in o and the technologies of o ’. Such knowledgeable contacts certainly exist, but given the language and cultural barriers between countries, it is more likely than not that their knowledge domain extends to pairs of countries that are geographically or linguistically close to each other. In columns (2) to (4), we tease out the influence of such experts by controlling for the technology similarity between ω and the geographic or linguistic neighbors of o . We define geographic neighbors for a region as those with the closest 10 percentile distance, and language neighbors as those sharing a common official language. We find that the coefficients associated with proximity to language and geographic neighbors do not matter on themselves; more importantly, they do not diminish the role of similarity with technology neighbors.

B.5.3 Using Citation as an Alternative Measure

Researchers often use patent citations as an alternative measure of technology similarity. Although we have chosen the text-based measure of similarity as our preferred approach, Table B.12 replicates the baseline result in Table B.9 using citations, showing its robustness to this alternative measure.

B.6 Additional Details on Tariff Variation and Technological Choice

B.6.1 MFN-Binding Tariffs

In Section 4.2, we instrument the effectively applied tariffs with the exogenous tariff changes from binding MFN tariffs. To construct this instrument, we first construct ‘pseudo’ MFN tariffs using the applied tariffs in the initial periods and *changes* in MFN rates at the country-product level:

$$\widehat{\tau}_{ISOd,ISOo,HS6,t}^{MFN} \equiv \tau_{ISOd,ISOo,HS6,initial}^{Applied} + \sum_{t'=initial+1}^t \widehat{\Delta\tau}_{ISOd,ISOo,HS6,t'}^{MFN}$$

Table B.12: Firm-Level Correlation between Trade and Citation

	$\mathbb{I}[\text{Import}_{\omega ot} > 0]$			
	(1)	(2)	(3)	(4)
$\mathbb{I}[\text{Citation}_{\omega ot} > 0]$	0.011*** (0.001)	0.011*** (0.001)	0.009*** (0.001)	0.009*** (0.001)
$\text{Competition}_{\omega ot}$	0.003 (0.006)	0.003 (0.006)	0.006 (0.006)	0.010* (0.006)
FE ω - t	Yes	Yes	Yes	Yes
FE ω - o	Yes	Yes	Yes	Yes
FE o - t	Yes	Yes	Yes	Yes
$X_{i(\omega)ot}$		Yes		
FE $p(\omega)$ - $i(\omega)$ - o - t			Yes	Yes
Exclude Foreign Firms				Yes
Observations	3,326,764	3,326,764	3,262,280	2,431,940
R^2	0.748	0.748	0.766	0.729

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered by firm. $i(\omega)$ and $p(\omega)$ represent the industry and province of firm ω , respectively. $X_{i(\omega)ot}$ includes the applied tariffs of industry i 's inputs and outputs. Industry i 's input tariff is defined by the average applied tariff on imports of all goods from world region o , weighted by each input's share in industry i 's production. Industries are classified at the three-digit level of China's Industry Classification. $\text{Competition}_{\omega ot}$ is the competition score between firm ω and region o , constructed in Appendix B.3.

where the changes are defined as the interaction between the changes in MFN tariffs and an indicator for whether the MFN tariffs of d applies to o in both period t and $t - 1$:

$$\widehat{\Delta\tau}_{ISOd,ISOo,HS6,t}^{MFN} \equiv \mathbb{I} \left[\tau_{ISOd,ISOo,HS6,t}^{Applied} = \tau_{ISOd,HS6,t}^{MFN} \right] \times \mathbb{I} \left[\tau_{ISOd,ISOo,HS6,t-1}^{Applied} = \tau_{ISOd,HS6,t-1}^{MFN} \right] \\ \times \left(\tau_{ISOd,HS6,t}^{MFN} - \tau_{ISOd,HS6,t-1}^{MFN} \right).$$

Then, we aggregate both the applied and 'pseudo' MFN tariffs into the region-product level by weighing on initial trade shares:

$$\tau_{do,HS6,t}^{MFN} \equiv \sum_{ISOd \in d} \sum_{ISOo \in o} \widehat{\tau}_{ISOd,ISOo,HS6,t}^{MFN} \times \frac{X_{ISOd,ISOo,HS6,initial}}{\sum_{ISOd' \in d} \sum_{ISOo' \in o} X_{ISOd',ISOo',HS6,initial}} \\ \tau_{do,HS6,t}^{Applied} \equiv \sum_{ISOd \in d} \sum_{ISOo \in o} \tau_{ISOd,ISOo,HS6,t}^{Applied} \times \frac{X_{ISOd,ISOo,HS6,initial}}{\sum_{ISOd' \in d} \sum_{ISOo' \in o} X_{ISOd',ISOo',HS6,initial}}.$$

The (log of) MFN-binding tariff rates $\tau_{do,HS6,t}^{MFN}$ are then used as the instrument for the (log of) effectively applied tariff rates $\tau_{do,HS6,t}^{Applied}$. Since our regressions control for d-o-HS6 fixed effects, this IV exploits the changes in the MFN rates for identification. When running the regressions, we exclude the largest exporter for each importer-product pair to address the concern that countries choose MFN tariffs with the major exporters in mind.

Table B.13: Additional Evidence on Tariff Shocks and Technology Similarity

	ln Citation _{<i>d</i>,<i>HS6</i>,<i>t</i>}		Similarity _{<i>d</i>,<i>HS6</i>,<i>t</i>}	
	OLS (1)	IV (2)	OLS (3)	IV (4)
$\ln \tau_{d,HS6,t}^{MFN}$	-0.151** (0.076)		-0.010*** (0.001)	
$\ln \tau_{d,HS6,t}^{Applied}$		-0.184** (0.093)		-0.013*** (0.001)
Competition _{<i>d</i>,<i>HS6</i>,<i>t</i>}	0.511*** (0.142)	0.518*** (0.142)	0.723*** (0.002)	0.723*** (0.002)
FE <i>d</i> - <i>o</i> - <i>HS6</i>	Yes	Yes	Yes	Yes
FE <i>o</i> - <i>HS6</i> - <i>t</i>	Yes	Yes	Yes	Yes
FE <i>d</i> - <i>HS6</i> - <i>t</i>	Yes	Yes	Yes	Yes
Exclude Consumption Goods			Yes	Yes
Observations	4,201,742	4,201,742	4,338,370	4,338,370
R ²	0.937	0.937	0.995	0.995

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the *d*-*o*-*HS6* level. Columns 1 and 3 report reduced-form regressions using OLS, while columns 2 and 4 report the second stage of 2SLS regressions using $\ln \tau_{d,HS6,t}^{MFN}$ as an instrument for $\ln \tau_{d,HS6,t}^{Applied}$. Competition_{*d*,*HS6*,*t*} is the competition score between country *d* and the (*o*, *HS6*) pair, constructed in Appendix B.3.

B.6.2 Citation as Alternative Measure and Excluding Final Goods

Table B.13 provides robustness checks for our results in Section 4.2. In the first two columns, we show that we obtain similar findings using citation as an alternative measure of bilateral similarity. In the last two columns, we show that the estimates are essentially the same when goods for final consumers are excluded. This is reassuring because our model centers on firms' choice of intermediate suppliers and technologies. Our definition of final consumption goods rely on the Classification by the Broad Economic Categories (BEC) and its mapping with HS products provided by the UNSD.

Appendix C Quantification

C.1 Proof of Proposition 7

We prove the proposition by guess and verification. Suppose that the distribution of production cost is characterized by (31), and the ex-post technology distribution is characterized by (33). Then, under Assumption 2, by taking the log of (5), we have

$$\ln C_d^i(\theta) = \ln\left(\frac{\Xi^i}{A_d^i}\right) + \gamma^{iL} \ln(w_d) - \zeta^{-1} \sum_j \gamma^{ij} \ln \left(\sum_o \int [\tau_{do}^j]^{-\zeta} \cdot \exp[-\zeta(k_{A,o}^j + m_A^j(\tilde{\theta} - n_{A,o}^j)^2) - \frac{1}{2}\zeta\bar{t}(\theta - \tilde{\theta})^2] \cdot \frac{1}{\sqrt{2\pi}(\sigma^j)^2} \exp[-\frac{1}{2}(\frac{\tilde{\theta} - \mu_o^j}{\sigma^j})^2] d\tilde{\theta} \right)$$

where

$$\begin{aligned} & k_{A,o}^j + m_A^j(\tilde{\theta} - n_{A,o}^j)^2 + \frac{1}{2}\bar{t}(\theta - \tilde{\theta})^2 \\ &= [m_A^j + \bar{t}](\tilde{\theta} - \frac{m_A^j n_{A,o}^j + \frac{1}{2}\bar{t}\theta}{m_A^j + \frac{1}{2}\bar{t}})^2 + k_{A,o}^j + \frac{m_A^j \frac{1}{2}\bar{t}(\theta - n_{A,o}^j)^2}{m_A^j + \frac{1}{2}\bar{t}}, \end{aligned}$$

which collects the quadratic term with respect to $\tilde{\theta}$ and then θ .

Since $\tilde{\theta} \sim N(\mu_o^j, [\sigma^j]^2)$, $\tilde{\theta} - \frac{m_A^j n_{A,o}^j + \frac{1}{2}\bar{t}\theta}{m_A^j + \frac{1}{2}\bar{t}} \sim N(\mu_o^j - \frac{m_A^j n_{A,o}^j + \frac{1}{2}\bar{t}\theta}{m_A^j + \frac{1}{2}\bar{t}}, [\sigma^j]^2)$, apply Lemma A.8 stated below and we have

$$\begin{aligned} & \int \exp[-\zeta(k_{A,o}^j + m_A^j(\tilde{\theta} + n_{A,o}^j)^2 + \frac{1}{2}\bar{t}(\theta - \tilde{\theta})^2)] \cdot \frac{1}{\sqrt{2\pi}(\sigma^j)^2} \exp[-\frac{1}{2}(\frac{\tilde{\theta} - \mu_o^j}{\sigma^j})^2] d\tilde{\theta} \\ &= \int \exp[-\zeta((m_A^j + \frac{1}{2}\bar{t})(\tilde{\theta} - \frac{m_A^j n_{A,o}^j + \frac{1}{2}\bar{t}\theta}{m_A^j + \frac{1}{2}\bar{t}})^2 + k_{A,o}^j + \frac{m_A^j \frac{1}{2}\bar{t}(\theta - n_{A,o}^j)^2}{m_A^j + \frac{1}{2}\bar{t}})] \cdot \frac{1}{\sqrt{2\pi}(\sigma^j)^2} \exp[-\frac{1}{2}(\frac{\tilde{\theta} - \mu_o^j}{\sigma^j})^2] d\tilde{\theta} \\ &= \exp[-\zeta(k_{A,o}^j + \frac{m_A^j \frac{1}{2}\bar{t}(\theta - n_{A,o}^j)^2}{m_A^j + \frac{1}{2}\bar{t}})] \cdot \mathbb{E} \exp[-\zeta(m_A^j + \frac{1}{2}\bar{t})(\tilde{\theta} - \frac{m_A^j n_{A,o}^j + \frac{1}{2}\bar{t}\theta}{m_A^j + \frac{1}{2}\bar{t}})^2] \\ &= \exp[-\zeta(k_{A,o}^j + \frac{m_A^j \frac{1}{2}\bar{t}(\theta - n_{A,o}^j)^2}{m_A^j + \frac{1}{2}\bar{t}})] \cdot [1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2]^{-1/2} \cdot \exp(\frac{-\zeta(m_A^j + \frac{1}{2}\bar{t})}{1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2} [\mu_o^j - \frac{m_A^j n_{A,o}^j + \frac{1}{2}\bar{t}\theta}{m_A^j + \frac{1}{2}\bar{t}}]^2) \\ &= \exp(-\zeta k_{A,o}^j - \frac{1}{2} \log[1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2] - \frac{\zeta m_A^j (\mu_o^j - n_{A,o}^j)^2}{1 + 2\zeta m_A^j (\sigma^j)^2} - \frac{\frac{1}{2}\zeta\bar{t}[1 + 2\zeta m_A^j (\sigma^j)^2]}{1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2} [\theta - \frac{\mu_o^j + 2\zeta m_A^j n_{A,o}^j (\sigma^j)^2}{1 + 2\zeta m_A^j (\sigma^j)^2}]^2) \end{aligned}$$

Therefore,

$$\begin{aligned} \ln C_d^i(\theta) &= \ln\left(\frac{\Xi^i}{A_d^i}\right) + \gamma^{iL} \ln(w_d) - \zeta^{-1} \sum_j \gamma^{ij} \ln(\sum_o [\tau_{do}^j]^{-\zeta} \exp[-\zeta(k_{B,o}^j + m_B^j(\theta - n_{B,o}^j)^2)]), \\ &= \ln\left(\frac{\Xi^i}{A_d^i}\right) + \gamma^{iL} \ln(w_d) - \zeta^{-1} \sum_j \gamma^{ij} \ln(\sum_o \exp[-\zeta(\ln \tau_{do}^j + k_{B,o}^j + m_B^j(\theta - n_{B,o}^j)^2)]), \end{aligned} \tag{C.1}$$

where

$$\begin{aligned}
k_{B,o}^j &= k_{A,o}^j + \frac{1}{2\zeta} \log[1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2] + \frac{m_A^j(\mu_o^j - n_{A,o}^j)^2}{1 + 2\zeta m_A^j(\sigma^j)^2}, \\
m_B^j &= \frac{\frac{1}{2}\bar{t}[1 + 2\zeta m_A^j(\sigma^j)^2]}{1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2}, \\
n_{B,o}^j &= \frac{\mu_o^j + 2\zeta m_A^j(\sigma^j)^2 n_{A,o}^j}{1 + 2\zeta m_A^j(\sigma^j)^2}.
\end{aligned}$$

Consider

$$\begin{aligned}
\frac{d \ln C_d^i(\theta)}{d\theta} &= \sum_j \gamma^{ij} \sum_o \chi_{do}^j(\theta) 2m_B^j[\theta - n_{B,o}^j] \\
\frac{d^2 \ln C_d^i(\theta)}{d\theta^2} &= \sum_j \gamma^{ij} \sum_o ([\chi_{do}^j]'(\theta) 2m_B^j[\theta - n_{B,o}^j] + \chi_{do}^j(\theta) 2m_B^j),
\end{aligned}$$

where

$$\chi_{do}^j(\theta) \equiv \frac{\exp[-\zeta(\ln \tau_{do}^j + k_{B,o}^j + m_B^j(\theta - n_{B,o}^j)^2)]}{\sum_{o'} \exp[-\zeta(\ln \tau_{do'}^j + k_{B,o'}^j + m_B^j(\theta - n_{B,o'}^j)^2)]},$$

with $\sum_o \chi_{do}^j(\theta) = 1$.

We consider a second-order approximation with respect to θ around a fixed $\hat{\theta}_d^i$,¹⁵ which gives

$$\begin{aligned}
&\ln C_d^i(\theta) \\
&\approx \ln C_d^i(\hat{\theta}_d^i) + \sum_j \gamma^{ij} \sum_o \chi_{do}^j(\hat{\theta}_d^i) 2m_B^j(\hat{\theta}_d^i - n_{B,o}^j)(\theta - \hat{\theta}_d^i) + \frac{1}{2} \sum_j \gamma^{ij} \sum_o \chi_{do}^j(\hat{\theta}_d^i) 2m_B^j(\theta - \hat{\theta}_d^i)^2 \\
&= (\sum_j \gamma^{ij} m_B^j)[(\theta - \hat{\theta}_d^i)^2 + 2 \sum_{j,o} \frac{\gamma^{ij} m_B^j \chi_{do}^j(\hat{\theta}_d^i)}{\sum_{j'} \gamma^{ij'} m_B^{j'}} (\hat{\theta}_d^i - n_{B,o}^j)(\theta - \hat{\theta}_d^i)] + \ln C_d^i(\hat{\theta}_d^i) \\
&= (\sum_j \gamma^{ij} m_B^j)[\theta - \sum_{j,o} \hat{\chi}_{do}^{ij} n_{B,o}^j]^2 - (\sum_j \gamma^{ij} m_B^j)[\sum_{j,o} \hat{\chi}_{do}^{ij} (\hat{\theta}_d^i - n_{B,o}^j)]^2 + \ln C_d^i(\hat{\theta}_d^i), \tag{C.2}
\end{aligned}$$

where

$$\hat{\chi}_{do}^{ij} \equiv \frac{\gamma^{ij} m_B^j \chi_{do}^j(\hat{\theta}_d^i)}{\sum_{j'} \gamma^{ij'} m_B^{j'}},$$

with $\sum_{o,j} \hat{\chi}_{do}^{ij} = 1$.

¹⁵The derivation works for any $\hat{\theta}_d^i$. In the numerical implementation, we choose $\hat{\theta}_d^i$ to be the mean of the ex-post technology distribution, μ_d^i .

This verifies the functional form in (31) with m_A^i , $n_{A,d}^i$ and $k_{A,d}^i$ satisfying

$$\begin{aligned} m_A^i &= \sum_j \gamma^{ij} m_B^j = \sum_j \gamma^{ij} \frac{\frac{1}{2}\bar{t}[1 + 2\zeta m_A^j (\sigma^j)^2]}{1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2}, \\ n_{A,d}^i &= \sum_{j,o} \hat{\chi}_{do}^{ij} n_{B,o}^j = \sum_{j,o} \hat{\chi}_{do}^{ij} \frac{\mu_o^j + 2\zeta m_A^j (\sigma^j)^2 n_{A,o}^j}{1 + 2\zeta m_A^j (\sigma^j)^2}, \\ k_{A,d}^i &= \ln C_d^i(\hat{\theta}_d^i) - m_A^i [\sum_{j,o} \hat{\chi}_{do}^{ij} (\hat{\theta}_d^i - n_{B,o}^j)]^2. \end{aligned}$$

Consider the innovation decision in (o, j) given by (15). Under Assumption 2, this is equivalent to

$$\max_{\theta} \exp[-\frac{1}{2}\bar{\phi}(\bar{\theta} - \theta)^2] \cdot [C_o^j(\theta)]^{1-\eta}.$$

Taking log and apply the quadratic approximation to $\ln C_o^j(\theta)$, this is

$$\max_{\theta} (1 - \eta) m_A^j (\theta - n_{A,o}^j)^2 - \frac{1}{2}\bar{\phi}(\bar{\theta} - \theta)^2.$$

The first-order condition implies that

$$(1 - \eta) m_A^j (\theta - n_{A,o}^j) = \frac{1}{2}\bar{\phi}(\bar{\theta} - \theta),$$

which gives the policy function (32)

$$\theta = g_o^j(\bar{\theta}) \equiv \alpha_o^j + \beta^j \bar{\theta},$$

where

$$\begin{aligned} \alpha_o^j &= \frac{(\eta - 1) m_A^j}{\frac{1}{2}\bar{\phi} + (\eta - 1) m_A^j} n_{A,o}^j, \\ \beta^j &= \frac{\frac{1}{2}\bar{\phi}}{\frac{1}{2}\bar{\phi} + (\eta - 1) m_A^j}. \end{aligned}$$

Since the ex-ante technology distribution $\bar{\Theta}_o^j$ is Normal with mean $\bar{\mu}_o^j$ and variance $(\bar{\sigma}^j)^2$, the ex-post technology distribution Θ_o^j is also Normal with mean μ_o^j and variance $(\sigma^j)^2$, where

$$\mu_o^j = \alpha_o^j + \beta^j \bar{\mu}_o^j \quad \text{and} \quad \sigma^j = \beta^j \bar{\sigma}^j.$$

This completes the proof of Proposition 7.

Lemma A.8. Suppose $X \sim N(\mu, \sigma^2)$. Then for $m < \frac{1}{2\sigma^2}$,

$$\mathbb{E}[\exp(mX^2)] = \exp\left(\frac{m\mu^2}{1 - 2m\sigma^2}\right)(1 - 2m\sigma^2)^{-1/2}.$$

Proof.

$$\begin{aligned}
& \mathbb{E} \exp(mX^2) \\
&= \int \exp(mx^2) \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) dx \\
&= \int \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}[(1-2m\sigma^2)x^2 - 2\mu x + \mu^2]\right) dx \\
&= \int \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1-2m\sigma^2}{2\sigma^2}\left[x - \frac{1}{1-2m\sigma^2}\mu\right]^2 + \frac{m}{1-2m\sigma^2}\mu^2\right) dx \\
&= \exp\left(\frac{m\mu^2}{1-2m\sigma^2}\right) (1-2m\sigma^2)^{-1/2} \int \frac{1}{\sqrt{2\pi\sigma^2(1-2m\sigma^2)^{-1}}} \exp\left(-\frac{1}{2\sigma^2(1-2m\sigma^2)^{-1}}\left[x - \frac{1}{1-2m\sigma^2}\mu\right]^2\right) dx \\
&= \exp\left(\frac{m\mu^2}{1-2m\sigma^2}\right) (1-2m\sigma^2)^{-1/2},
\end{aligned}$$

where the last line applies that the integrand is a probability density function. $\square$

C.2 Algorithm to Solve the Equilibrium

Building on Proposition 7, we develop the following algorithm to solve the model.

Step 1. Given wages $\{w_d\}$ and parameters on geography $\{\tau_{do}^j\}$, preference η , production technology $\{\gamma^{ij}, \gamma^{iL}, \Xi^i, A_d^i, \zeta, \bar{\ell}, \bar{\phi}\}$, and the ex-ante technology distribution $\{\bar{\mu}_d^i, \bar{\sigma}^i\}$, we solve for $\{k_{A,d}^i, m_{A,d}^i, n_{A,d}^i\}$ and $\{\mu_d^i, \sigma^i\}$ to obtain the cost functions $\{C_o^j(\cdot)\}$ and the ex-post technology distributions. This involves simultaneously solving the following system of equations:

$$m_A^i = \sum_j \gamma^{ij} m_B^j \quad (C.3)$$

$$n_{A,d}^i = \sum_{j,o} \hat{\chi}_{do}^{ij} n_{B,o}^j \quad (C.4)$$

$$k_{A,d}^i = \ln C_d^i(\mu_d^i) - m_A^i \left[\sum_{j,o} \hat{\chi}_{do}^{ij} (\mu_d^i - n_{B,o}^j) \right]^2 \quad (C.5)$$

$$\mu_d^i = \alpha_d^i + \beta^i \bar{\mu}_d^i \quad (C.6)$$

$$\sigma^i = \beta^i \bar{\sigma}^i \quad (C.7)$$

where

$$m_B^j = \frac{\frac{1}{2}\bar{t}[1 + 2\zeta m_A^j(\sigma^j)^2]}{1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2}, \quad (\text{C.8})$$

$$n_{B,o}^j = \frac{1}{1 + 2\zeta m_A^j(\sigma^j)^2} \mu_o^j + \frac{2\zeta m_A^j(\sigma^j)^2}{1 + 2\zeta m_A^j(\sigma^j)^2} n_{A,o}^j \quad (\text{C.9})$$

$$k_{B,o}^j = k_{A,o}^j + \frac{1}{2\zeta} \log[1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2] + \frac{m_A^j(\mu_o^j - n_{A,o}^j)^2}{1 + 2\zeta m_A^j(\sigma^j)^2} \quad (\text{C.10})$$

$$\chi_{do}^{ij} \equiv \frac{\gamma^{ij} m_B^j}{\sum_{j'} \gamma^{ij'} m_B^{j'}} \times \frac{\exp[-\zeta(\ln \tau_{do}^j + k_{B,o}^j + m_B^j(\mu_d^i - n_{B,o}^j)^2)]}{\sum_{o'} \exp[-\zeta(\ln \tau_{do'}^j + k_{B,o'}^j + m_B^j(\mu_d^i - n_{B,o'}^j)^2)]} \quad (\text{C.11})$$

$$\ln C_d^i(\mu_d^i) = \ln\left(\frac{\Xi^i}{A_d^i}\right) + \gamma^{iL} \ln(w_d) - \zeta^{-1} \sum_j \gamma^{ij} \ln\left(\sum_o \exp[-\zeta(\ln \tau_{do}^j + k_{B,o}^j + m_B^j(\mu_d^i - n_{B,o}^j)^2)]\right) \quad (\text{C.12})$$

$$\alpha_d^i = \frac{(\eta - 1)m_A^i}{\frac{1}{2}\bar{\phi} + (\eta - 1)m_A^i} n_{A,d}^i = (1 - \beta^i) n_{A,d}^i \quad (\text{C.13})$$

$$\beta^i = \frac{\frac{1}{2}\bar{\phi}}{\frac{1}{2}\bar{\phi} + (\eta - 1)m_A^i}. \quad (\text{C.14})$$

Here, (C.3) and (C.8) form a contraction mapping for $\{m_A^i\}$ with $\{\sigma^i, \beta^i\}$ determined by (C.7) and (C.14). With $\{m_A^i\}$ and $\{\sigma^j\}$ solved, (C.4) and (C.5) form another contraction mapping for $\{n_{A,d}^i, k_{A,d}^i\}$ when $\bar{t}$ is not too large, with $\{\mu_d^i, n_{B,o}^j, k_{B,o}^j, \chi_{do}^{ij}, \ln C_d^i(\mu_d^i), \alpha_d^i\}$ determined by (C.6), (C.9), (C.10), (C.11), (C.12) and (C.13), which can be evaluated directly.

Step 2. With objects solved in Step 1, the cost function $\{C_o^j(\theta)\}$ can be constructed from (C.2), and we can then explicitly evaluate the sourcing decisions of intermediate firms $\chi_{do}^j(\theta, \tilde{\theta})$ and final-good producers $\pi_{do}^j(\theta)$ for all $\theta, \tilde{\theta} \in \mathbb{T}$.

For any firm in (d, i) with technology location θ to source input j , by (C.1),

$$\begin{aligned} [\tau_{do}^j \Lambda_o^j(\theta)]^{-\zeta} &= (\tau_{do}^j)^{-\zeta} \int [C_o^j(\tilde{\theta}) t(\theta, \tilde{\theta})]^{-\zeta} d\Theta_o^j(\tilde{\theta}) \\ &= \exp[-\zeta(\ln \tau_{do}^j + k_{B,o}^j + m_B^j(\theta - n_{B,o}^j)^2)]. \end{aligned}$$

Then, the probability density of sourcing from firms in country o with $\tilde{\theta}$ in (6) is

$$\chi_{do}^j(\theta, \tilde{\theta}) = \frac{[\tau_{do}^j C_o^j(\tilde{\theta}) t(\theta, \tilde{\theta})]^{-\zeta}}{\sum_{o'} [\tau_{do'}^j \Lambda_{o'}^j(\theta)]^{-\zeta}} = \frac{\exp[-\zeta(\ln \tau_{do}^j + k_{A,o}^j + m_A^j(\tilde{\theta} - n_{A,o}^j)^2 + \frac{1}{2}\bar{t}(\theta - \tilde{\theta})^2)]}{\sum_{o'} \exp[-\zeta(\ln \tau_{do'}^j + k_{B,o'}^j + m_B^j(\theta - n_{B,o'}^j)^2)]}. \quad (\text{C.15})$$

For final-good producers in country d to consume sector- j goods, by Lemma A.8,

$$\begin{aligned}
(\bar{\Lambda}_o^j)^{1-\eta} &= \int C_o^j(\tilde{\theta})^{1-\eta} d\Theta_o^j(\tilde{\theta}) \\
&= \int \exp[(1-\eta)(k_{A,o}^j + m_A^j(\tilde{\theta} - n_{A,o}^j)^2)] \cdot \frac{1}{\sqrt{2\pi(\sigma^j)^2}} \exp[-\frac{1}{2}(\frac{\tilde{\theta} - \mu_o^j}{\sigma^j})^2] d\tilde{\theta} \\
&= \exp[(1-\eta)k_{A,o}^j] \times [1 - 2(1-\eta)m_A^j(\sigma^j)^2]^{-1/2} \times \exp[\frac{(1-\eta)m_A^j(\mu_o^j - n_{A,o}^j)^2}{1 - 2(1-\eta)m_A^j(\sigma^j)^2}] \\
&= \exp[(1-\eta) \underbrace{(k_{A,o}^j - \frac{1}{2(1-\eta)} \log[1 - 2(1-\eta)m_A^j(\sigma^j)^2] + \frac{m_A^j(\mu_o^j - n_{A,o}^j)^2}{1 - 2(1-\eta)m_A^j(\sigma^j)^2})}_{\equiv k_{C,o}^j}]]
\end{aligned}$$

Then, the expenditure density allocated to goods from country o with $\tilde{\theta}$ in (10) is

$$\pi_{do}^j(\tilde{\theta}) = \frac{[\tau_{do}^{Uj} C_o^j(\tilde{\theta})]^{1-\eta}}{\sum_{o'} [\tau_{do'}^{Uj} \bar{\Lambda}_{o'}^j]^{1-\eta}} = \frac{\exp[(1-\eta)(\ln \tau_{do}^{Uj} + k_{A,o}^j + m_A^j(\tilde{\theta} - n_{A,o}^j)^2)]}{\sum_{o'} \exp[(1-\eta)(\ln \tau_{do'}^{Uj} + k_{C,o'}^j)]}. \quad (\text{C.16})$$

Step 3. With the sourcing decisions specified, we can combine the market-clearing conditions (18) to (21) to arrive at a system of equations for $\{X_o^j(\theta)\}$ and $\{M_o^j(\theta)\}$, taking as given $\{w_d\}$. We discretize the domain of θ , in which case the system of equations is linear in $\{X_o^j(\theta)\}$ and $\{M_o^j(\theta)\}$ and can be easily solved.

Specifically, since the policy function (32) is invertible, summing over the market-clearing conditions (20), (21) and (22), we get

$$P_d Q_d = w_d L_d + \sum_i \int \Pi_d^i(g_d^i(\bar{\theta}); \bar{\theta}) d\bar{\Theta}_d^i(\bar{\theta}) + \sum_i \int K_d^i(g_d^i(\bar{\theta}); \bar{\theta}) d\bar{\Theta}_d^i(\bar{\theta}) \quad (\text{C.17})$$

$$= w_d L_d + \sum_i \int [1 - \phi(\theta; (g_d^i)^{-1}(\theta)) + \phi(\theta; (g_d^i)^{-1}(\theta))] \frac{1}{\eta} X_d^i(\theta) d\Theta_d^i(\theta) \quad (\text{C.18})$$

$$= w_d L_d + \sum_i \int \frac{1}{\eta} X_d^i(\theta) d\Theta_d^i(\theta). \quad (\text{C.19})$$

Substituting (C.15), (C.16) and (C.19) back to (18) and (19), we get

$$\begin{aligned}
X_o^j(\tilde{\theta}) &= \sum_d \sum_i \int \rho_d^j \pi_{do}^j(\tilde{\theta}) [(\frac{1}{\eta} + \gamma^{iL}(1 - \frac{1}{\eta})) X_d^i(\theta) + \gamma^{iL} M_d^i(\theta)] d\Theta_d^i(\theta), \\
M_o^j(\tilde{\theta}) &= \sum_d \sum_i \int \gamma^{ij} \chi_{do}^j(\theta, \tilde{\theta}) [(1 - \frac{1}{\eta}) X_d^i(\theta) + M_d^i(\theta)] d\Theta_d^i(\theta).
\end{aligned}$$

To numerically approximate this system of equations, we discretize the domain of θ into $\theta \in \tilde{\mathbb{T}} \equiv \{\vartheta_1, \vartheta_2, \dots, \vartheta_{N_\theta}\}$ and have

$$d\theta \in \{[\vartheta_1, \vartheta_2 - \frac{\vartheta_2 - \vartheta_1}{2}), [\vartheta_2 - \frac{\vartheta_2 - \vartheta_1}{2}, \vartheta_2 + \frac{\vartheta_3 - \vartheta_2}{2}), \dots, [\vartheta_{N_\theta} - \frac{\vartheta_{N_\theta} - \vartheta_{N_\theta-1}}{2}, \vartheta_{N_\theta})\}.$$

This transforms the system of equations into

$$\begin{aligned} X_o^j(\tilde{\theta})|d\tilde{\theta}| &= \sum_d \sum_i \sum_{\theta} \rho_d^j \pi_{do}^j(\tilde{\theta})|d\tilde{\theta}| \times [(\frac{1}{\eta} + \gamma^{iL}(1 - \frac{1}{\eta}))X_d^i(\theta)|d\theta| + \gamma^{iL}M_d^i(\theta)|d\theta|], \\ M_o^j(\tilde{\theta})|d\tilde{\theta}| &= \sum_d \sum_i \sum_{\theta} \gamma^{ij} \chi_{do}^j(\theta, \tilde{\theta})|d\tilde{\theta}| \times [(1 - \frac{1}{\eta})X_d^i(\theta)|d\theta| + M_d^i(\theta)|d\theta|], \end{aligned}$$

where $|\cdot|$ denotes the length of an interval. This is a linear system of equation for $X_d^i(\theta)|d\theta|$ and $M_d^i(\theta)|d\theta|$, two vectors of real numbers of length $N \times S \times N_{\theta}$. In matrix form, this is

$$\begin{bmatrix} \mathbf{X} \\ \mathbf{M} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_{X \rightarrow X} & \mathbf{B}_{M \rightarrow X} \\ \mathbf{B}_{X \rightarrow M} & \mathbf{B}_{M \rightarrow M} \end{bmatrix} \begin{bmatrix} \mathbf{X} \\ \mathbf{M} \end{bmatrix}, \quad (\text{C.20})$$

where $\mathbf{B}$'s are matrixes of coefficients depending on $\{k_{A,d}^i, m_A^i, n_{A,d}^i, \mu_d^i, \sigma^i\}$ and parameters $\{\rho_d^i, \tau_{do}^j, \tau_{do}^{Uj}, \gamma^{ij}, \gamma^{iL}, \zeta, \eta\}$.¹⁶

The linear system (C.20) is homogeneous of degree one. By adding a normalized equation that total expenditures equal one, it can be solved to obtain $\{X_o^j(\theta)\}$ and $\{M_o^j(\theta)\}$.

Step 4. Finally, given the expenditures $\{X_o^j(\theta)\}$ and $\{M_o^j(\theta)\}$, the ex-post technology distribution $\{\mu_d^i, \sigma^i\}$, and parameters $\{L_d, \gamma^{iL}, \eta\}$, we can evaluate whether the labor-market clearing condition, equation (22), is satisfied, i.e.,

$$w_d = \frac{1}{L_d} \sum_i \sum_{\theta} \gamma^{iL} [M_d^i(\theta)|d\theta| + (1 - \frac{1}{\eta})X_d^i(\theta)|d\theta|].$$

If yes, then we have found an equilibrium; if not, update wages $\{w_d\}$ and return to step 1.

Other statistics in equilibrium. Once the model is solved, we can characterize the equilibrium explicitly with a number of statistics. These would then allow us to evaluate both consumer welfare and economic efficiency.

We define consumer welfare for each country d as

$$U_d \equiv \frac{w_d L_d + \Pi_d}{P_d},$$

where $w_d L_d$ are the total outputs (GDP) of country d , Π_d are the total profits earned by domestic firms, and P_d the aggregate price index.

The total profits earned by domestic firms can be calculated as

$$\Pi_d \equiv \sum_i \int \Pi_d^i(\theta) d\Theta_d^i(\theta) = \sum_i \int \exp[-\frac{1}{2}\bar{\phi}(\frac{\theta - \alpha_d^i}{\beta^i} - \theta)^2] \cdot \frac{1}{\eta} X_d^i(\theta) d\Theta_d^i(\theta).$$

To calculate the price index, recall that the sectoral-level price indexes are

$$(P_d^j)^{1-\eta} = \Gamma(1 + \frac{1-\eta}{\lambda}) \cdot [\frac{\eta}{\eta-1}]^{1-\eta} \cdot \sum_o \exp[(1-\eta)(\ln \tau_{do}^{Uj} + k_{C,o}^j)].$$

¹⁶Note that (C.20) is homogeneous of degree 1. This can be verified by summing over o, j , and $\tilde{\theta}$ on both sides.

This gives the aggregate price index for each country as

$$\begin{aligned}
\ln P_d &= \sum_j \rho_d^j (\ln P_d^j - \ln \rho_d^j) \\
&= \sum_j \rho_d^j \left[\frac{1}{1-\eta} \ln \left(\sum_o \exp[(1-\eta)(\ln \tau_{do}^{Uj} + k_{C,o}^j)] \right) \right. \\
&\quad \left. + \frac{1}{1-\eta} \ln \left(\Gamma \left(1 + \frac{1-\eta}{\lambda} \right) \cdot \left[\frac{\eta}{\eta-1} \right]^{1-\eta} \right) - \ln \rho_d^j \right] \\
&= \sum_j \frac{\rho_d^j}{1-\eta} \ln \left(\sum_o \exp[(1-\eta)(\ln \tau_{do}^{Uj} + k_{C,o}^j)] \right) + \sum_j \ln \left(\frac{\Gamma \left(1 + \frac{1-\eta}{\lambda} \right)^{\frac{1}{1-\eta}} \cdot \left(\frac{\eta}{\eta-1} \right)}{\rho_d^j} \right) \rho_d^j.
\end{aligned}$$

In this economy, firms adopt new technologies in order to reduce the costs of being incompatible with suppliers. In equilibrium, the total costs due to technology incompatibility incurred in each country can be calculated as

$$t_d \equiv \sum_i \sum_o \sum_j \int \int \frac{t(\theta, \tilde{\theta}) - 1}{t(\theta, \tilde{\theta})} \cdot \hat{M}_{do}^{ij}(\theta, \tilde{\theta}) d\Theta_d^i(\theta) d\Theta_o^j(\tilde{\theta}), \quad (C.21)$$

where the imports from $(o, j, \tilde{\theta})$ by (d, i, θ) are

$$\hat{M}_{do}^{ij}(\theta, \tilde{\theta}) \equiv \left[M_d^i(\theta) + \left(1 - \frac{1}{\sigma} \right) X_d^i(\theta) \right] \cdot \gamma^{ij} \cdot \chi_{do}^j(\theta, \tilde{\theta}).$$

Correspondingly, the total technology adaptation costs spent by firms in each country are

$$K_d \equiv \sum_i \int K_d^i(\theta) d\Theta_d^i(\theta) = \sum_i \int \left(1 - \exp \left[-\frac{1}{2} \tilde{\phi} \left(\frac{\theta - \alpha_d^i}{\beta^i} - \theta \right)^2 \right] \right) \cdot \frac{1}{\eta} X_d^i(\theta) d\Theta_d^i(\theta).$$

C.3 Fitting the Posterior Technology Distributions

Before discussing the details of the calibration procedure, in this subsection, we explain how we solve the problem in (36) as the first step. Under the assumption that model similarity follows $\text{sim}(\mu_d^i, \mu_o^j) = \exp(-(\mu_d^i - \mu_o^j)^2)$, the fitting problem for $\{\mu_d^i\}$ solves

$$\begin{aligned}
\min_{\{\mu_d^i\}} \sum_{d,i,o,j} \left[\ln(\text{sim}(\mu_d^i, \mu_o^j)) - \ln(\text{sim}_{do}^{ij,Data}) \right]^2 \\
= \min_{\{\mu_d^i\}} \sum_{d,i,o,j} \left[(\mu_d^i - \mu_o^j)^2 - D_{do}^{ij} \right]^2
\end{aligned}$$

where $\text{sim}_{do}^{ij,Data}$ is the empirical technology similarity between country-sector pairs (d, i) and (o, j) , averaged over the period 2012-2014, and $D_{do}^{ij} \equiv -\ln(\text{sim}_{do}^{ij,Data})$ is a measure of bilateral distance. Instead of solving this original problem, which is known to be difficult for locating the global optimum, we transform it into a classical multidimensional scaling problem. This transformation recognizes the fact that if D_{do}^{ij} is a true bilateral distance matrix in Euclidean space, then by writing the summand of the objective into

$$(\mu_d^i - \bar{\mu})^2 + (\mu_o^j - \bar{\mu})^2 - 2(\mu_d^i - \bar{\mu})(\mu_o^j - \bar{\mu}) - D_{do}^{ij},$$

one can first demean the rows and columns of D_{do}^{ij} , and fit

$$\begin{aligned} \min_{\{\tilde{\mu}_d^i\}} \sum_{d,i,o,j} [\tilde{\mu}_d^i \tilde{\mu}_o^j - B_{do}^{ij}]^2, \\ \text{s.t.} \quad \sum_{d,i} \tilde{\mu}_d^i = 0 \end{aligned} \quad (\text{C.22})$$

where $\tilde{\mu}_d^i = \mu_d^i - \bar{\mu}$, and B_{do}^{ij} is the corresponding entry in the demeaned matrix B , defined as $B = -\frac{1}{2}CDC$. Here, $D = (D_{do}^{ij})$ is the bilateral distance matrix, and $C = I - \frac{1}{NS}J_{NS}$ is the centering matrix, with J_{NS} being an $NS \times NS$ matrix of ones.

The solution to the transformed problem (C.22) is uniquely given by the eigenvector of B associated with the largest eigenvalue, up to a scaling factor.¹⁷ We further scale this eigenvector so that the standard deviation of the off-diagonal elements of the generated similarity matrix between country-sectors matches the data counterpart.

C.4 Algorithm for Calibration

This subsection discusses the details of calibration to recover the primitive of the model. Aside the parameters calibrated externally, we jointly determine the remaining parameters on technology distributions $\{\bar{\mu}_o^j, \bar{\sigma}^j\}$, technology adaptation costs $\bar{\phi}$, input incompatibility costs $\bar{t}$, and those determining production and trade $\{\tau_{do}^j, \tau_{do}^{Uj}, A_d^i\}$, leaning on the equilibrium conditions of the model.

The ex-ante technology distribution of countries are by assumption not observed. To calibrate $\{\bar{\mu}_d^i, \bar{\sigma}^i\}$, we use two pieces of information: the ex-post technology distribution, and the one-to-one mapping from the ex-ante to the ex-post distributions characterized in Proposition 7. As the mapping depends on all model primitives (such as trade costs) and the equilibrium wage, the ex-ante distributions cannot be recovered independent of the rest of the model. Instead, we recover the ex-ante distribution in two steps.

In the first step, we choose the parameters governing the ex-post distributions, $\{\mu_d^i, \sigma^i\}$, to match bilateral similarities between country-sectors, as in Appendix C.3. This step can be carried out without knowing the primitives of the model. Conditional on their own technology, firms' sourcing decisions only depend on the ex-post distributions. We can therefore calibrate the primitives of the model governing trade using only the ex-post distributions and other data. In the second step, we calibrate $\bar{\phi}$ and recover the ex-ante distributions using equation (33).

Trade costs and distribution of production techniques. With the ex-post technology distributions $\{\Theta_o^j\}$ at hand, we design a nested algorithm to jointly calibrate the parameters $\{A_d^i\}$, which determines the productivity of (d, i) , to match the output share of (d, i) in industry i , and calibrate $\{\tau_{do}^j, \tau_{do}^{Uj}\}$ to match the trade shares of intermediate and final goods, respectively. We lay out the algorithm as follows before discussing several details.

The nested algorithm. We describe the nested calibration algorithm below:

(A) Choose a $\bar{t}$

(a) Choose a set of parameters $\{\tau_{do}^j, \tau_{do}^{Uj}, A_d^i\}$

¹⁷See e.g., Chapter 12 of Borg and Groenen (2007).

- (b) Solve the equilibrium given the parameters and the ex-post technology distributions $\{\mu_d^i, \sigma^i\}$
 - (c) Evaluate the trade shares of intermediate and final goods at the equilibrium. If they match their data counterparts, proceed to Step (B); if not, adjust $\{\tau_{do}^j, \tau_{do}^{Uj}, A_d^i\}$ and return to Step (A)(b).
- (B) Simulate 10,000 Chinese firms from each of the 19 sectors and regress the extensive margin of importing from each country o on the similarity between the technology of a firm and the technology of country o , controlling for firm and country-industry fixed effects.
- (C) Compare the model-based regression coefficient in Step 2 to its data counterpart (Column 3 of Table B.9). If they are close enough, exit; if not, adjust $\bar{t}$ and return to Step (A).

In Step (A)(b), we need to solve the trade equilibrium *conditional on* the ex-post distribution and parameters. This differs from the algorithm developed in Section C.2, which solves the general equilibrium (in which technology choice is endogenous). We modify the algorithm accordingly. Specifically, the system of equations from Step 1 of the algorithm in Section C.2 can be now simplified into

$$\begin{aligned}
m_A^i &= \sum_j \gamma^{ij} m_B^j \\
n_{A,d}^i &= \sum_{j,o} \hat{\chi}_{do}^{ij} n_{B,o}^j \\
k_{A,d}^i &= \ln C_d^i(\mu_d^i) - m_A^i [\sum_{j,o} \hat{\chi}_{do}^{ij} (\mu_d^i - n_{B,o}^j)]^2
\end{aligned}$$

where

$$\begin{aligned}
m_B^j &= \frac{\frac{1}{2}\bar{t}[1 + 2\zeta m_A^j (\sigma^j)^2]}{1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2}, \\
n_{B,o}^j &= \frac{1}{1 + 2\zeta m_A^j (\sigma^j)^2} \mu_o^j + \frac{2\zeta m_A^j (\sigma^j)^2}{1 + 2\zeta m_A^j (\sigma^j)^2} n_{A,o}^j \\
k_{B,o}^j &= k_{A,o}^j + \frac{1}{2\zeta} \log[1 + 2\zeta(m_A^j + \frac{1}{2}\bar{t})(\sigma^j)^2] + \frac{m_A^j (\mu_o^j - n_{A,o}^j)^2}{1 + 2\zeta m_A^j (\sigma^j)^2} \\
\hat{\chi}_{do}^{ij} &\equiv \frac{\gamma^{ij} m_B^j}{\sum_{j'} \gamma^{ij'} m_B^{j'}} \times \frac{\exp[-\zeta(\ln \tau_{do}^j + k_{B,o}^j + m_B^j (\mu_d^i - n_{B,o}^j)^2)]}{\sum_{o'} \exp[-\zeta(\ln \tau_{do'}^j + k_{B,o'}^j + m_B^j (\mu_d^i - n_{B,o'}^j)^2)]} \\
\ln C_d^i(\mu_d^i) &= \ln\left(\frac{\Xi^i}{A_d^i}\right) + \gamma^{iL} \ln(w_d) - \zeta^{-1} \sum_j \gamma^{ij} \ln\left(\sum_o \exp[-\zeta(\ln \tau_{do}^j + k_{B,o}^j + m_B^j (\mu_d^i - n_{B,o}^j)^2)]\right)
\end{aligned}$$

Given wages $\{w_d\}$ and parameters $\{\tau_{do}^j, \tau_{do}^{Uj}, A_d^i\}$, this is a contraction mapping with $\bar{t}$ being not too large and can be efficiently solved.

The remaining steps to solve the equilibrium follow exactly with

$$\begin{aligned}
\chi_{do}^j(\theta, \tilde{\theta}) &= \frac{\exp[-\zeta(\ln \tau_{do}^j + k_{A,o}^j + m_A^j(\tilde{\theta} - n_{A,o}^j)^2 + \frac{1}{2}\bar{t}(\theta - \tilde{\theta})^2)]}{\sum_{o'} \exp[-\zeta(\ln \tau_{do'}^j + k_{B,o'}^j + m_B^j(\theta - n_{B,o'}^j)^2)]} \\
\pi_{do}^j(\tilde{\theta}) &= \frac{\exp[(1-\eta)(\ln \tau_{do}^{Uj} + k_{A,o}^j + m_A^j(\tilde{\theta} - n_{A,o}^j)^2)]}{\sum_{o'} \exp[(1-\eta)(\ln \tau_{do'}^{Uj} + k_{C,o'}^j)]} \\
X_o^j(\tilde{\theta})|d\tilde{\theta}| &= \sum_d \sum_i \sum_{\tilde{\theta}} \rho_d^j \pi_{do}^j(\tilde{\theta})|d\tilde{\theta}| \times [(\frac{1}{\eta} + \gamma^{iL}(1 - \frac{1}{\eta}))X_d^i(\theta)|d\theta| + \gamma^{iL}M_d^i(\theta)|d\theta|] \\
M_o^j(\tilde{\theta})|d\tilde{\theta}| &= \sum_d \sum_i \sum_{\tilde{\theta}} \gamma^{ij} \chi_{do}^j(\theta, \tilde{\theta})|d\tilde{\theta}| \times [(1 - \frac{1}{\eta})X_d^i(\theta)|d\theta| + M_d^i(\theta)|d\theta|].
\end{aligned} \tag{C.23}$$

The equilibrium is reached when wages $\{w_d\}$ satisfy the labor-market clearing condition given by

$$w_d = \frac{1}{L_d} \sum_i \sum_{\tilde{\theta}} \gamma^{iL} [M_d^i(\theta)|d\theta| + (1 - \frac{1}{\eta})X_d^i(\theta)|d\theta|].$$

In equilibrium, we calculate the trade shares in final and intermediate goods as

$$\begin{aligned}
\frac{\hat{M}_{do}^j}{\sum_{o'} \hat{M}_{do'}^j} \text{ with } \hat{M}_{do}^j &\equiv \sum_{\tilde{\theta}} \sum_i \sum_{\tilde{\theta}} \gamma^{ij} \chi_{do}^j(\theta, \tilde{\theta})|d\tilde{\theta}| \times [(1 - \frac{1}{\eta})X_d^i(\theta)|d\theta| + M_d^i(\theta)|d\theta|] \\
\frac{\hat{X}_{do}^j}{\sum_{o'} \hat{X}_{do'}^j} \text{ with } \hat{X}_{do}^j &\equiv \sum_{\tilde{\theta}} \sum_i \sum_{\tilde{\theta}} \rho_d^j \pi_{do}^j(\tilde{\theta})|d\tilde{\theta}| \times [(\frac{1}{\eta} + \gamma^{iL}(1 - \frac{1}{\eta}))X_d^i(\theta)|d\theta| + \gamma^{iL}M_d^i(\theta)|d\theta|]
\end{aligned}$$

We match these two statistics to their data counterparts constructed from WIOTs, both averaged over 2012-2014.

In Step (B), for each value of $\bar{t}$, we simulate 190,000 Chinese firms ($d = CHN$), 10,000 from each sector i with θ drawn from the calibrated ex-post distribution Θ_d^i . With parameters $\{\tau_{do}^j, \tau_{do}^{Uj}, A_d^i\}$ calibrated, we can explicitly determine the input sourcing and patent citation patterns for each firm. Specifically, a firm with technology location θ from (d, i) would source input from (o, j) with probability $\tilde{\chi}_{do}^j(\theta)$ given by

$$\tilde{\chi}_{do}^j = \int \chi_{do}^j(\theta, \tilde{\theta}) d\Theta_o^j(\tilde{\theta}),$$

with $\chi_{do}^j(\theta, \tilde{\theta})$ given by (C.23), and the average similarity between the technology of the firm and that of (o, j) is

$$\frac{1}{S} \sum_j \text{sim}(\theta, \mu_o^j) = \frac{1}{S} \sum_j \exp(-(\theta - \mu_o^j)^2).$$

Then, we regress the realized extensive margin of importing from each country o on the average similarity constructed above, controlling for firm and country-industry fixed effects. We calibrate $\bar{t} = 12.2$ to match this coefficient to 0.021, the extensive-margin import-similarity correlation obtained from the regression results in Column 3 of Table B.9.

Notice that the above algorithm does not rely on the value of parameter $\bar{\phi}$. This is because that conditional on the ex-post distribution, varying $\bar{\phi}$ only affects the split of firms'

markup between profit to the representative consumer and the expense on adaptation. In other words, $\bar{\phi}$ only affects the welfare of agents but not other equilibrium prices and allocations. For this reason, we can calibrate $\bar{\phi}$ separately.

Calibrate $\bar{\phi}$ and the ex-ante distributions. With all other model primitives calibrated, the last step is to calibrate $\bar{\phi}$, the parameter on adaptation costs. We calibrate it by matching the elasticity of similarity to tariffs (Column 3 of Table 1). As shown in Proposition 2, conditional on trade shares and γ^{ij} , this elasticity identifies $\bar{\phi}$. To obtain the elasticities in the model, we replicate the variation in MFN tariffs observed in the data, scaling the model's trade cost $\{\tau_{do}^j\}$ by the standard deviation of the MFN tariff for each (d, j) over time. We then treat the calibrated equilibrium and the counterfactual as two periods and regress the model's technology similarity between d and (o, j) on the logarithms of τ_{do}^j , controlling for d - o - j , o - j - t , and d - t fixed effects, mirroring the empirical specification. We adjust $\bar{\phi}$ until the model's elasticity matches the empirical coefficient of -0.007 , yielding a calibrated value of $\bar{\phi} = 0.58$.

With $\bar{\phi}$ specified, we can then recover the ex-ante technology distribution by equations (32) and (33), i.e.,

$$\begin{aligned}\bar{\mu}_d^i &= \frac{1}{\beta^i} \cdot \mu_d^i - \frac{1 - \beta^i}{\beta^i} \cdot n_{A,d}^i \\ \bar{\sigma}^i &= \frac{1}{\beta^i} \cdot \sigma^i \\ \text{with } \beta^i &= \frac{\frac{1}{2}\bar{\phi}}{\frac{1}{2}\bar{\phi} + (\eta - 1)m_A^i}.\end{aligned}$$

This completes the calibration.